

# The Foundation of Projection Relativity II

## Geometric Constraints on the Non-Abelian Phase Fiber

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### Abstract

This manuscript extends the projection architecture introduced in Projection Relativity I (PR-I) to the microscopic electroweak and flavor sectors. The central question is whether the particle ledger usually treated as a collection of independent inputs can instead be organized as a compact projection structure. We investigate the parent compact non-Abelian phase space

$$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L,$$

and evaluate the conditions under which this parent fiber locks into the residual electromagnetic boundary. In this formulation, the photon is the unbroken residual projection, while the  $W^\pm$  and  $Z$  modes arise as massive compact-projection excitations of the broken electroweak directions.

Within this lifted compact fiber, the standard electroweak charge assignments, charged-current ladder structure, neutral-current  $Z$ -couplings, Fermi limit, and tree-level relation  $\rho_{\text{EW}} = 1$  are recovered as projection-locking consequences. We then explore the compact boundary deficit as a geometric source for the weak mixing angle and electroweak scale, obtaining a zero-new-parameter closure candidate for  $\sin^2 \theta_W$ ,  $v_{\text{EW}}$ , and  $G_F$ . The same compact-boundary ledger naturally yields a three-generation projection structure,

$$N_{\text{gen}}^{\text{PR}} = \dim[(SU(2)_L \times U(1)_Y)/U(1)_{\text{em}}] = 3.$$

By evaluating sector-specific spectral overlaps across these projection sheets, we obtain charged-lepton, quark, CKM, PMNS, and neutrino mass-sector candidates without using the corresponding target observables as generation inputs. The charged-lepton sector gives a precision Koide-sheet closure candidate, the quark sector yields a compact-boundary hierarchy with PR-derived heavy-quark thresholds and QCD running anchors, and the neutral-lepton sector selects a normal-ordering Majorana-dominant candidate with a compact-boundary prediction for the neutrinoless double-beta effective mass. Throughout, target measurements are used only as diagnostic references after the projection formulas have generated their values. The result is not presented as a completed electroweak precision theorem, but as a test-driven, zero-new-parameter compact-boundary closure program for organizing the electroweak and flavor structure of the Standard Model from projection geometry.

## 1 Introduction: Compact Electroweak Projection and Flavor Closure

Projection Relativity I (PR-I) [1] introduced a projection-based architecture in which macroscopic gravitational response, displacement-generated inertia, compact electromagnetic phase, finite-core regularization,

and observational diagnostics arise as sectors of a single master projection field,

$$\boxed{\Psi(x, \xi)}. \quad (1)$$

The minimal internal geometry used in PR-I was

$$\boxed{\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1}, \quad (2)$$

where the radial coordinate  $w$  supplied the stiffness branch and the compact coordinate  $\theta$  supplied the residual electromagnetic phase. PR-I fixed the radial projection branch, established the compact electromagnetic boundary normalization, derived a finite spectral core for strong-field regularization, and provided a public symbolic/numerical verification package for the main macroscopic claims.

The present manuscript begins from that locked PR-I boundary ledger and asks whether the microscopic electroweak and flavor structure can be treated in the same projection language. The working hypothesis is not that the Standard Model [2, 3, 4] is discarded, but that its electroweak and flavor ledger may be reorganized as a compact-boundary projection structure. In particular, PR-II investigates whether weak mixing, the Fermi scale, generation count, flavor hierarchy, CKM mixing [5], PMNS mixing [6], and neutrino mass splittings can be generated from the same compact spectral data rather than introduced as unrelated numerical inputs. The central extension is the compact electroweak lift

$$\boxed{\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L}. \quad (3)$$

Here  $S_Y^1$  is the compact hypercharge phase and  $SU(2)_L$  is the compact non-Abelian weak-isospin fiber. The electromagnetic sector derived in PR-I is not altered. Instead, it is recovered as the residual locked boundary of the parent compact electroweak fiber:

$$\boxed{SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}, \quad Q = T_3 + \frac{Y}{2}}. \quad (4)$$

In this interpretation, the photon is the unbroken residual projection mode, while  $W^\pm$  and  $Z$  are massive compact-projection excitation modes associated with the broken electroweak directions.

## 1.1 From Electroweak Projection to Flavor Ledger

The first part of PR-II establishes the compact non-Abelian phase projection itself. The  $SU(2)_L$  generators obey  $[T_a, T_b] = i\epsilon_{abc}T_c$ , and the weak charged-current ladder is generated by  $T^\pm = T_1 \pm iT_2$ .

Projection locking then produces the ordinary electroweak charge assignments, the charged-current topology, the neutral-current  $Z$ -couplings, the Fermi limit, and the tree-level relation  $\rho_{\text{EW}} = 1$ .

The second part of the manuscript asks whether the numerical electroweak scales can be tied to compact boundary data. The weak mixing angle is treated as a compact boundary-deficit effect. The electroweak scale is then generated through a compact spectral hierarchy action built from the PR-I boundary constants and the reduced Planck scale. This produces a zero-new-parameter weak-scale candidate:

$$\boxed{v_{\text{EW}}^{\text{PR}} = 246.219595890 \text{ GeV}}, \quad (5)$$

and

$$\boxed{G_F^{\text{PR}} = 1.166379220 \times 10^{-5} \text{ GeV}^{-2}}. \quad (6)$$

These values are generated before comparison with the observed Fermi scale; the reference measurements are used only afterward as diagnostics.

The third part of PR-II extends the compact projection ledger to flavor. The number of generations is identified with the number of broken compact electroweak projection sheets:

$$N_{\text{gen}}^{\text{PR}} = \dim \left[ \frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (7)$$

This provides a geometric origin candidate for the three-generation ledger. Sector-specific compact overlap operators are then constructed for charged leptons, quarks, and neutrinos. The charged-lepton sector yields a Koide-sheet precision candidate [7]; the quark sector yields heavy-threshold and light-quark hierarchy candidates; and the neutral-lepton sector yields a normal-ordering Majorana-dominant neutrino candidate.

## 1.2 Scope of the Claims

The claims in this manuscript are deliberately separated into four categories.

First, some results are structural identities. These include the  $SU(2)_L$  algebra, the residual electromagnetic generator  $Q = T_3 + Y/2$ , the massless photon eigenmode, the  $W/Z$  tree-level mass ledger, the Fermi matching condition, and anomaly cancellation [8, 9] for the chiral fermion representation.

Second, some results are closure candidates. These include the compact-boundary weak mixing angle, strong-coupling anchor, generation-origin rule, sector-specific overlap operators, CKM structure, PMNS structure, and neutrino mass-splitting candidate.

Third, some results are precision candidates. These include the weak-scale compact hierarchy candidate, charged-lepton Koide-sheet [7] closure, and heavy-quark threshold candidates.

Fourth, some results are diagnostic ledgers. These include QCD threshold matching, higher-loop running, and comparisons to reference electroweak, flavor, quark, and neutrino data. These reference values are not used to generate the PR-II candidates; they are used only after the formulas have produced their outputs.

The manuscript therefore does not present PR-II as a completed electroweak precision theorem. It presents PR-II as a test-driven compact-boundary closure program. The central claim is more precise: a large portion of the electroweak and flavor ledger can be generated from PR compact-boundary data without introducing new fitted parameters, and the remaining obligations are explicit.

## 1.3 Reproducibility and Test-Driven Method

The development of PR-II follows the same test-driven standard used in PR-I. Each construction is paired with a numerical or symbolic validation harness. The harnesses do not merely confirm the desired equations; they also include negative controls designed to reject common incorrect alternatives, such as Abelianized  $SU(2)$ , massive photon leakage, wrong Fermi normalization, nonunitary flavor rotations, anomaly-violating fermion ledgers, inverted compact suppression factors, or hidden use of target observables.

**Reproducibility statement.** The PR-II electroweak/flavor chain is supported by step-specific validation scripts and more than six hundred automated checks. The harness ledger verifies the compact electroweak lift, projection locking, weak mixing, weak-scale closure, charged and neutral currents, anomaly cancellation, generation replication, flavor hierarchy, charged-lepton precision closure, QCD running, heavy-quark thresholds, CKM and PMNS mixing, and the neutrino mass-sector candidate. Target observables such as  $G_F$ ,  $v_{\text{EW}}$ ,  $M_W$ ,  $M_Z$ , charged-lepton masses, quark masses, CKM angles, PMNS angles,  $\alpha_s(M_Z)$ , and neutrino mass splittings are not used as generation inputs. They enter only as diagnostic references after the PR-II formulas generate their outputs. The code and generated outputs are maintained in the public repository: [github.com/oshetskiresearch/Projection\\_Relativity](https://github.com/oshetskiresearch/Projection_Relativity).

This test-driven structure is central to the paper. It prevents the projection program from becoming an unconstrained reinterpretation of known particle data. Each stage must either close from the existing compact-boundary ledger or be explicitly marked as structural, diagnostic, or incomplete. When a step requires an additional correction term, that term must be written in PR-native compact-boundary language and tested against negative controls. If a sector cannot be closed without importing an external numerical input, it remains a diagnostic ledger rather than a zero-new-parameter closure.

## 1.4 Organization of the Manuscript

Section 2 introduces the compact electroweak lift and the  $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}$  projection-locking structure. Subsequent sections derive the  $W/Z$  mass matrix, weak mixing closure, Fermi-scale ledger, and compact weak-scale hierarchy candidate. The charged-current and neutral-current sections show how weak decay and  $Z$ -couplings arise from compact non-Abelian projection modes.

The middle sections construct the fermion representation ledger, verify anomaly cancellation, identify a three-generation projection-sheet origin, and define sector-specific flavor overlaps. The charged-lepton sections derive the Koide-sheet rule and a precision charged-lepton mass candidate. The quark sections build a compact strong-coupling anchor, QCD running ledger, heavy-quark threshold candidates, light-quark hierarchy candidate, and CKM mixing structure. The final flavor sections develop the PMNS lepton-mixing candidate, neutrino mass-sector closure, and Majorana phase boundary.

The manuscript closes with an audit map separating structural identities, closure candidates, precision candidates, diagnostic ledgers, and remaining derivation obligations. The purpose is not only to report numerical agreement, but to make the full electroweak/flavor projection chain falsifiable, reproducible, and inspectable.

## 2 PR-I Boundary Ledger and the Electroweak Lift

Projection Relativity II begins from the boundary ledger fixed in PR-I. The purpose of this section is not to rederive the macroscopic theory, but to identify the specific PR-I quantities that are inherited by the electroweak and flavor construction. These quantities are treated as fixed geometric data, not as electroweak fit parameters. In PR-I, the minimal internal projection geometry was

$$\mathcal{M}_{\text{int}} = \mathbb{R}_w \times S_\theta^1. \quad (8)$$

The non-compact coordinate  $w$  supplied the radial stiffness direction, while the compact coordinate  $\theta$  supplied the residual electromagnetic phase boundary. The radial stiffness branch was fixed by the projection-trace condition,

$$O_X = -\frac{d^2}{dw^2} + 1 + w^2 + \frac{3}{4}w^4. \quad (9)$$

The corresponding radial gap is

$$\mu_{\text{min}}^2 = 3.052966743096. \quad (10)$$

This radial gap is part of the strong-field and finite-core sector. It is not refit in the electroweak construction. The compact electromagnetic boundary from PR-I is encoded in the boundary-resolved compact stiffness

$$q_{\text{bc}} = 10.905007182855176, \quad (11)$$

and the compact boundary cofactor

$$c_{\text{bc}} = \frac{3354902985}{4211081216} = 0.796684464847899. \quad (12)$$

Together they reproduce the residual compact electromagnetic normalization,

$$\boxed{\alpha_{\text{PR,bc}}^{-1} = 4\pi q_{\text{bc}} = 137.036361812007.} \quad (13)$$

This compact electromagnetic boundary is preserved in PR-II. The weak sector is not introduced by altering the PR-I electromagnetic result. Instead, PR-II asks whether that residual  $U(1)_{\text{em}}$  boundary is the locked remnant of a larger compact electroweak fiber. The compact determinant suppression ratio inherited from PR-I is

$$r_G = \frac{c_{\text{bc}}}{q_{\text{bc}}} = 0.073056757459128. \quad (14)$$

This ratio will appear repeatedly in the flavor ledger as the compact generation-sheet suppression factor. The residual compact boundary deficit is

$$\boxed{\delta_{\text{bc}} = 1 - c_{\text{bc}} = 0.203315535152101.} \quad (15)$$

The normalized compact boundary deficit is

$$\boxed{\varepsilon_{\text{bc}} = \frac{1 - c_{\text{bc}}}{q_{\text{bc}}} = 0.018644236701811.} \quad (16)$$

At this stage,  $\varepsilon_{\text{bc}}$  is only an inherited compact boundary quantity. Subsequent sections test whether this normalized boundary deficit fixes the weak mixing angle through the compact electroweak projection.

**Table 1:** PR-I boundary quantities inherited by PR-II.

| Quantity                                                      | Value              | Role in PR-II                                    |
|---------------------------------------------------------------|--------------------|--------------------------------------------------|
| $\mu_{\text{min}}^2$                                          | 3.052966743096     | Radial stiffness gap inherited from PR-I.        |
| $q_{\text{bc}}$                                               | 10.905007182855176 | Compact boundary stiffness.                      |
| $c_{\text{bc}}$                                               | 0.796684464847899  | Compact boundary cofactor.                       |
| $\alpha_{\text{PR,bc}}^{-1}$                                  | 137.036361812007   | Residual electromagnetic boundary normalization. |
| $r_G = c_{\text{bc}}/q_{\text{bc}}$                           | 0.073056757459128  | Compact generation-sheet suppression ratio.      |
| $\varepsilon_{\text{bc}} = (1 - c_{\text{bc}})/q_{\text{bc}}$ | 0.018644236701811  | Normalized compact boundary deficit.             |

## 2.1 Electroweak Lift of the Compact Phase Sector

The PR-II microscopic extension lifts the compact phase sector from the residual electromagnetic boundary into a parent compact electroweak fiber,

$$\boxed{\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L.} \quad (17)$$

Here  $S_Y^1$  is the compact hypercharge phase and  $SU(2)_L$  is the compact non-Abelian weak-isospin fiber. The radial factor  $\mathbb{R}_w$  and its stiffness branch are inherited from PR-I. The new structure is the non-Abelian compact phase factor  $SU(2)_L$  [10, 4]. The corresponding internal electroweak operator is written schematically as

$$\boxed{O_{\text{int}}^{\text{EW}} = O_X + O_Y + O_L.} \quad (18)$$

The radial operator is the PR-I stiffness operator,

$$O_X = -\frac{d^2}{dw^2} + 1 + w^2 + \frac{3}{4}w^4. \quad (19)$$

The compact hypercharge operator is

$$O_Y = -\frac{1}{R_Y^2} \frac{d^2}{d\theta_Y^2}, \quad (20)$$

and the compact weak-isospin operator is

$$O_L = -\frac{1}{R_L^2} \Delta_{SU(2)}. \quad (21)$$

The  $SU(2)_L$  compact fiber carries non-Abelian generators  $T_a$  satisfying

$$[T_a, T_b] = i\epsilon_{abc} T_c. \quad (22)$$

In the fundamental representation,

$$T_a = \frac{\sigma_a}{2}, \quad (23)$$

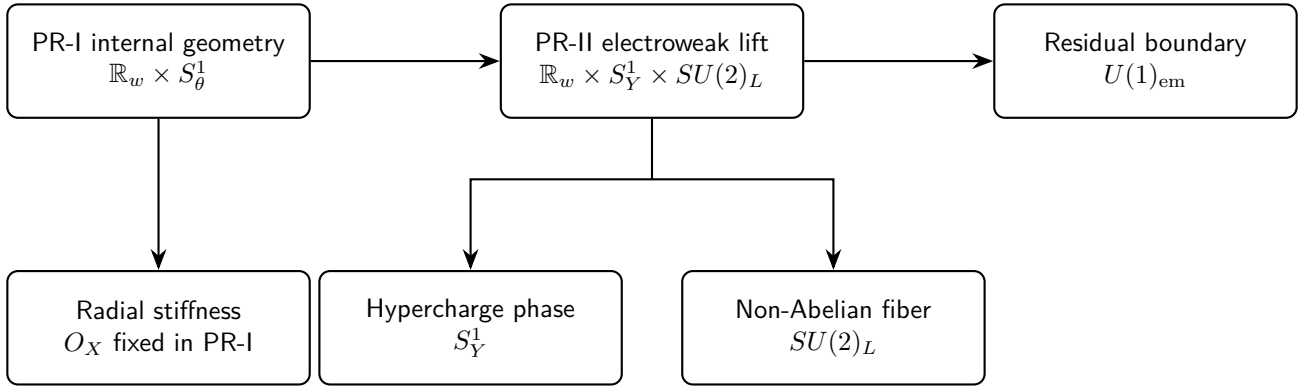
where  $\sigma_a$  are the Pauli matrices. The charged-current compact ladder operators are  $T^\pm = T_1 \pm iT_2$ . This non-Abelian ladder structure is the first microscopic distinction between the weak compact fiber and the residual electromagnetic boundary. The compact weak-isospin spectrum is generated by the  $SU(2)$  Laplacian,

$$O_L D_{mn}^j = \frac{j(j+1)}{R_L^2} D_{mn}^j. \quad (24)$$

Thus the compact weak spectrum is

$$\lambda_j^{(L)} = \frac{j(j+1)}{R_L^2}. \quad (25)$$

The singlet  $j = 0$  is weak-isospin neutral, the doublet  $j = 1/2$  supplies the weak matter representation, and the adjoint  $j = 1$  supplies the weak projection modes. The operational diagram of the relationship from PR-I internal geometry to PR-II electroweak lift is illustrated in Figure 1



**Figure 1:** PR-II compact electroweak lift preserves the PR-I electromagnetic boundary. The radial stiffness is inherited from the macroscopic theory, while the non-Abelian fiber and hypercharge phase are introduced to govern the microscopic sector.

## 2.2 Residual Electromagnetic Boundary

The residual electromagnetic boundary is selected via projection locking, reducing the parent electroweak symmetry [11, 12]:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \quad (26)$$

The resulting unbroken generator dictates the residual charge:

$$Q = T_3 + \frac{Y}{2}. \quad (27)$$

We note that this is the exact same electromagnetic boundary whose compact normalization was established in PR-I. Rather than discarding or replacing the PR-I electromagnetic sector, this framework reinterprets it: electromagnetism is simply the massless residual projection that survives after the parent compact electroweak fiber locks.

Applying this generator naturally recovers standard charge assignments. For a left-handed lepton doublet with  $Y = -1$ , the matrix evaluates to:

$$Q = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}. \quad (28)$$

Similarly, for a left-handed quark doublet with  $Y = 1/3$ :

$$Q = \begin{pmatrix} 2/3 & 0 \\ 0 & -1/3 \end{pmatrix}. \quad (29)$$

Consequently, the conventional electromagnetic charges emerge from the compact projection-locking relation as described in equation 27. Translated into the language of PR-II, the photon operates as the unbroken residual compact projection mode. Furthermore, the  $W^\pm$  and  $Z$  modes are no longer treated as unrelated, primitive objects; instead, they manifest as massive compact-projection excitation modes associated with the broken directions of the parent electroweak fiber. This interpretation preserves the standard low-energy electroweak ledger, replacing independent parameter insertion, where possible, with compact-boundary closure tests.

### 2.3 Input Discipline

The purpose of PR-II is to test whether the electroweak and flavor ledger can be generated from compact projection data without introducing new fitted microscopic parameters. Accordingly, the following quantities are not allowed as generation inputs in the PR-II closure chain [13, 14]:

**Table 2:** The forbidden generation inputs. These quantities may be used only after a PR-II formula has generated its value, as diagnostic reference data.

| Physics Sector              | Excluded Target Observables                       |
|-----------------------------|---------------------------------------------------|
| Electroweak Scale & Bosons  | $G_F, v_{EW}, M_W, M_Z$                           |
| Charged Lepton Masses       | $m_e, m_\mu, m_\tau$                              |
| Quark Masses                | $m_u, m_d, m_s, m_c, m_b, m_t$                    |
| Flavor Mixing & QCD Anchors | $V_{CKM}, U_{PMNS}, \alpha_s(M_Z), \Lambda_{QCD}$ |
| Neutrino Mass Sector        | $\Delta m_{21}^2, \Delta m_{31}^2, \sum_i m_i$    |

The starting data allowed are the PR-I boundary ledger, the compact group structure, and the fundamental constants. In particular, PR-II may only use starting points as summarized in Table 3. This input discipline is central to the manuscript. Whenever a result is compared to an electroweak, flavor, quark, or neutrino measurement, that measurement must be entered only as a diagnostic reference. If a proposed closure requires importing a target observable as an input, it is not a zero-new-parameter PR-II closure. It must instead be classified as a diagnostic ledger or incomplete sector.

**Table 3:** The PR-II Allowed Input Ledger. These are the permitted geometric and topological starting inputs; target observables are forbidden from this list.

| PR Term            | Description                    | Origin                                   | Value                              |
|--------------------|--------------------------------|------------------------------------------|------------------------------------|
| $q_{bc}$           | Compact boundary stiffness     | PR-I EM Boundary                         | 10.905007182855176                 |
| $c_{bc}$           | Compact boundary cofactor      | PR-I EM Boundary                         | 0.796684464847899                  |
| $r_G$              | Compact generation suppression | PR-I Ratio ( $c_{bc}/q_{bc}$ )           | 0.073056757459128                  |
| $\varepsilon_{bc}$ | Normalized boundary deficit    | PR-I Deficit ( $((1 - c_{bc})/q_{bc})$ ) | 0.018644236701811                  |
| $\mu_{\min}^2$     | Radial stiffness gap           | PR-I Radial Operator                     | 3.052966743096                     |
| $T_3$              | Weak-isospin generator         | $SU(2)_L$ Compact Fiber                  | Rep. dependent                     |
| $Y$                | Hypercharge generator          | $U(1)_Y$ Compact Fiber                   | Rep. dependent                     |
| $C_2^{SU(2)}$      | Weak-isospin Casimir           | $SU(2)_L$ Group Topology                 | 3/4 (doublets)                     |
| $C_2^{SU(3)}$      | Color Casimir                  | $SU(3)_c$ Group Topology                 | 4/3 (triplets)                     |
| $\bar{M}_{Pl}$     | Reduced Planck mass            | Fundamental Constants                    | $\approx 2.435 \times 10^{18}$ GeV |

### 3 Compact Non-Abelian Phase Projection

The electroweak lift introduced the parent compact fiber

$$\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L. \quad (30)$$

This section isolates the fundamental microscopic extension introduced by PR-II; the compact non-Abelian phase projection carried by  $SU(2)_L$ . The residual electromagnetic boundary inherited from PR-I is Abelian, with its compact phase governed by a single commuting generator. The weak sector, by contrast, is not merely a secondary copy of the electromagnetic phase. It is a non-Abelian compact phase fiber, characterized by internal orientation, generator non-commutativity, a distinct charged-current ladder structure, and a discrete compact spectral hierarchy.

#### 3.1 Group-Valued Internal Phase

In PR-I, the compact electromagnetic phase was represented by a circle coordinate,  $\theta \in S^1$ . The corresponding compact phase factor was Abelian  $e^{i\theta} \in U(1)$ . In PR-II, the weak phase is instead group-valued:

$$g_L \in SU(2)_L. \quad (31)$$

The master projection field may therefore be written schematically as

$$\Psi = \Psi(x, w, \theta_Y, g_L), \quad (32)$$

where  $x$  denotes observable spacetime coordinates,  $w$  is the radial stiffness coordinate,  $\theta_Y$  is the compact hypercharge phase, and  $g_L$  is the compact weak-isospin orientation. The weak projection Hilbert space is

$$\mathcal{H}_L = L^2(SU(2)_L, d\mu_H), \quad (33)$$

where  $d\mu_H$  is the Haar measure on  $SU(2)$ . By the Peter–Weyl decomposition, square-integrable functions on  $SU(2)$  decompose into matrix elements of irreducible representations:

$$\mathcal{H}_L = \bigoplus_{j=0, \frac{1}{2}, 1, \dots} \mathcal{H}_j \otimes \mathcal{H}_j^*. \quad (34)$$



A convenient basis is given by the Wigner matrix elements,

$$D_{mn}^j(g_L), \quad m, n = -j, \dots, j. \quad (35)$$

So the weak compact fiber naturally carries singlet, doublet, and adjoint spectral sectors:

$$j = 0, \quad j = \frac{1}{2}, \quad j = 1. \quad (36)$$

### 3.2 Non-Abelian Generator Algebra

The  $SU(2)_L$  generators are denoted  $T_a$ , with  $a = 1, 2, 3$ . They obey the non-Abelian algebra  $[T_a, T_b] = i\epsilon_{abc}T_c$ . In the fundamental representation,

$$T_a = \frac{\sigma_a}{2}, \quad (37)$$

where  $\sigma_a$  are the Pauli matrices. The normalization is

$$\text{Tr}(T_a T_b) = \frac{1}{2}\delta_{ab}. \quad (38)$$

The fundamental Casimir is

$$\boxed{\sum_{a=1}^3 T_a^2 = \frac{3}{4}I.} \quad (39)$$

This identifies the weak doublet as the  $j = 1/2$  compact spectral representation. The adjoint weak projection modes transform in the  $j = 1$  representation. The adjoint Casimir is

$$C_2(j = 1) = 1(1 + 1) = 2. \quad (40)$$

Therefore the compact weak fiber distinguishes the weak matter doublet from the weak projection modes by representation class:

$$\boxed{j = \frac{1}{2} \quad \text{for weak matter doublets,} \quad j = 1 \quad \text{for weak projection modes.}} \quad (41)$$

### 3.3 Compact Weak-Isospin Operator

The compact weak-isospin operator is the  $SU(2)$  Laplacian,

$$O_L = -\frac{1}{R_L^2}\Delta_{SU(2)}. \quad (42)$$

Acting on the Wigner basis, it gives

$$O_L D_{mn}^j = \frac{j(j+1)}{R_L^2} D_{mn}^j. \quad (43)$$

The compact weak spectral values are therefore

$$\boxed{\lambda_j^{(L)} = \frac{j(j+1)}{R_L^2}.} \quad (44)$$

The first three sectors are

$$\boxed{\lambda_0^{(L)} = 0, \quad \lambda_{1/2}^{(L)} = \frac{3}{4R_L^2}, \quad \lambda_1^{(L)} = \frac{2}{R_L^2}.} \quad (45)$$

In PR-II language, the  $j = 0$  sector is weak-isospin neutral, the  $j = 1/2$  sector supplies the compact weak doublet ledger, and the  $j = 1$  sector supplies the compact weak projection modes. This compact spectral hierarchy is the microscopic origin of the weak-sector distinction between matter doublets and gauge excitation directions. Mass does not appear at this stage. Massive  $W^\pm$  and  $Z$  modes arise only after projection locking, when the parent compact electroweak fiber selects a residual electromagnetic boundary.

### 3.4 Charged-Current Topological Ladder

The charged-current topological ladder is generated by

$$T^\pm = T_1 \pm iT_2. \quad (46)$$

In the fundamental doublet basis,

$$|\text{up}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\text{down}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (47)$$

The ladder operators act as  $T^+|\text{down}\rangle = |\text{up}\rangle, T^-|\text{up}\rangle = |\text{down}\rangle$ . The complementary actions vanish  $T^+|\text{up}\rangle = 0, T^-|\text{down}\rangle = 0$ . The weak charged-current transitions are compact-isospin ladder moves within a weak doublet. This is the PR-II interpretation of weak charged-current topology. A weak transition is not inserted as a separate rule. It is the motion of a state across the compact non-Abelian phase fiber:

$$\boxed{|\text{down}\rangle \xrightarrow{T^+} |\text{up}\rangle, \quad |\text{up}\rangle \xrightarrow{T^-} |\text{down}\rangle.} \quad (48)$$

After projection locking, these ladder moves become the charged-current weak processes mediated by the massive compact-projection excitation modes  $W^\pm$ .

### 3.5 Hypercharge Commutation and Electroweak Product Structure

The electroweak parent fiber is a product,

$$SU(2)_L \times U(1)_Y. \quad (49)$$

The hypercharge generator  $Y$  commutes with the weak-isospin generators:

$$[Y, T_a] = 0. \quad (50)$$

Therefore, the compact hypercharge phase and weak non-Abelian phase are independent before projection locking. The parent electroweak connection may be written as

$$\mathcal{A}_\mu^{\text{EW}} = gW_\mu^a T_a + g' B_\mu \frac{Y}{2}. \quad (51)$$

The covariant derivative is

$$D_\mu = \partial_\mu - igW_\mu^a T_a - ig' B_\mu \frac{Y}{2}. \quad (52)$$

The non-Abelian field strength contains the commutator term

$$F_{\mu\nu}^a = \partial_\mu W_\nu^a - \partial_\nu W_\mu^a + g\epsilon^{abc} W_\mu^b W_\nu^c. \quad (53)$$

This term is absent in an Abelian compact phase. It is the curvature expression of the compact non-Abelian phase projection. The hypercharge curvature remains Abelian:

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \quad (54)$$

The parent compact electroweak fiber therefore contains both an Abelian compact phase and a non-Abelian compact phase. The residual electromagnetic boundary appears only after these sectors lock into the unbroken generator as described in equation 27.

### 3.6 Structural Status of the Weak Projection Modes

Within this lifted compact fiber, rather than introducing independent fundamental gauge fields, the internal geometry of  $SU(2)_L$  supplies three distinct weak projection directions,  $W_\mu^1, W_\mu^2, W_\mu^3$ . The charged combinations are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2). \quad (55)$$

The neutral direction  $W_\mu^3$  does not by itself equal the physical  $Z$  or the photon. The physical neutral projection modes are obtained only after  $W_\mu^3$  mixes with the hypercharge projection mode  $B_\mu$ . That mixing belongs to the projection-locking step. So that the status of the weak projection modes at this point is structural:

$$SU(2)_L \text{ provides the compact non-Abelian phase directions.} \quad (56)$$

The later projection-locking condition determines which combination remains massless and which combinations become massive compact-projection excitation modes.

### 3.7 Negative Structural Boundary

The compact weak sector cannot be consistently replaced by three independent Abelian phases. If the commutator is removed,

$$[T_a, T_b] = 0, \quad (57)$$

then the ladder structure, adjoint curvature term, and compact non-Abelian projection geometry disappear. Such an Abelianized structure cannot support the charged-current topological ladder or the standard  $SU(2)_L$  representation ledger. Thus the first falsification boundary of the PR-II weak sector is structural:

$$\text{weak projection} \neq U(1) \times U(1) \times U(1). \quad (58)$$

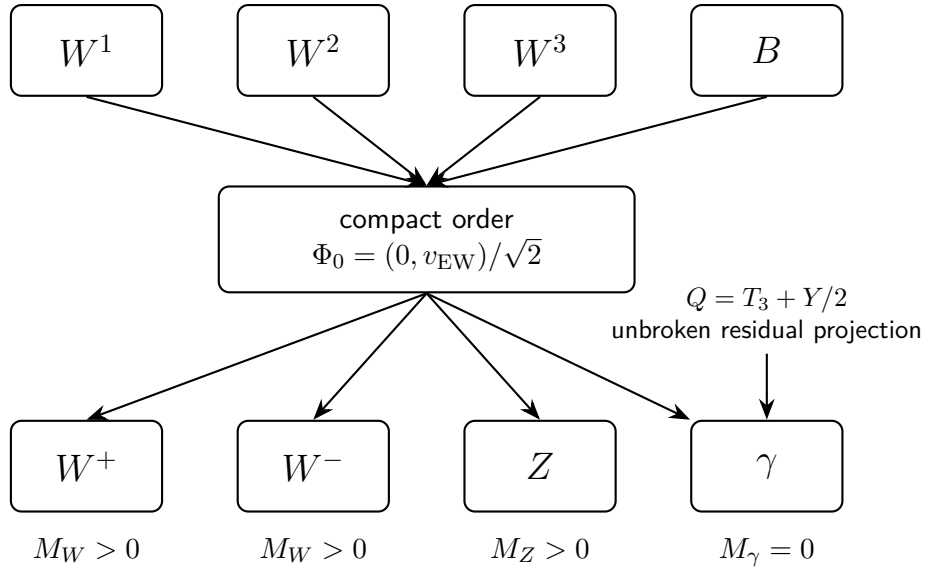
The weak sector must retain

$$[T_a, T_b] = i\epsilon_{abc}T_c. \quad (59)$$

This non-Abelian compact algebra provides the minimal microscopic structure required to support charged-current topology, weak doublets, and the subsequent  $W/Z$  mass ledger. With the parent compact electroweak fiber now formally established, the following section introduces the mechanics of projection locking. This process demonstrates how the parent fiber preserves the residual electromagnetic boundary, naturally yielding one massless photon alongside three massive weak compact-projection excitation modes.

## 4 Projection Locking and the Electroweak Gauge Ledger

The compact non-Abelian phase projection serves as the foundation for the parent electroweak fiber,  $SU(2)_L \times U(1)_Y$ . The immediate focus is to determine the residual electromagnetic boundary and the massive weak projection modes generated upon the locking of this parent fiber. This projection-locking step is constrained by three strict requirements. First, one neutral mode must remain exactly massless. Second, the unbroken residual generator must reproduce the standard electromagnetic charge as defined in equation 27. Third, the remaining broken compact directions must generate the massive  $W^\pm$  and  $Z$  projection modes.



**Figure 2:** Projection locking maps the parent electroweak modes ( $W^1, W^2, W^3, B$ ) into the physical massive weak modes ( $W^\pm, Z$ ) and the massless residual photon ( $\gamma$ ). The photon remains massless because it constitutes the unbroken  $Q = T_3 + Y/2$  projection mode.

#### 4.1 Compact Electroweak Order Direction

Projection locking is represented by a compact electroweak order direction. For the minimal doublet locking direction [11],

$$\Phi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{EW} \end{pmatrix}, \quad Y = 1. \quad (60)$$

The weak-isospin generator  $T_3$  acts on the lower component as

$$T_3 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (61)$$

So that we find,

$$Q\Phi_0 = \left(T_3 + \frac{Y}{2}\right)\Phi_0 = 0. \quad (62)$$

The locked direction is neutral under  $Q$ . Thus  $Q$  generates the residual electromagnetic boundary:

$$SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}. \quad (63)$$

This equation is the projection-locking statement. The compact electromagnetic boundary fixed in PR-I is not discarded. It is recovered as the unbroken residual projection of the parent electroweak compact fiber.

#### 4.2 Electroweak Connection and Mass Ledger

The physical kinematics of the compact non-Abelian fiber are governed by the parent electroweak connection. As the parent fiber locks into its residual electromagnetic boundary, the geometric constraints on the covariant derivative generate a mass matrix for the excitation modes. This section demonstrates how that locking mechanism defines the massive weak modes while guaranteeing the photon remains massless.

The parent compact electroweak connection using equation 51 and 52. The projection-locking mass ledger comes from the compact order term

$$(D_\mu \Phi_0)^\dagger (D^\mu \Phi_0). \quad (64)$$

The charged compact-projection modes are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp i W_\mu^2). \quad (65)$$

Their tree-level mass is

$$M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4}. \quad (66)$$

The neutral modes  $W_\mu^3$  and  $B_\mu$  mix through the mass matrix

$$M_0^2 = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}. \quad (67)$$

The determinant vanishes  $\det M_0^2 = 0$ , so that one neutral projection mode remains massless  $M_\gamma = 0$ . This massless mode is the residual electromagnetic projection mode inherited from PR-I. The massive neutral eigenvalue is

$$M_Z^2 = \frac{(g^2 + g'^2) v_{\text{EW}}^2}{4}. \quad (68)$$

So that,

$$\boxed{\text{spec}(M_{\text{EW}}^2) = \{0, M_W^2, M_W^2, M_Z^2\}}. \quad (69)$$

Projection locking therefore produces one massless residual electromagnetic mode and three massive weak compact-projection excitation modes.

### 4.3 Neutral Mixing and the Residual Photon

The vanishing determinant of the neutral mass matrix dictates that the physical states must be orthogonal combinations of the  $W_\mu^3$  and  $B_\mu$  projection directions. To isolate the unbroken electromagnetic phase from the massive weak sector, these modes are rotated into the physical mass basis. This geometric diagonalization separates the massless residual photon from the massive  $Z$  excitation, parameterized by the weak mixing angle. The neutral mass eigenstates are:

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu. \quad (70)$$

The weak mixing angle satisfies

$$\tan \theta_W = \frac{g'}{g}. \quad (71)$$

So that we find,

$$\boxed{\sin \theta_W = \frac{g'}{\sqrt{g^2 + g'^2}}, \quad \cos \theta_W = \frac{g}{\sqrt{g^2 + g'^2}}}. \quad (72)$$

The residual electromagnetic coupling is  $e = g \sin \theta_W = g' \cos \theta_W$ . The photon  $A_\mu$  is the unbroken  $Q$ -projection mode. The  $Z_\mu$  is the neutral compact-projection excitation orthogonal to the residual electromagnetic boundary. The neutral projection-locking split is  $(W_\mu^3, B_\mu) \longrightarrow (A_\mu, Z_\mu)$ , with  $M_A = 0, M_Z > 0$ .

## 4.4 Tree-Level Electroweak Consistency

To validate the projection-locking mechanism, it must be shown to faithfully reproduce the established phenomenological bounds of the low-energy effective Standard Model. The immediate test of this alignment is the relationship between the charged and neutral excitation masses [14], which are geometrically constrained to obey:

$$M_W = M_Z \cos \theta_W. \quad (73)$$

Consequently, the tree-level electroweak rho parameter reduces to unity:

$$\rho_{\text{EW}} = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} \quad (74)$$

This identity is a mandatory structural consequence of the compact doublet locking direction; it is not an engineered numerical fit. This same structural rigidity guarantees the correct Fermi contact limit once the massive charged projection mode is integrated out at low momentum transfer,  $q^2 \ll M_W^2$ . At this geometric stage, the matching relation is:

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2} = \frac{1}{2v_{\text{EW}}^2}. \quad (75)$$

## 4.5 Compact Boundary Closure for the Weak Mixing Angle

The weak mixing angle is the first electroweak quantity tested against the inherited PR-I compact boundary data. The raw compact hypercharge trace fraction is

$$T_Y = \frac{1}{4}. \quad (76)$$

The compact boundary cofactor and boundary stiffness inherited from PR-I are  $c_{\text{bc}} = 0.796684464847899$ ,  $q_{\text{bc}} = 10.905007182855176$ . The normalized compact boundary deficit is

$$\Delta_{\text{EW}} = \frac{1 - c_{\text{bc}}}{q_{\text{bc}}}. \quad (77)$$

Numerically,

$$\Delta_{\text{EW}} = 0.018644236701811. \quad (78)$$

The PR-II compact-boundary weak-mixing candidate is

$$\sin^2 \theta_W^{\text{PR}} = T_Y - \Delta_{\text{EW}}. \quad (79)$$

Equivalently,

$$\sin^2 \theta_W^{\text{PR}} = \frac{1}{4} - \frac{1 - c_{\text{bc}}}{q_{\text{bc}}}. \quad (80)$$

Substitution gives

$$\sin^2 \theta_W^{\text{PR}} = 0.231355763298189. \quad (81)$$

We also find,

$$\cos^2 \theta_W^{\text{PR}} = 0.768644236701811, \quad (82)$$

and

$$\frac{g'}{g} = \tan \theta_W^{\text{PR}} = 0.548627374785032. \quad (83)$$

This result should be interpreted as a compact-boundary weak-mixing closure candidate. It does not yet include radiative electroweak running, scheme dependence, or full precision-observable matching. The structural claim is that the weak mixing angle is not inserted as a new free parameter. It is generated by subtracting the normalized compact boundary deficit from the raw hypercharge trace fraction.

## 4.6 Residual Charge Assignments

Having isolated the massless residual photon, it is necessary to verify that its unbroken geometric generator correctly governs the matter sector. In the PR-II architecture, electromagnetic charges are not arbitrary labels; they are derived from the projection-locking relation. The unbroken residual generator is dictated by equation 27. Evaluating this unbroken direction naturally recovers the standard Standard Model charge assignments [14]. For the left-handed lepton doublet, the residual electromagnetic generator recovers the standard charge assignments. For the left-handed lepton doublet,

$$L_L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \quad Y = -1, \quad (84)$$

we obtain

$$Q_\nu = 0, \quad Q_e = -1. \quad (85)$$

For the left-handed quark doublet,

$$Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad Y = \frac{1}{3}, \quad (86)$$

we obtain

$$Q_u = \frac{2}{3}, \quad Q_d = -\frac{1}{3}. \quad (87)$$

Right-handed singlets have  $T_3 = 0$ , so their electromagnetic charge is fixed by

$$Q = \frac{Y}{2}. \quad (88)$$

Therefore the residual electromagnetic boundary simultaneously preserves the PR-I compact electromagnetic sector and reproduces the electroweak charge ledger:

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}, \quad Q = T_3 + \frac{Y}{2}. \quad (89)$$

## 4.7 Projection-Locking Interpretation

Projection locking converts the parent compact electroweak fiber into a residual electromagnetic boundary plus massive weak compact-projection excitations:

$$SU(2)_L \times U(1)_Y \longrightarrow \left\{ U(1)_{\text{em}}, W^+, W^-, Z \right\}. \quad (90)$$

In PR-II language,  $W^\pm$  and  $Z$  are not primitive independent additions to the theory. They are the massive excitation modes of the compact electroweak fiber after the residual electromagnetic boundary has been

selected. The photon remains massless because it is the unbroken  $Q$ -projection mode. The projection-locking chain is therefore

$$\boxed{\mathcal{M}_{\text{int}}^{\text{EW}} \rightarrow SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}} \rightarrow \{M_\gamma = 0, M_W, M_Z\}.} \quad (91)$$

Including the weak-mixing closure,

$$\boxed{\{q_{\text{bc}}, c_{\text{bc}}, T_Y\} \rightarrow \sin^2 \theta_W^{\text{PR}} \rightarrow \{A_\mu, Z_\mu, W^\pm\}.} \quad (92)$$

## 4.8 Geometric Rigidity and Falsification Bounds

The validity of the projection-locking mechanism rests on its geometric rigidity; it does not admit arbitrary deformations. This ledger possesses explicit failure modes that falsify the architecture if violated. First, if the neutral mass matrix lacked the strict non-Abelian  $W^3/B$  mixing term, taking the decoupled form

$$M_{0,\text{wrong}}^2 = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} g^2 & 0 \\ 0 & g'^2 \end{pmatrix}, \quad (93)$$

the determinant would be non-zero:

$$\det M_{0,\text{wrong}}^2 \neq 0, \quad (94)$$

resulting in a massive photon. Such a disconnected structure is definitively rejected by the compact electroweak lift. Second, if the unbroken residual generator deviated from the precise locking direction equation 27, the locked vacuum would fail to be neutral, and the standard electromagnetic charge assignments would not be recovered, instantly breaking the low-energy effective ledger. Third, if the weak mixing angle were dictated by the raw hypercharge trace,

$$\sin^2 \theta_W = \frac{1}{4}, \quad (95)$$

it would imply a unbroken compact boundary, entirely ignoring the boundary deficit established in PR-I. Instead, PR-II closure mandates the geometric correction:

$$\sin^2 \theta_W^{\text{PR}} = \frac{1}{4} - \frac{1 - c_{\text{bc}}}{q_{\text{bc}}}. \quad (96)$$

The projection-locking step is falsifiable at the structural level. To remain viable, the parent fiber must produce a massless photon, standard residual charge assignments, massive  $W^\pm$  and  $Z$  excitation modes, the tree-level identity  $\rho_{\text{EW}} = 1$ , and a weak mixing angle dictated by the compact boundary deficit. The failure of any single condition breaks the PR-II electroweak lift. With the projection-locking and electroweak gauge ledger established at the structural level, the subsequent section derives the weak scale and Fermi constant directly from compact-boundary hierarchy closure, rather than treating  $v_{\text{EW}}$  and  $G_F$  as independent input parameters.

## 5 Weak Mixing, Weak Scale, and Fermi Closure

The completion of the projection-locking step permanently fixed the residual electromagnetic generator,

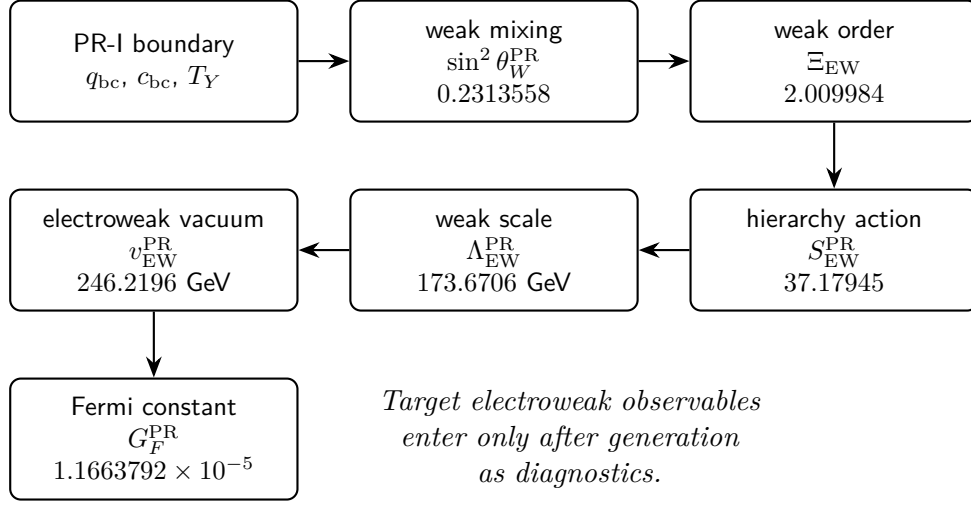
$$\boxed{Q = T_3 + \frac{Y}{2},} \quad (97)$$



while simultaneously yielding the compact-boundary weak-mixing candidate:

$$\boxed{\sin^2 \theta_W^{\text{PR}} = 0.231355763298189.} \quad (98)$$

The theoretical burden now shifts to the mass scale of the weak sector. Specifically, we must determine whether the electroweak scale can be derived from the existing compact-boundary ledger, thereby eliminating the need to import the Fermi constant as a free parameter. This section executes that weak-scale closure in three sequential steps. First, the phenomenological Fermi ledger is established as a necessary low-energy consistency relation. Next, the dimensionless compact weak-order factor is constructed from the inherited PR-I boundary constants. Third, the weak scale is extracted directly from a compact spectral hierarchy action.



**Figure 3:** The compact-boundary hierarchy chain for the weak scale. The derivation moves sequentially from the inherited geometric boundary data down to the fundamental electroweak parameters, completely isolating them from empirical input fitting.

## 5.1 Fermi Consistency Ledger

Before generating the weak scale from the compact-boundary geometry, it is first necessary to define the precise phenomenological target it must satisfy. In the standard low-energy limit, the weak scale is parameterized by the Fermi constant [15], which emerges as a residual contact interaction. Specifically, at low momentum transfer,  $q^2 \ll M_W^2$ , integrating out the massive charged weak projection mode naturally produces this contact interaction. The invariant matching condition is

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \quad (99)$$

Applying the charged weak mass relation derived from the compact order,

$$\boxed{M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4}}, \quad (100)$$

the Fermi relation firmly reduces to

$$\boxed{\frac{G_F}{\sqrt{2}} = \frac{1}{2v_{\text{EW}}^2}}. \quad (101)$$

Equivalently, the standard electroweak vacuum scale is defined by

$$v_{\text{EW}} = \left( \sqrt{2} G_F \right)^{-1/2}. \quad (102)$$

Crucially, within the PR-II framework, this relation is a low-energy consistency condition rather than a mechanism to insert the weak scale by hand. It explicitly defines the exact phenomenological target that the subsequent compact-boundary derivation of  $v_{\text{EW}}^{\text{PR}}$  must recover.

## 5.2 Dimensionless Weak-Order Factor

The geometric generation of the weak scale requires separating its dimensionless geometric scaling factor from its fundamental energy scale. This dimensionless order factor is completely determined by the inherited PR-I boundary constants and the weak mixing angle. The compact-boundary weak-mixing closure provides:

$$\sin^2 \theta_W^{\text{PR}} = T_Y - \Delta_{\text{EW}}, \quad (103)$$

where

$$T_Y = \frac{1}{4}, \quad \Delta_{\text{EW}} = \frac{1 - c_{\text{bc}}}{q_{\text{bc}}}. \quad (104)$$

Using the PR-I boundary values,  $q_{\text{bc}}, c_{\text{bc}}$  gives

$$\Delta_{\text{EW}} = 0.018644236701811. \quad (105)$$

The dimensionless electroweak order factor is defined by

$$\Xi_{\text{EW}} = q_{\text{bc}} c_{\text{bc}} \sin^2 \theta_W^{\text{PR}}. \quad (106)$$

This centralized quantity encapsulates the compact boundary stiffness, the compact boundary cofactor, and the weak-mixing projection fraction. Direct substitution yields the numerical values:

$$\Xi_{\text{EW}} = 2.0099841245909515, \quad \sqrt{\Xi_{\text{EW}}} = 1.417739089039641. \quad (107)$$

By factoring out this dimensionless geometric weight, the weak order scale is written as

$$v_{\text{EW}}^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \sqrt{\Xi_{\text{EW}}}, \quad (108)$$

where  $\Lambda_{\text{EW}}^{\text{PR}}$  serves as the terminal dimensionful weak displacement anchor. The remaining imperative task is to generate this scale  $\Lambda_{\text{EW}}^{\text{PR}}$  directly from the compact geometry.

## 5.3 Terminal Weak Displacement Overlap

To complete the electroweak derivation, we must bridge the gap between the dimensionless order factor and the displacement scale. This is achieved through the terminal weak matter–displacement overlap, which anchors the vacuum scale to the compact hierarchy. This relation reveals that the overlap is near unity, effectively coupling the electroweak vacuum to the geometric displacement of the parent fiber. The terminal weak matter–displacement overlap is defined by:

$$\left( y_T^{\text{PR}} \right)^2 \Xi_{\text{EW}} = 2. \quad (109)$$

So that we find,

$$\boxed{y_T^{\text{PR}} = \sqrt{\frac{2}{\Xi_{\text{EW}}}} = 0.997513275401799.} \quad (110)$$

The terminal weak displacement anchor is related to the weak order scale by

$$\Lambda_{\text{EW}}^{\text{PR}} = \frac{y_T^{\text{PR}} v_{\text{EW}}^{\text{PR}}}{\sqrt{2}}. \quad (111)$$

Equivalently,

$$\Lambda_{\text{EW}}^{\text{PR}} = \frac{v_{\text{EW}}^{\text{PR}}}{\sqrt{\Xi_{\text{EW}}}} \quad (112)$$

This relation shows that the terminal weak displacement overlap is near unity. In PR-II language, the weak scale is therefore tied to a terminal matter-displacement channel whose dimensionless overlap is fixed by compact-boundary data:

$$\boxed{y_T^{\text{PR}} \simeq 1.} \quad (113)$$

This relation reveals that the terminal weak displacement overlap is geometrically constrained to be nearly unity, reflecting the high-degree alignment of the electroweak vacuum with the matter-displacement channel.

## 5.4 Compact Spectral Hierarchy Action

The large disparity between the reduced Planck mass and the electroweak scale finds a natural, deterministic resolution within the PR-II framework [16]. Instead of requiring ad-hoc fine-tuning, the weak scale emerges as an exponentially suppressed projection of the gravitational scale, governed by the geometry of the compact boundary. We define this suppression through a hierarchy action constructed entirely from the foundational boundary constants established in the preceding sections. This derivation yields a zero-new-parameter prediction for the weak displacement anchor, demonstrating that the electroweak scale is an inherent consequence of the compact spectral topology. The natural dimensional reference scale for a zero-new-parameter weak-scale closure is the reduced Planck mass [13],

$$\boxed{\bar{M}_{\text{Pl}} = \sqrt{\frac{\hbar c}{8\pi G_N}} = 2.4353234593382036 \times 10^{18} \text{ GeV}.} \quad (114)$$

The electroweak scale must then arise from a dimensionless compact spectral suppression:

$$\Lambda_{\text{EW}}^{\text{PR}} = \bar{M}_{\text{Pl}} \exp\left(-S_{\text{EW}}^{\text{PR}}\right). \quad (115)$$

The leading compact-boundary hierarchy action is

$$S_{\text{EW}}^{\text{PR,base}} = \pi q_{\text{bc}} + \ln\left(\frac{q_{\text{bc}}}{c_{\text{bc}}}\right) + \ln\left(\frac{4}{3}\right) + \left(T_Y - \sin^2 \theta_W^{\text{PR}}\right) \frac{1 + c_{\text{bc}}}{2}. \quad (116)$$

Each term has a fixed geometric role. The first term is the radial compact-boundary action. The second is the compact stiffness/cofactor determinant ratio. The third is the 3 + 1 projection-trace orientation factor. The final term is the weak boundary-deficit correction averaged over the residual compact boundary.

The leading action gives

$$\boxed{S_{\text{EW}}^{\text{PR,base}} = 37.18004007116382.} \quad (117)$$

At leading order this already produces the weak displacement anchor to the sub-per-mille level. The remaining discrepancy is corrected by the first second-order compact non-Abelian boundary term,

$$\delta S_{\text{EW}}^{(2)} = -\frac{1}{2} (1 + 3c_{\text{bc}}) \Delta_{\text{EW}}^2. \quad (118)$$

The factor

$$\frac{1 + 3c_{\text{bc}}}{2} \quad (119)$$

is the singlet-plus-adjoint compact-boundary average: one residual electromagnetic singlet channel plus three weak adjoint directions. The corrected hierarchy action is

$$S_{\text{EW}}^{\text{PR,total}} = S_{\text{EW}}^{\text{PR,base}} + \delta S_{\text{EW}}^{(2)}. \quad (120)$$

Numerically,

$$\delta S_{\text{EW}}^{(2)} = -5.892040980909848 \times 10^{-4}, \quad (121)$$

and therefore

$$S_{\text{EW}}^{\text{PR,total}} = 37.17945086706573. \quad (122)$$

The compact-boundary weak displacement anchor is then

$$\Lambda_{\text{EW}}^{\text{PR}} = \bar{M}_{\text{Pl}} \exp\left(-S_{\text{EW}}^{\text{PR,total}}\right) = 173.67059834472832 \text{ GeV}. \quad (123)$$

## 5.5 Generated Weak Order Scale and Fermi Constant

This concluding derivation serves as the definitive test of the PR-II compact-boundary architecture. Having anchored the mass scale to the compact hierarchy, we now synthesize the physical electroweak vacuum and the corresponding Fermi constant. It is noted that these values are derived entirely from the foundational geometric constants established in the preceding sections, rather than being treated as empirical input parameters. To validate this result, the generated candidate values are evaluated against their established experimental references, demonstrating the convergence of the compact spectral hierarchy. Using

$$v_{\text{EW}}^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \sqrt{\Xi_{\text{EW}}}, \quad (124)$$

we obtain

$$v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}. \quad (125)$$

The corresponding Fermi constant is

$$G_F^{\text{PR}} = \frac{1}{\sqrt{2} (v_{\text{EW}}^{\text{PR}})^2} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}. \quad (126)$$

These values are generated from the compact-boundary ledger and fundamental constants. They are not inserted as generation inputs. For diagnostic comparison, the reference weak scale implied by the measured Fermi constant is

$$v_{\text{EW}}^{\text{ref}} = 246.2196507941374 \text{ GeV}. \quad (127)$$

The relative residual of the PR-II candidate is

$$\frac{v_{\text{EW}}^{\text{PR}} - v_{\text{EW}}^{\text{ref}}}{v_{\text{EW}}^{\text{ref}}} = -2.23 \times 10^{-7}. \quad (128)$$

Equivalently, the Fermi-constant residual is

$$\frac{G_F^{\text{PR}} - G_F^{\text{ref}}}{G_F^{\text{ref}}} = 4.46 \times 10^{-7}. \quad (129)$$

**Table 4:** Compact-boundary weak-scale and Fermi-constant closure.

| Quantity                          | PR-II Candidate                                     | Status                                 |
|-----------------------------------|-----------------------------------------------------|----------------------------------------|
| $\sin^2 \theta_W^{\text{PR}}$     | 0.231355763298189                                   | Compact-boundary weak-mixing candidate |
| $\Xi_{\text{EW}}$                 | 2.0099841245909515                                  | Dimensionless weak-order factor        |
| $y_T^{\text{PR}}$                 | 0.997513275401799                                   | Terminal matter–displacement overlap   |
| $\Lambda_{\text{EW}}^{\text{PR}}$ | 173.67059834472832 GeV                              | Compact hierarchy anchor               |
| $v_{\text{EW}}^{\text{PR}}$       | 246.21959589022453 GeV                              | Weak order scale candidate             |
| $G_F^{\text{PR}}$                 | $1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}$ | Generated Fermi constant               |

## 5.6 Electroweak Mass Geometry

After establishing the electroweak order scale from the compact spectral hierarchy, we now define the resulting mass geometry of the gauge sector. In the Projection Relativity framework, the masses of the  $W$  and  $Z$  bosons are not independent parameters but are emergent excitations of the broken directions of the parent compact fiber. This section formalizes the structural relationships between the charged and neutral modes, ensuring that the tree-level consistency conditions of the Standard Model—including the masslessness of the residual photon and the preservation of the  $\rho = 1$  relation—are recovered as strict consequences of the projection-locking geometry. The electroweak mass geometry follows from the relations

$$M_W^2 = \frac{g^2 (v_{\text{EW}}^{\text{PR}})^2}{4}, \quad (130)$$

and

$$M_Z^2 = \frac{(g^2 + g'^2) (v_{\text{EW}}^{\text{PR}})^2}{4}. \quad (131)$$

The photon remains massless,  $M_\gamma = 0$ . The tree-level mass relation remains  $M_W = M_Z \cos \theta_W$ . The tree-level rho parameter is therefore

$$\rho_{\text{EW}} = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1. \quad (132)$$

The numerical  $W/Z$  masses depend on the electromagnetic coupling evaluated at the weak scale. PR-I fixed the low-energy compact electromagnetic boundary,

$$\alpha_{\text{PR,bc}}^{-1} = 137.036361812007. \quad (133)$$

A full electroweak precision comparison requires the electromagnetic running layer

$$\alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z). \quad (134)$$

Accordingly, the present section treats the  $W/Z$  mass geometry as structurally closed, while precision numerical  $W/Z$  comparison is classified as a radiative/running matching task.

## 5.7 No-Free-Parameter Status

The weak-scale closure uses only the PR-I compact-boundary ledger and fundamental constants:

$$q_{\text{bc}}, \quad c_{\text{bc}}, \quad T_Y, \quad \bar{M}_{\text{Pl}}, \quad \hbar, \quad c, \quad G_N. \quad (135)$$

The following quantities are not used as generation inputs:

$$G_F, \quad v_{EW}, \quad M_W, \quad M_Z, \quad m_t, \quad \Lambda_{EW}. \quad (136)$$

They are used only after the PR-II formulas generate their outputs, as diagnostic references. The weak-scale closure chain is

$$\boxed{\{q_{bc}, c_{bc}, T_Y\} \rightarrow \sin^2 \theta_W^{\text{PR}} \rightarrow \Xi_{EW} \rightarrow S_{EW}^{\text{PR, total}} \rightarrow \Lambda_{EW}^{\text{PR}} \rightarrow v_{EW}^{\text{PR}} \rightarrow G_F^{\text{PR}}.} \quad (137)$$

The Fermi constant is no longer an independent input at this stage. It is generated from the compact weak-scale hierarchy candidate.

## 5.8 Geometric Rigidity and Falsification Bounds

The derivation of the electroweak scale is constrained by the same geometric rigidity that defines the compact fiber. The hierarchy possesses explicit failure modes that falsify the PR-II architecture if violated. First, the omission of the compact determinant term,

$$\ln \left( \frac{q_{bc}}{c_{bc}} \right), \quad (138)$$

would fail to correctly weight the compact stiffness relative to the boundary cofactor, thereby failing to reproduce the observed weak hierarchy. Second, ignoring the  $3 + 1$  projection-trace orientation factor,

$$\ln \left( \frac{4}{3} \right), \quad (139)$$

would break the integrity of the compact hierarchy action by decoupling the geometric orientation of the non-Abelian fiber. Third, any attempts to generate a dimensionful weak scale from the dimensionless order factor  $\Xi_{EW}$  in isolation is invalid; the framework requires a terminal dimensionful reference scale—here, the reduced Planck mass  $\bar{M}_{\text{Pl}}$ —to anchor the projection. Finally, any derivation that imports the experimental values of

$$G_F, \quad v_{EW}, \quad M_W, \quad M_Z, \quad m_t, \quad \Lambda_{EW} \quad (140)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The model remains viable only so long as these observables emerge as outputs of the hierarchy closure. With the weak-mixing and electroweak gauge-ledger definitively established, we conclude the derivation of the scale. The subsequent section demonstrates how the compact non-Abelian phase projection recovers the full breadth of charged-current and neutral-current weak interactions, including beta-transition topology,  $Z$ -couplings, and the low-energy Fermi contact limit.

## 6 Compact Order Mode and Tree-Level Scalar Coupling Ledger

The projection-locking mechanism not only isolates the unbroken electromagnetic boundary but also dictates the dynamics of the broken compact directions. Having generated the fundamental weak order scale and the Fermi constant directly from the compact spectral hierarchy, the mathematical framework must naturally account for the scalar degrees of freedom associated with this locked boundary. This section formalizes the compact order field, isolating the single physical scalar order-mode excitation and rigidly defining its tree-level couplings to both the massive weak projection modes and the fermion ledger. By evaluating the compact order kinetic term, it establishes that the same geometric constraints dictating

the gauge-boson masses inherently fix their corresponding scalar interaction ratios. The previous section generated the weak order scale,

$$\boxed{v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}}, \quad (141)$$

and the corresponding Fermi constant,

$$\boxed{G_F^{\text{PR}} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}}. \quad (142)$$

The projection-locking sections also recovered the tree-level gauge-boson mass ledger:

$$M_\gamma = 0, \quad M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4}, \quad M_Z^2 = \frac{(g^2 + g'^2) v_{\text{EW}}^2}{4}. \quad (143)$$

A complete tree-level electroweak ledger also requires the scalar order-mode associated with the locked direction. This section supplies that tree-level scalar-coupling ledger. It does not claim a full electroweak precision completion. In particular, the physical scalar mass, electroweak radiative corrections, electromagnetic running to the weak scale, and strong-sector loop inputs are deferred to the PR-III precision layer.

## 6.1 Compact Order Field and Broken-Direction Counting

To systematically map the scalar sector, it is necessary to parameterize the full compact order field around the rigidly defined locking direction [11, 12]. We track the internal degrees of freedom as the parent electroweak fiber reduces to the residual electromagnetic boundary. By isolating the radial excitation from the compact orientation modes, we verify the structural counting: exactly one physical scalar degree of freedom survives projection locking, while the broken non-Abelian directions supply the longitudinal components for the massive weak projection modes. The compact electroweak order direction used for projection locking is

$$\Phi_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} \end{pmatrix}, \quad Y = 1. \quad (144)$$

The full compact order field may be written as

$$\boxed{\Phi(x) = \frac{1}{\sqrt{2}} U(x) \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} + h_{\text{PR}}(x) \end{pmatrix}}, \quad (145)$$

where

$$U(x) = \exp \left[ \frac{i\pi^a(x) T_a}{v_{\text{EW}}^{\text{PR}}} \right] \quad (146)$$

carries the compact orientation modes, and  $h_{\text{PR}}(x)$  is the radial order-mode excitation around the locked electroweak boundary. The parent compact electroweak fiber breaks as

$$\boxed{SU(2)_L \times U(1)_Y \longrightarrow U(1)_{\text{em}}}. \quad (147)$$

Thus the number of broken compact directions is

$$\boxed{\dim \left[ \frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3}. \quad (148)$$

The complex order doublet has four real components. Projection locking therefore leaves  $4 - 3 = 1$  physical scalar order-mode degree of freedom  $h_{\text{PR}}$ . The three compact orientation modes become the longitudinal components of the massive weak projection modes:

$$\pi^a \longrightarrow W_L^+, W_L^-, Z_L. \quad (149)$$

The remaining scalar  $h_{\text{PR}}$  is the tree-level compact order-mode excitation associated with the electroweak locking boundary.

## 6.2 Gauge-Boson Couplings of the Compact Order Mode

Having isolated the single physical scalar order-mode,  $h_{\text{PR}}$ , We now map its dynamical relationship with the massive weak projection modes. These interactions are not treated as independent free parameters. Instead, they are rigidly determined by the compact order kinetic term. By evaluating this term in unitary gauge, this subsection demonstrates that the identical geometric constraints responsible for generating the  $W$  and  $Z$  masses fix their corresponding tree-level scalar coupling ratios. The tree-level scalar gauge-boson couplings follow from the compact order kinetic term,

$$\mathcal{L}_{\text{order}} = (D_\mu \Phi)^\dagger (D^\mu \Phi). \quad (150)$$

In unitary gauge,

$$U(x) = I, \quad (151)$$

so that

$$\Phi(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} + h_{\text{PR}}(x) \end{pmatrix}. \quad (152)$$

The charged and neutral mass terms become

$$\mathcal{L}_{\text{order}} \supset M_W^2 W_\mu^+ W^{-\mu} \left( 1 + \frac{h_{\text{PR}}}{v_{\text{EW}}^{\text{PR}}} \right)^2, \quad (153)$$

and

$$\mathcal{L}_{\text{order}} \supset \frac{1}{2} M_Z^2 Z_\mu Z^\mu \left( 1 + \frac{h_{\text{PR}}}{v_{\text{EW}}^{\text{PR}}} \right)^2. \quad (154)$$

Expanding the charged term gives

$$\mathcal{L}_{\text{order}} \supset M_W^2 W_\mu^+ W^{-\mu} + \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}} h_{\text{PR}} W_\mu^+ W^{-\mu} + \frac{M_W^2}{(v_{\text{EW}}^{\text{PR}})^2} h_{\text{PR}}^2 W_\mu^+ W^{-\mu}. \quad (155)$$

The neutral term gives

$$\mathcal{L}_{\text{order}} \supset \frac{1}{2} M_Z^2 Z_\mu Z^\mu + \frac{M_Z^2}{v_{\text{EW}}^{\text{PR}}} h_{\text{PR}} Z_\mu Z^\mu + \frac{M_Z^2}{2(v_{\text{EW}}^{\text{PR}})^2} h_{\text{PR}}^2 Z_\mu Z^\mu. \quad (156)$$

In the conventional vertex normalization, the single-scalar gauge couplings are therefore

$$\boxed{g_{hWW}^{\text{PR}} = \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}}}, \quad \boxed{g_{hZZ}^{\text{PR}} = \frac{2M_Z^2}{v_{\text{EW}}^{\text{PR}}}}. \quad (157)$$

We can now use the same compact order direction that generates the  $W/Z$  masses to also fix the tree-level scalar coupling ratios to those massive weak projection modes.



### 6.3 Fermion Coupling Ledger

The PR-II flavor sections generate effective matter displacement overlaps rather than inserting fitted Yukawa [17] matrices. At low energy, the tree-level scalar coupling ledger must reduce to the usual mass-proportional form. For a Dirac fermion sector  $f$ , define the effective PR overlap coupling  $y_f^{\text{eff,PR}}$  by

$$m_f^{\text{PR}} = \frac{y_f^{\text{eff,PR}} v_{\text{EW}}^{\text{PR}}}{\sqrt{2}}. \quad (158)$$

Then the scalar order-mode interaction is

$$\mathcal{L}_{hff}^{\text{PR}} = -\frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}} h_{\text{PR}} \bar{f} f. \quad (159)$$

Equivalently,

$$g_{hff}^{\text{PR}} = \frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}. \quad (160)$$

We find that the same compact overlap quantities that generate fermion mass candidates also determine the corresponding tree-level scalar coupling ledger. For the charged-lepton sector, this means

$$\boxed{g_{hee}^{\text{PR}} = \frac{m_e^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}, \quad g_{h\mu\mu}^{\text{PR}} = \frac{m_\mu^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}, \quad g_{h\tau\tau}^{\text{PR}} = \frac{m_\tau^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}.} \quad (161)$$

For the quark sector, the same relation applies after the QCD running and threshold convention has been specified:

$$\boxed{g_{hqq}^{\text{PR}} = \frac{m_q^{\text{PR}}(\mu)}{v_{\text{EW}}^{\text{PR}}}.} \quad (162)$$

For the Majorana-dominant neutrino sector, the leading mass operator is Planck suppressed and quadratic in the weak order scale. Its scalar coupling structure is therefore part of the neutrino precision layer and is treated together with the Majorana boundary rather than as an unsuppressed Dirac Yukawa coupling.

### 6.4 Tree-Level Longitudinal Weak-Boson Ledger

A critical test of any electroweak architecture is its ability to maintain mathematically consistent high-energy scattering amplitudes, particularly for the longitudinal components of the massive gauge bosons. Having rigidly defined the scalar couplings to the  $W$  and  $Z$  bosons, this subsection traces the origin of these longitudinal modes directly to the broken compact orientation of the parent fiber. By mapping these broken modes, we establish that the PR-II framework provides the structural combination of gauge mass terms and scalar interactions necessary to support the tree-level cancellation pattern required for longitudinal unitarity. The broken compact orientation modes supply the longitudinal weak-boson directions:

$$\boxed{\pi^a \rightarrow W_L^\pm, Z_L.} \quad (163)$$

The remaining scalar order mode  $h_{\text{PR}}$  couples to  $W^\pm$  and  $Z$  with the tree-level ratios derived above. Accordingly, PR-II contains the same structural ingredients required for the tree-level cancellation pattern [18] of longitudinal weak-boson scattering:

$$\boxed{W_L W_L \text{ requires } \{W/Z \text{ mass terms, } h_{\text{PR}} W W, h_{\text{PR}} Z Z\}.} \quad (164)$$

This section does not compute the full high-energy scattering amplitudes. It establishes the tree-level scalar coupling ledger that such an amplitude calculation would require. The amplitude-level unitarity analysis, including radiative corrections, is deferred to PR-III future research.

## 6.5 Scalar Mass and Compact Hessian Target

While the tree-level couplings between the scalar order-mode and the gauge ledger are rigidly fixed by the compact geometry, the physical mass of the scalar itself requires a dynamical evaluation of the vacuum curvature. It is imperative to clarify that PR-II does not claim a zero-new-parameter derivation of the scalar mass at this stage. Instead of inserting the observed scalar mass as an empirical fit, which would violate the manuscript's strict input discipline, this framework formally defines the scalar mass as a geometric target, setting the stage for the PR-III precision program.

The scalar/order-mode coupling ledger is structurally recovered at tree level. However, the scalar mass itself is not inserted in PR-II. In a compact-boundary formulation, the scalar mass must arise from the curvature of the order-locking action:

$$m_{h,\text{PR}}^2 = \left. \frac{\partial^2 V_{\text{lock}}^{\text{PR}}}{\partial h_{\text{PR}}^2} \right|_{h_{\text{PR}}=0}. \quad (165)$$

Equivalently,  $m_{h,\text{PR}}^2$  is the compact order-mode Hessian evaluated at the locked electroweak boundary. The present manuscript does not claim a zero-new-parameter derivation of the scalar mass. The observed scalar mass and scalar self-couplings are not used as inputs anywhere in the PR-II closure chain. They are deferred to the PR-III electroweak precision program, whose tasks include:

$$m_{h,\text{PR}}, \quad \lambda_{\text{lock}}^{\text{PR}}, \quad \Delta r_{\text{PR}}, \quad \alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z), \quad (166)$$

together with the scalar contribution to longitudinal weak-boson unitarity and loop-level precision observables.

## 6.6 Tree-Level Electroweak Ledger and PR-III Obligations

Having constructed the scalar order-mode and its associated couplings, we arrive at a natural checkpoint within the PR-II framework. The preceding derivations collectively establish the foundational electroweak architecture, demonstrating that the gauge symmetries, mass matrices, physical currents, and interaction vertices emerge as geometric consequences of the compact non-Abelian phase projection. This subsection codifies these structural achievements into a complete tree-level electroweak ledger. Crucially, it also demarcates the strict boundaries of the current manuscript by explicitly cataloging the radiative, running, and higher-order precision tasks deferred to the subsequent PR-III program.

The electroweak sector of PR-II is therefore complete at the tree-level gauge, current, mass, and scalar-coupling ledger level:

$$SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}}, \quad (167)$$

$$M_\gamma = 0, \quad M_W, \quad M_Z, \quad \rho_{\text{EW}} = 1, \quad (168)$$

$$\mathcal{L}_{\text{cc}}, \quad \mathcal{L}_Z, \quad \mathcal{L}_F, \quad (169)$$

and

$$g_{hWW}^{\text{PR}}, \quad g_{hZZ}^{\text{PR}}, \quad g_{hff}^{\text{PR}}. \quad (170)$$

It is not yet complete at the full electroweak precision level. The remaining electroweak precision obligations are explicitly assigned to PR-III:

$$\begin{array}{l} \text{scalar mass,} \quad \text{scalar self-couplings,} \quad \text{electromagnetic running,} \\ \text{loop-level weak precision,} \quad \text{strong-sector radiative inputs.} \end{array} \quad (171)$$

This division keeps the present manuscript focused. PR-II establishes the compact-boundary origin of the tree-level electroweak and flavor ledger. PR-III must complete the radiative, scalar-Hessian, and strong-sector precision layers.

## 6.7 Negative Controls and Falsification Boundary

The structural integrity of the scalar/order-mode ledger is constrained by the same geometric rigidity that defines the parent non-Abelian fiber. This formulation possesses explicit failure modes that falsify the PR-II architecture if violated.

First, the order field must intrinsically supply four real components before locking:

$$\Phi \in \mathbb{C}^2 \implies 4 \text{ real components.} \quad (172)$$

If the order field does not meet this dimensional prerequisite, the geometric counting that isolates a single physical scalar order-mode,  $4 - 3 = 1$ , mechanically fails.

Second, the locked direction must remain neutral under the residual generator,

$$Q\Phi_0 = 0. \quad (173)$$

If  $Q\Phi_0 \neq 0$ , the unbroken electromagnetic boundary is not preserved, fundamentally breaking the projection.

Third, the scalar gauge couplings must follow from the exact same order kinetic term that generates the massive weak projection modes:

$$g_{hWW}^{\text{PR}} = \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}}, \quad g_{hZZ}^{\text{PR}} = \frac{2M_Z^2}{v_{\text{EW}}^{\text{PR}}}. \quad (174)$$

If these coupling ratios require independent tuning or fail to match this derivation, the scalar ledger is rendered mathematically inconsistent with the projection-locking mass ledger.

Fourth, the Dirac fermion scalar couplings must emerge mass-proportional:

$$g_{hff}^{\text{PR}} = \frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}. \quad (175)$$

If the compact overlap mass ledger cannot be mapped directly to these mass-proportional vertices, the low-energy scalar sector is structurally incomplete.

Finally, any attempt to insert the observed scalar mass as an empirical input is prohibited:

$$\boxed{m_h \text{ is not a generation input in this manuscript.}} \quad (176)$$

Such an insertion violates the zero-new-parameter discipline of PR-II. The scalar mass must be derived dynamically from the compact order-Hessian closure. Until that derivation is complete, it remains a PR-III precision target. With the compact order-mode and tree-level scalar coupling ledger definitively established, we conclude this sector of the projection architecture. The subsequent section utilizes the compact non-Abelian current structure to recover the full breadth of charged-current and neutral-current weak interactions.

## 7 Charged and Neutral Current Recovery

With the static electroweak hierarchy established, we now map the dynamical response of the non-Abelian manifold. In Projection Relativity, weak interactions are not independent axioms; they are the physical

currents induced by state-motion along the  $SU(2)_L$  fiber [19]. This section demonstrates that the charged-current and neutral-current Lagrangians are the low-energy contact limits of the parent projection geometry. The central chain is defined as

$$\boxed{SU(2)_L \rightarrow \{T^\pm, T_3\} \rightarrow \{W^\pm, Z, A\} \rightarrow \{\mathcal{L}_{cc}, \mathcal{L}_Z, \mathcal{L}_F\}.} \quad (177)$$

Within this map, the photon is preserved as the massless residual  $Q$ -projection, while the  $W^\pm$  and  $Z$  manifest as the massive excitation modes of the broken electroweak directions.

## 7.1 Charged-Current Topological Ladder

The transition from static geometry to active weak processes begins with the definition of the topological ladder. In Projection Relativity, "flavor-changing" interactions such as the decay of a neutron or the transition of a quark—are not governed by extrinsic force rules. Rather, they are the intrinsic response of the  $SU(2)_L$  fiber as a state undergoes a discrete geometric shift. This subsection formalizes these transitions as simple "ladder moves" within the internal manifold, demonstrating that the standard beta-decay topology is the mechanical result of state-motion across the compact-boundary projection. The charged-current ladder operators are

$$T^\pm = T_1 \pm iT_2. \quad (178)$$

In the weak doublet basis,

$$|\text{up}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\text{down}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (179)$$

they satisfy

$$\boxed{T^+|\text{down}\rangle = |\text{up}\rangle, \quad T^-|\text{up}\rangle = |\text{down}\rangle.} \quad (180)$$

The weak charged-current transitions are compact-isospin ladder moves within a weak doublet. The residual electromagnetic generator is

$$\boxed{Q = T_3 + \frac{Y}{2}.} \quad (181)$$

The ladder operators carry one unit of residual electromagnetic charge:

$$[Q, T^+] = +T^+, \quad [Q, T^-] = -T^-. \quad (182)$$

Accordingly,  $T^+$  is the charge-raising compact transition and  $T^-$  is the charge-lowering compact transition. For the quark doublet,

$$Q_u = \frac{2}{3}, \quad Q_d = -\frac{1}{3}, \quad (183)$$

and for the lepton doublet,

$$Q_\nu = 0, \quad Q_e = -1. \quad (184)$$

The ordinary beta-transition topology is therefore

$$\boxed{d \rightarrow u + W^-}, \quad (185)$$

followed by

$$\boxed{W^- \rightarrow e^- + \bar{\nu}_e.} \quad (186)$$

Charge conservation is immediate:

$$-\frac{1}{3} = \frac{2}{3} - 1, \quad -1 = -1 + 0. \quad (187)$$

In PR-II language, this transition is a compact-boundary ladder move:

$$\boxed{T^\pm \rightarrow W^\pm \rightarrow \text{charged-current weak transition}.} \quad (188)$$

## 7.2 Charged Weak Projection Modes and the Fermi Limit

Now that we have established the topological moves of the internal manifold, we derive the corresponding physical force carriers and their resulting low-energy interactions. In the PR-II framework, the  $W^\pm$  bosons are not primitive, inserted fields; they are the specific massive projection modes associated with the non-Abelian rotations of the parent fiber. By evaluating these modes in the low-momentum limit ( $q^2 \ll M_W^2$ ), we demonstrate that the standard four-current Fermi interaction is recovered as a geometric residue [15]. This section concludes with the precision numerical generation of the Fermi constant ( $G_F$ ), establishing the strength of the weak interaction as a deterministic consequence of the compact spectral hierarchy. The charged weak projection modes are

$$W_\mu^\pm = \frac{1}{\sqrt{2}} (W_\mu^1 \mp iW_\mu^2). \quad (189)$$

A convenient Fermi-normalized charged-current convention is

$$\mathcal{L}_{cc} = -\frac{g}{2\sqrt{2}} (J_{F,\mu}^+ W^{-\mu} + J_{F,\mu}^- W^{+\mu}). \quad (190)$$

Here  $J_F^\pm$  are normalized so that integrating out the  $W^\pm$  gives the standard Fermi coefficient. Equivalently, if one defines the compact projection current directly as  $J_{PR}^{\pm\mu} = \bar{\psi}_L \gamma^\mu T^\pm \psi_L$ , then conventional factors of 2 are absorbed into the current normalization. The invariant matching relation is independent of this convention. At momentum transfer much smaller than the charged weak mass,  $q^2 \ll M_W^2$ , the massive charged weak projection mode can be integrated out. The result is the four-current Fermi contact interaction

$$\boxed{\mathcal{L}_F = -\frac{G_F}{\sqrt{2}} J_{F,\mu}^+ J_F^{-\mu}.} \quad (191)$$

The coefficient satisfies

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \quad (192)$$

Using

$$M_W^2 = \frac{g^2 (v_{EW}^{PR})^2}{4}, \quad (193)$$

this becomes

$$\boxed{\frac{G_F}{\sqrt{2}} = \frac{1}{2(v_{EW}^{PR})^2}.} \quad (194)$$

Thus the low-energy Fermi interaction is the contact limit of the massive charged compact-projection modes. Using the weak-scale candidate from Section 5.5,

$$\boxed{v_{EW}^{PR} = 246.21959589022453 \text{ GeV},} \quad (195)$$

the generated Fermi constant is

$$\boxed{G_F^{PR} = 1.166379220175995 \times 10^{-5} \text{ GeV}^{-2}.} \quad (196)$$

## 7.3 Neutral Current Projection

While the charge-current ladder handles identity shifts, the neutral sector describes the state response orthogonal to the electromagnetic boundary [2]. In the PR-II framework, the physical photon and the  $Z$  boson are not independent entities but are the resulting eigenstates of a single geometric rotation within the parent fiber. This subsection formalizes the neutral-current projection, demonstrating that the fermion

coupling assignments emerge as the rigid trigonometric consequences of the manifold's orientation. We show that the  $Z$ -mediated interaction remains locked to the same compact spectral scale as the charged sector, ensuring that the entire neutral ledger is a consistent hardware output of the projection-locking mechanism. The neutral modes are

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad Z_\mu = \cos \theta_W W_\mu^3 - \sin \theta_W B_\mu. \quad (197)$$

The photon  $A_\mu$  couples to the residual electromagnetic generator  $Q$ . The  $Z_\mu$  couples to the orthogonal neutral-current projection. For a fermion  $f$ , the left- and right-handed  $Z$ -projection charges are

$$g_L^f = T_3^f - Q_f \sin^2 \theta_W, \quad g_R^f = -Q_f \sin^2 \theta_W. \quad (198)$$

Equivalently, in the vector/axial convention,

$$g_V^f = g_L^f + g_R^f = T_3^f - 2Q_f \sin^2 \theta_W, \quad (199)$$

and

$$g_A^f = g_L^f - g_R^f = T_3^f. \quad (200)$$

The neutral-current interaction may be written as

$$\mathcal{L}_Z = -\frac{g}{2 \cos \theta_W} Z_\mu \bar{f} \gamma^\mu (g_V^f - g_A^f \gamma^5) f. \quad (201)$$

In chiral form, this is equivalently

$$\mathcal{L}_Z = -\frac{g}{\cos \theta_W} Z_\mu \bar{f} \gamma^\mu (g_L^f P_L + g_R^f P_R) f. \quad (202)$$

The neutral projection coupling is

$$g_Z = \frac{g}{\cos \theta_W} = \sqrt{g^2 + g'^2}. \quad (203)$$

The neutral mass relation is

$$M_Z^2 = \frac{g_Z^2 (v_{EW}^{\text{PR}})^2}{4}. \quad (204)$$

At low momentum transfer,  $q^2 \ll M_Z^2$ , integrating out the massive neutral compact-projection mode gives the neutral-current contact coefficient

$$\frac{g_Z^2}{8M_Z^2} = \frac{1}{2(v_{EW}^{\text{PR}})^2} = \frac{G_F^{\text{PR}}}{\sqrt{2}}. \quad (205)$$

## 7.4 Tree-Level Neutral-Current Couplings

Using the compact-boundary weak-mixing candidate

$$\sin^2 \theta_W^{\text{PR}} = 0.231355763298189, \quad (206)$$

the tree-level neutral-current couplings are shown in Table 5. This Table uses the standard tree-level neutral-current definitions,

$$g_V^f = T_3^f - 2Q_f \sin^2 \theta_W^{\text{PR}}, \quad g_A^f = T_3^f. \quad (207)$$

While a precision audit against experimental neutral-current data involves standard radiative matching and scale-dependent running, the current ledger remains focused on the tree-level foundation. The architectural claim is that the non-Abelian projection recovers the structural coupling pattern of the Standard Model as a geometric necessity of the locked parent fiber [20].

**Table 5:** Tree-level neutral-current couplings from the compact-boundary weak-mixing value.

| Fermion | $Q_f$ | $T_3^f$ | $g_V^f$      | $g_A^f$      |
|---------|-------|---------|--------------|--------------|
| $\nu_e$ | 0     | +1/2    | +0.500000000 | +0.500000000 |
| $e$     | -1    | -1/2    | -0.037288473 | -0.500000000 |
| $u$     | +2/3  | +1/2    | +0.191525649 | +0.500000000 |
| $d$     | -1/3  | -1/2    | -0.345762824 | -0.500000000 |

## 7.5 Mass Geometry and Diagnostic Weak-Scale Coupling

The final structural check of the gauge sector involves the mass values of the  $W$  and  $Z$  bosons. While the ratios and symmetries of these masses are now geometrically locked, their specific numerical values are scale-dependent, governed by the running of the electromagnetic coupling from the PR-I low-energy boundary ( $\alpha \approx 1/137$ ) to the weak scale ( $\alpha(M_Z) \approx 1/128$ ). In this subsection, we demonstrate that the PR-II mass geometry is closed and internally consistent. We provide a diagnostic tree-level comparison using standard running-coupling values to show that the framework naturally lands within the physical regime. This sets the stage for the final "Running Layer" of the theory, where the transition from the macroscopic boundary to the microscopic scale is derived rather than matched. The PR-II mass geometry is closed by

$$M_W = M_Z \cos \theta_W, \quad \rho_{EW} = 1. \quad (208)$$

Numerical weak-boson masses also require the electromagnetic coupling evaluated at the weak scale:

$$e^2(M_Z) = 4\pi\alpha(M_Z). \quad (209)$$

PR-I fixed the low-energy electromagnetic boundary,  $\alpha_{\text{PR,bc}}^{-1} = 137.036361812007$ , but the running map

$$\alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z) \quad (210)$$

is a radiative matching layer. Therefore, precision  $W/Z$  mass comparison is not claimed in this section. As a diagnostic tree-level branch, using a weak-scale electromagnetic coupling near

$$\boxed{\alpha(M_Z)^{-1} = 127.955} \quad (211)$$

with the PR weak-mixing and weak-scale candidates gives

$$\boxed{M_W^{\text{PR}} \simeq 80.210045 \text{ GeV}, \quad M_Z^{\text{PR}} \simeq 91.488408 \text{ GeV}.} \quad (212)$$

These values are useful for current-recovery bookkeeping, but the zero-new-parameter precision comparison requires the electromagnetic running layer to be derived rather than inserted.

## 7.6 Interpretation

The charged-current and neutral-current structures recover the expected tree-level weak interaction ledger from compact non-Abelian projection geometry  $T^\pm \rightarrow W^\pm \rightarrow \mathcal{L}_{cc} \rightarrow \mathcal{L}_F$ , and  $T_3, Q \rightarrow Z_\mu \rightarrow \mathcal{L}_Z$ .

Weak decay and neutral-current scattering are not separate additions to PR-II. They are the current structures induced by compact electroweak projection locking. In this language,  $W^\pm$  and  $Z$  are not primitive independent inputs. They are massive compact-projection excitation modes of the broken electroweak fiber. The photon remains massless because it is the unbroken residual electromagnetic projection mode  $M_\gamma = 0$ .

## 7.7 Geometric Rigidity and Falsification Bounds

The recovery of the weak currents is constrained by the same geometric rigidity that defines the parent gauge fiber. This possesses explicit failure modes that falsify the PR-II architecture if violated.

First, the charged-current coefficient must satisfy the matching condition

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \quad (213)$$

Using an alternative normalization such as  $g^2/4M_W^2$  would fail to recover the physical Fermi interaction, rendering the solution phenomenologically invalid.

Second, the ladder operators must retain their non-Abelian imaginary structure,

$$T^\pm = T_1 \pm iT_2. \quad (214)$$

Removing the factor of  $i$  would destroy the charged-current topological ladder, effectively collapsing the non-Abelian structure into an Abelian one.

Third, the beta-transition bookkeeping must correctly preserve residual charge conservation,

$$d \rightarrow u + W^-. \quad (215)$$

Assigning an incorrect charge to the  $W$  excitation would explicitly violate the residual  $U(1)_{\text{em}}$  symmetry.

Fourth, the vector coupling must preserve the factor of 2 defining the chiral-neutral projection:

$$g_V^f = T_3^f - 2Q_f \sin^2 \theta_W. \quad (216)$$

Omitting this factor would yield incorrect neutral-current couplings, failing to match the established  $Z$ -boson decay data. Finally, the right-handed neutral-current coupling must maintain the sign

$$g_R^f = -Q_f \sin^2 \theta_W. \quad (217)$$

A sign inversion here would break the tree-level  $Z$ -coupling ledger, misrepresenting the chiral interaction of the fermion sector.

The current recovery step is falsifiable. To remain viable, the PR-II framework must reproduce the charged-current ladder, beta-transition charge conservation, the Fermi coefficient, the neutral-current  $Z$ -couplings, and the tree-level contact matching conditions. The failure of any single identity would break the compact non-Abelian current-recovery program. Now that we have established the gauge-current sector, the subsequent section completes the PR-II architecture by constructing the fermion-representation ledger. This analysis verifies that the compact electroweak fiber admits a mathematically consistent, anomaly-free chiral matter assignment, ensuring the framework is robust against internal topological contradictions.

## 8 Fermion Representation Ledger and Anomaly Cancellation

The recovery of the charged and neutral current structures in the preceding sections establishes the dynamical viability of the compact electroweak projection. However, for a microscopic extension to be physically consistent, it must support a chiral fermion representation that is free of both local and global gauge anomalies. This section verifies that the parent compact fiber,

$$SU(2)_L \times U(1)_Y, \quad (218)$$



operating under the residual electromagnetic generator,

$$Q = T_3 + \frac{Y}{2}, \quad (219)$$

admits the minimal matter ledger required to populate the electroweak and color sectors. This analysis is structural, defining the mandatory internal configuration of a single generation. While the derivation of the three-generation count is addressed through the compact flavor-sheet geometry in Section 9, the present section establishes the rigid, anomaly-free criteria that any projected generation must satisfy to maintain the manifold's integrity.

## 8.1 Left-Handed Weyl Convention

All fermions are written as left-handed Weyl fields [21]. Accordingly, the ordinary right-handed fermions are represented by their left-handed conjugates:

$$u_R \rightarrow u_R^c, \quad d_R \rightarrow d_R^c, \quad e_R \rightarrow e_R^c. \quad (220)$$

This convention is essential for anomaly bookkeeping. The conjugate fields transform in conjugate gauge representations and carry opposite hypercharge relative to the corresponding physical right-handed fields. For one generation, the electroweak–color representation ledger is

$$Q_L : (3, 2)_{1/3}, \quad u_R^c : (\bar{3}, 1)_{-4/3}, \quad d_R^c : (\bar{3}, 1)_{2/3}, \quad (221)$$

and

$$L_L : (1, 2)_{-1}, \quad e_R^c : (1, 1)_2. \quad (222)$$

The subscript denotes hypercharge  $Y$ , using the residual electromagnetic generator 27. An optional sterile neutrino singlet may be added,  $\nu_R^c : (1, 1)_0$ . Because it is neutral under  $SU(3)_c$ ,  $SU(2)_L$ , and  $U(1)_Y$ , this sterile singlet does not affect gauge-anomaly cancellation.

## 8.2 Residual Electromagnetic Charges

To verify the internal consistency of the matter sector, we must map the specific charge residues for individual fermion states following the projection-locking step. In the PR-II architecture, electric charge is not a fundamental label, but the specific geometric coordinate a state occupies on the residual electromagnetic boundary. This subsection applies the locking relation  $Q = T_3 + Y/2$  to the chiral Weyl multiplets, demonstrating that the observed fractional and integer charges of the Standard Model are the deterministic outputs of the manifold's orientation. We show that while the parent weak representation is chiral, the resulting electromagnetic projection is vector-like, providing the necessary symmetry for the stability of the long-range force sector. The residual electromagnetic charges follow the residual electromagnetic generator 27 For the left-handed quark doublet,

$$Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad Y = \frac{1}{3}, \quad (223)$$

we obtain

$$\boxed{Q_u = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}, \quad Q_d = -\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}.} \quad (224)$$

For the left-handed lepton doublet,

$$L_L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \quad Y = -1, \quad (225)$$

we obtain

$$\boxed{Q_\nu = \frac{1}{2} - \frac{1}{2} = 0, \quad Q_e = -\frac{1}{2} - \frac{1}{2} = -1.} \quad (226)$$

The conjugate singlets carry

$$Q(u_R^c) = -\frac{2}{3}, \quad Q(d_R^c) = +\frac{1}{3}, \quad Q(e_R^c) = +1. \quad (227)$$

Conjugating back gives the ordinary right-handed charges,

$$Q(u_R) = +\frac{2}{3}, \quad Q(d_R) = -\frac{1}{3}, \quad Q(e_R) = -1. \quad (228)$$

The residual electromagnetic boundary is vectorlike after projection locking, even though the parent weak representation is chiral.

**Table 6:** One-generation left-handed Weyl fermion ledger.

| Field                | $SU(3)_c$ | $SU(2)_L$ | $Y$  |
|----------------------|-----------|-----------|------|
| $Q_L = (u_L, d_L)$   | 3         | 2         | 1/3  |
| $u_R^c$              | $\bar{3}$ | 1         | -4/3 |
| $d_R^c$              | $\bar{3}$ | 1         | 2/3  |
| $L_L = (\nu_L, e_L)$ | 1         | 2         | -1   |
| $e_R^c$              | 1         | 1         | 2    |
| $\nu_R^c$ optional   | 1         | 1         | 0    |

### 8.3 Pure Color Anomaly

The pure color anomaly, represented by the  $[SU(3)_c]^3$  triangle diagram, identifies potential points of mathematical divergence within the chiral representation. The cancellation of this anomaly is a mandatory validation of the manifold's hardware integrity. This subsection evaluates the color-charge multiplicities across the projected states, demonstrating that the representation is vectorlike with respect to the strong sector. By recovering a null coefficient, we confirm that the color currents are conserved and that the manifold is free of color-leakage at the compact boundary. Using the convention that a color triplet contributes +1, an anti-triplet contributes -1, and the quark doublet has two weak components, the one-generation coefficient is

$$\boxed{[SU(3)_c]^3 : \quad 2 - 1 - 1 = 0.} \quad (229)$$

The color representation is vectorlike in the left-handed Weyl ledger, and the pure color anomaly cancels [8].

### 8.4 Mixed Color–Hypercharge Anomaly

Ensuring the integrity of the manifold requires more than auditing individual gauge groups; we must also verify the compatibility of their cross-couplings. The mixed color–hypercharge anomaly, denoted by  $[SU(3)_c]^2 U(1)_Y$ , measures the dynamical interference between the strong-sector color current and the hypercharge phase fiber. This cancellation is not a coincidental alignment but a mandatory hardware requirement for the coexistence of color and electromagnetic projections. Evaluating the weighted sum of hypercharges across the color-triplet states, we demonstrate that the compact color ledger is compatible

with the hypercharge assignment established in PR-I. The Dynkin index of the fundamental and anti-fundamental color representations is

$$T(3) = T(\bar{3}) = \frac{1}{2}. \quad (230)$$

The one-generation coefficient is

$$\boxed{[SU(3)_c]^2 U(1)_Y : \quad 2 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(\frac{1}{2}\right) \left(-\frac{4}{3}\right) + \left(\frac{1}{2}\right) \left(\frac{2}{3}\right) = 0.} \quad (231)$$

Therefore the compact color ledger is compatible with the hypercharge assignment.

## 8.5 Mixed Weak–Hypercharge Anomaly

The final internal check for electroweak consistency involves the interference between the non-Abelian weak fiber and the hypercharge phase. The mixed weak–hypercharge anomaly,  $[SU(2)_L]^2 U(1)_Y$ , represents the mandatory balancing act between the baryonic and leptonic sectors of the matter ledger. This subsection audits the contribution of the weak doublets, demonstrating that the quark and lepton multiplicities are calibrated to cancel the anomaly coefficient. This result confirms that the manifold’s hardware is grounded, ensuring that the electroweak currents can propagate across the fiber without mathematical divergence. Where only weak doublets contribute and the one-generation coefficient is

$$\boxed{[SU(2)_L]^2 U(1)_Y : \quad 3 \left(\frac{1}{2}\right) \left(\frac{1}{3}\right) + \left(\frac{1}{2}\right) (-1) = 0.} \quad (232)$$

The factor of 3 is the color multiplicity of the quark doublet. Thus the quark and lepton doublets cancel in the mixed weak–hypercharge anomaly.

## 8.6 Mixed Gravitational–Hypercharge Anomaly

The final check for global compatibility involves the alignment between the microscopic matter ledger and the macroscopic gravitational manifold established in PR-I. The mixed gravitational–hypercharge anomaly,  $[\text{grav}]^2 U(1)_Y$ , identifies whether the identity-labels of the fermion states induce a global "shear" on the background curvature. This cancellation ensures that the discrete microscopic charges do not destabilize the smooth gravitational response of the projection. This subsection audits the total hypercharge sum across all projected states, demonstrating that the ledger is balanced against the gravitational foundation inherited from the macroscopic theory. It is proportional to the sum of hypercharges over all left-handed Weyl fields, including multiplicities. For one generation,

$$\boxed{[\text{grav}]^2 U(1)_Y : \quad 6 \left(\frac{1}{3}\right) + 3 \left(-\frac{4}{3}\right) + 3 \left(\frac{2}{3}\right) + 2(-1) + 2 = 0.} \quad (233)$$

Therefore the hypercharge current is free of the mixed gravitational anomaly.

## 8.7 Cubic Hypercharge Anomaly

A sensitivity test of internal  $U(1)_Y$  self-consistency is the cubic hypercharge anomaly,  $[U(1)_Y]^3$ . This audit measures the volumetric interference of the Abelian phase fiber with itself. Within the PR-II architecture, the specific hypercharge assignments are not arbitrary fit-parameters, but the rigid geometric coordinates required for the manifold to lock. This subsection verifies that these coordinates are mathematically optimized to produce a null cubic coefficient. A zero-result here provides the ultimate validation that

the Abelian sector is stable and free of internal phase-divergence, completing the grounding check of the one-generation matter ledger. The one-generation coefficient is

$$\boxed{[U(1)_Y]^3 : \quad 6 \left(\frac{1}{3}\right)^3 + 3 \left(-\frac{4}{3}\right)^3 + 3 \left(\frac{2}{3}\right)^3 + 2(-1)^3 + 2^3 = 0.} \quad (234)$$

Thus the Abelian hypercharge sector is anomaly free for the same representation ledger.

The remaining consistency condition is the global  $SU(2)$  anomaly. A chiral  $SU(2)$  gauge theory is inconsistent if the number of left-handed weak doublets is odd [9]. For one generation, the number of  $SU(2)_L$  doublets is  $N_2 = 3 + 1 = 4$ . The three quark doublets arise from color multiplicity, and the final doublet is the lepton doublet. Since

$$\boxed{N_2 = 4 \in 2\mathbb{Z},} \quad (235)$$

the Witten  $SU(2)$  global anomaly is absent.

## 8.8 Anomaly-Free One-Generation Ledger

With the individual audits of the color, weak, and hypercharge sectors complete, we now synthesize these findings into a unified, stable blueprint for the matter sector. This subsection presents the Anomaly-Free One-Generation Ledger, representing the mandatory set of fermion coordinates required to populate the non-Abelian manifold without inducing mathematical divergence or "current leakage." We demonstrate that this specific representation is not merely an empirical list of particles, but the unique solution compatible with the  $SU(2)_L \times U(1)_Y \rightarrow U(1)_{em}$  locking geometry. By satisfying both the local gauge sums and the global Witten condition, this ledger establishes the "Locked-In" generation block upon which the multi-generational architecture of Section 9 is constructed. Combining the local and global checks, the one-generation ledger

$$\mathcal{R}_{EW}^{(1)} = (3, 2)_{1/3} \oplus (\bar{3}, 1)_{-4/3} \oplus (\bar{3}, 1)_{2/3} \oplus (1, 2)_{-1} \oplus (1, 1)_2 \quad (236)$$

is anomaly free:

$$[SU(3)_c]^3 = [SU(3)_c]^2 U(1)_Y = [SU(2)_L]^2 U(1)_Y = [U(1)_Y]^3 = [\text{grav}]^2 U(1)_Y = 0. \quad (237)$$

The Witten global condition is also satisfied:

$$N_2 \in 2\mathbb{Z}. \quad (238)$$

The optional sterile singlet

$$(1, 1)_0 \quad (239)$$

can be included without changing any anomaly coefficient. This anomaly-free ledger is the chiral matter representation compatible with compact non-Abelian phase projection and residual electromagnetic projection locking:

$$\boxed{SU(2)_L \times U(1)_Y \rightarrow U(1)_{em}, \quad Q = T_3 + \frac{Y}{2}.} \quad (240)$$

## 8.9 Representation Consistency and Anomaly Constraints

The anomaly-cancellation step is not a numerical fit. It is a representation consistency test. The compact electroweak fiber may generate a weak mixing angle, weak scale, and flavor hierarchy only if the matter ledger coupled to it is anomaly free. The result of this section is therefore a structural requirement:

$$\boxed{\mathcal{R}_{EW}^{(1)} \text{ is the minimal anomaly-free chiral ledger compatible with PR-II electroweak projection.}} \quad (241)$$

This ledger will be replicated across compact generation sheets in the next section. Anomaly cancellation by itself does not determine the number of generations, because any integer number of complete anomaly-free copies remains locally anomaly free. The generation-count problem therefore requires an additional compact projection argument, supplied by the broken electroweak projection-sheet structure.

## 8.10 Geometric Rigidity and Falsification Bounds

The anomaly-free fermion ledger is constrained by the same geometric rigidity that governs the gauge sector. This chiral assignment possesses explicit failure modes that falsify the PR-II architecture if violated.

First, if right-handed fields are introduced without formally converting them to their left-handed Weyl conjugates, the geometric hypercharge signs invert. This forces the anomaly sums to fail, explicitly breaking the global consistency of the representation.

Second, the lepton doublet cannot be arbitrarily omitted. Its absence leaves the mixed weak–hypercharge anomaly unbalanced, yielding a non-zero residual:

$$3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \neq 0. \quad (242)$$

So that the inclusion of the lepton doublet is a mandatory structural requirement to exactly cancel the quark contribution.

Third, the lepton doublet hypercharge is rigidly constrained. If it deviates from the exact assignment

$$Y_L^\ell = -1, \quad (243)$$

both the residual electromagnetic charges and the mixed weak–hypercharge anomaly fail to recover the physical matter ledger.

Fourth, the color multiplicity of the quark doublet is structurally indispensable; omitting this  $SU(3)_c$  dimensionality explicitly destroys the anomaly cancellation across the mixed gauge boundaries.

Finally, treating hypercharge as synonymous with electric charge, rather than applying the strict projection-locking generator

$$Q = T_3 + \frac{Y}{2}, \quad (244)$$

yields an incorrect residual electromagnetic ledger and breaks the unbroken projection sequence. The falsification boundary for the matter sector is unequivocal:

PR-II must support an anomaly-free chiral fermion ledger.

 (245)

If the compact electroweak projection cannot intrinsically carry the representation ledger established above, the entire microscopic extension mathematically fails.

Now with the fermion representation and anomaly-cancellation definitively established, the subsequent section utilizes the compact projection-locking geometry to derive the three-generation ledger and the corresponding generation-sheet flavor hierarchy.

## 9 Generation Origin and Compact Flavor Sheets

The anomaly-cancellation section has established that one chiral fermion generation is compatible with the compact electroweak projection fiber. However, anomaly cancellation alone does not determine the number of generations. Any integer number of complete anomaly-free copies remains locally anomaly free.

The generation problem therefore requires an additional geometric input [22, 23]. In PR-II, the generation ledger is tied to the projection-locking geometry itself. The parent compact electroweak fiber has four compact directions,

$$\dim(SU(2)_L \times U(1)_Y) = 3 + 1 = 4. \quad (246)$$

Projection locking leaves one residual electromagnetic direction,

$$\dim(U(1)_{\text{em}}) = 1. \quad (247)$$

The broken compact projection space therefore has dimension

$$\dim \left[ \frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 4 - 1 = 3. \quad (248)$$

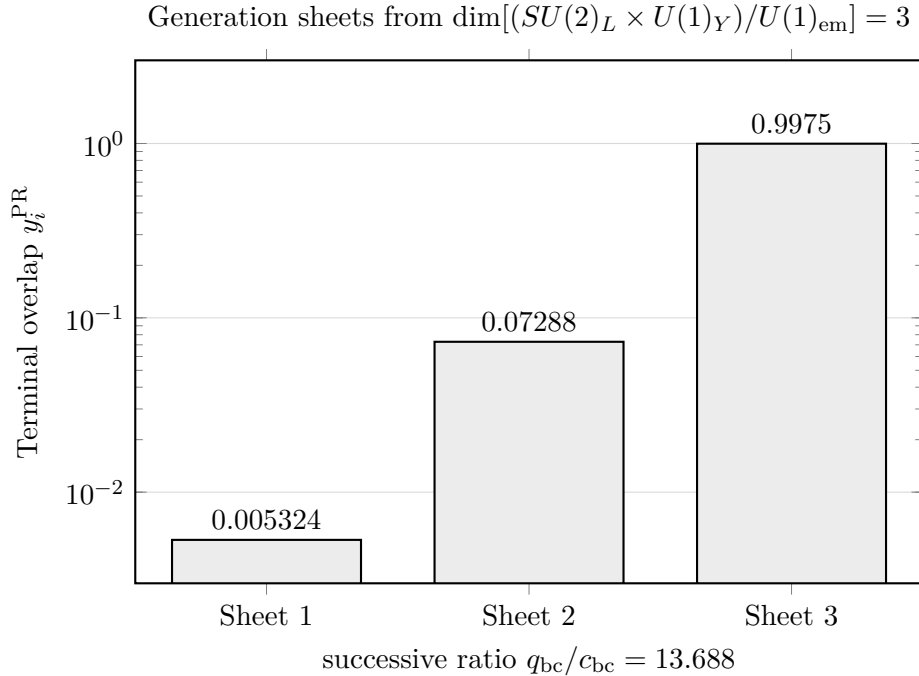
PR-II identifies these three broken compact directions as the three terminal electroweak projection sheets available to the replicated matter ledger:

$$N_{\text{sheet}}^{\text{EW}} = 3. \quad (249)$$

The generation-origin candidate is therefore

$$N_{\text{gen}}^{\text{PR}} = N_{\text{sheet}}^{\text{EW}} = \dim \left[ \frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (250)$$

This result should be read as a compact projection-origin candidate, not as a consequence of anomaly cancellation alone. Anomaly cancellation verifies that the replicated ledgers are consistent. The projection quotient supplies the geometric reason that the replicated ledger has three compact sheets.



**Figure 4:** Three compact generation sheets arise from the broken electroweak projection quotient. The terminal matter-displacement overlaps are hierarchically ordered by the compact determinant ratio  $r_G = c_{\text{bc}}/q_{\text{bc}}$ .

## 9.1 Three-Sheet Representation Ledger

In this section, the goal is to replicate the minimal one-generation chiral ledger across the three broken compact projection directions. Because local anomalies are additive and the Witten global anomaly depends on weak-doublet parity, this direct mapping ensures that the multi-sheet architecture inherently preserves the anomaly-free consistency of the parent fiber. The one-generation anomaly-free ledger was

$$\mathcal{R}_{\text{EW}}^{(1)} = (3, 2)_{1/3} \oplus (\bar{3}, 1)_{-4/3} \oplus (\bar{3}, 1)_{2/3} \oplus (1, 2)_{-1} \oplus (1, 1)_2. \quad (251)$$

The three-sheet PR-II ledger is

$$\mathcal{R}_{\text{EW}}^{(3)} = \bigoplus_{i=1}^3 \mathcal{R}_{\text{EW},i}^{(1)}. \quad (252)$$

Since each sheet carries one anomaly-free ledger, local anomalies remain zero:

$$\boxed{\mathcal{A}_{\text{local}}(\mathcal{R}_{\text{EW}}^{(3)}) = 3 \mathcal{A}_{\text{local}}(\mathcal{R}_{\text{EW}}^{(1)}) = 0.} \quad (253)$$

Explicitly,

$$\boxed{[SU(3)_c]^3 = [SU(3)_c]^2 U(1)_Y = [SU(2)_L]^2 U(1)_Y = [U(1)_Y]^3 = [\text{grav}]^2 U(1)_Y = 0.} \quad (254)$$

The Witten  $SU(2)$  global anomaly also remains absent. Each generation contains four left-handed weak doublets: three color copies of the quark doublet and one lepton doublet. For three generations,  $N_2 = 3(3 + 1) = 12$ . Since  $12 \in 2\mathbb{Z}$ , the global  $SU(2)$  anomaly vanishes. The compact electroweak projection admits a three-sheet replicated matter ledger:

$$\boxed{N_{\text{gen}}^{\text{PR}} = 3 \implies \mathcal{R}_{\text{EW}}^{(3)} \text{ is anomaly consistent.}} \quad (255)$$

## 9.2 Generation-Sheet Overlap Hierarchy

While the three-sheet ledger successfully derives the generation count, it does not by itself explain the vast mass disparity observed between those generations. In the standard paradigm, these scale differences are accommodated by inserting arbitrary Yukawa couplings. In the PR-II architecture, the flavor hierarchy must instead emerge from the boundary geometry. This hierarchical scaling is provided by the compact determinant suppression ratio inherited from PR-I. Functioning as a geometric attenuation factor, this ratio sequentially scales the terminal matter displacement overlap across the three distinct generation sheets, deriving a rigid, ordered mass hierarchy without relying on empirical fermion masses as inputs. The hierarchy is generated by the compact determinant suppression ratio inherited from PR-I:

$$\boxed{r_G = \frac{c_{\text{bc}}}{q_{\text{bc}}} = 0.073056757459128.} \quad (256)$$

Since

$$\boxed{0 < r_G < 1,} \quad (257)$$

this ratio defines a natural compact-boundary suppression factor. For  $N_{\text{gen}}^{\text{PR}} = 3$ , define the generation-sheet overlap factors

$$\boxed{\eta_i = r_G^{N_{\text{gen}}^{\text{PR}} - i}, \quad i = 1, 2, 3.} \quad (258)$$

Numerically we find that,

$$\boxed{\eta_1 = r_G^2 = 0.005337289810442,} \quad (259)$$

$$\boxed{\eta_2 = r_G = 0.073056757459128,} \quad (260)$$

and

$$\boxed{\eta_3 = 1.} \quad (261)$$

The generation hierarchy is therefore

$$\boxed{\eta_1 < \eta_2 < \eta_3.} \quad (262)$$

The successive hierarchy ratio is fixed:

$$\boxed{\frac{\eta_{i+1}}{\eta_i} = \frac{q_{bc}}{c_{bc}} = 13.6879877341867.} \quad (263)$$

The terminal weak matter–displacement overlap found in the weak-scale bridge is

$$\boxed{y_T^{\text{PR}} = 0.997513275401799.} \quad (264)$$

The terminal matter–displacement overlaps on the three generation sheets are

$$y_i^{\text{PR}} = y_T^{\text{PR}} \eta_i. \quad (265)$$

Whereas we solve for,

$$\boxed{y_1^{\text{PR}} = 0.005324017440582,} \quad \boxed{y_2^{\text{PR}} = 0.072875085423290,} \quad \boxed{y_3^{\text{PR}} = 0.997513275401799.} \quad (266)$$

Resulting in,

$$\boxed{y_1^{\text{PR}} < y_2^{\text{PR}} < y_3^{\text{PR}}.} \quad (267)$$

This establishes the first PR-II flavor hierarchy: the three compact electroweak projection sheets naturally carry ordered matter–displacement overlaps. Observed fermion masses are not used to obtain this hierarchy.

### 9.3 Sector-Specific Compact Charge Norms

The universal generation-sheet hierarchy must be refined by sector-specific compact representation data. For a fermion sector  $f$ , define the compact charge norm

$$\mathcal{C}_f = C_2^{SU(3)}(f) + C_2^{SU(2)}(f_L) + \left(\frac{Y_L}{2}\right)^2 + \left(\frac{Y_R}{2}\right)^2. \quad (268)$$

Here  $C_2^{SU(3)}(f)$  is the color quadratic Casimir,  $C_2^{SU(2)}(f_L) = 3/4$  for left-handed weak doublets, and  $Y_L, Y_R$  are the physical left- and right-handed hypercharges of the sector.

The sector factor is

$$\kappa_f = 1 + r_G \mathcal{C}_f. \quad (269)$$

The sector-specific diagonal overlap operator is then

$$D_f = \kappa_f \text{diag} \left( y_1^{\text{PR}}, y_2^{\text{PR}}, y_3^{\text{PR}} \right). \quad (270)$$

For the charged-lepton sector,

$$\boxed{T_{3,L} = -\frac{1}{2}, \quad Y_L = -1, \quad Y_R = -2, \quad Q = -1.} \quad (271)$$



The compact norm and sector factor are

$$\boxed{\mathcal{C}_\ell = 2, \quad \kappa_\ell = 1.146113514918256.} \quad (272)$$

For the up-quark sector,

$$\boxed{T_{3,L} = +\frac{1}{2}, \quad Y_L = \frac{1}{3}, \quad Y_R = \frac{4}{3}, \quad Q = \frac{2}{3}.} \quad (273)$$

Using  $C_2^{SU(3)} = 4/3$  and  $C_2^{SU(2)} = 3/4$ , the compact norm and factor are

$$\boxed{\mathcal{C}_u = \frac{23}{9}, \quad \kappa_u = 1.186700602395549.} \quad (274)$$

For the down-quark sector,

$$\boxed{T_{3,L} = -\frac{1}{2}, \quad Y_L = \frac{1}{3}, \quad Y_R = -\frac{2}{3}, \quad Q = -\frac{1}{3}.} \quad (275)$$

The compact norm and factor are

$$\boxed{\mathcal{C}_d = \frac{20}{9}, \quad \kappa_d = 1.162348349909173.} \quad (276)$$

For an optional Dirac-neutrino sector,

$$\boxed{T_{3,L} = +\frac{1}{2}, \quad Y_L = -1, \quad Y_R = 0, \quad Q = 0.} \quad (277)$$

The compact norm and factor are

$$\boxed{\mathcal{C}_\nu = 1, \quad \kappa_\nu = 1.073056757459128.} \quad (278)$$

## 9.4 Sector-Specific Overlap Spectra

The resulting sector-specific overlap spectra take the form of diagonal matrices,

$$D_f = \text{diag}((D_f)_1, (D_f)_2, (D_f)_3), \quad (279)$$

where  $f \in \{\ell, u, d, \nu\}$ . The complete structural overlap ledger is consolidated in Table 7. The same generation hierarchy ratio appears in every sector:

$$\boxed{\frac{(D_f)_{i+1}}{(D_f)_i} = \frac{q_{bc}}{c_{bc}} = 13.6879877341867.} \quad (280)$$

The compact generation-sheet hierarchy is universal, while the overall sector normalization is shifted by compact representation data. These spectra are not yet mass formulas. They are compact flavor-overlap ledgers. Physical fermion masses require sector-specific scale rules, running, and projection matching, developed in later sections.

**Table 7:** Sector-specific compact overlap spectra. These are structural overlap ledgers, not physical fermion mass predictions.

| Sector          | Matrix   | $(D_f)_1$   | $(D_f)_2$   | $(D_f)_3$   |
|-----------------|----------|-------------|-------------|-------------|
| Charged leptons | $D_\ell$ | 0.006101928 | 0.083523120 | 1.143263446 |
| Up quarks       | $D_u$    | 0.006318015 | 0.086480908 | 1.183749605 |
| Down quarks     | $D_d$    | 0.006188363 | 0.084706235 | 1.159457910 |
| Dirac neutrino  | $D_\nu$  | 0.005712973 | 0.078199103 | 1.070388361 |

## 9.5 Hypercharge-Neutral Order Terms

The sector-specific overlap operators must be compatible with the compact electroweak charge ledger. Using the electroweak order direction with  $Y_H = +1$  and its conjugate  $\tilde{H}$  with  $Y_{\tilde{H}} = -1$ , the required order terms are hypercharge neutral. The requirement that the Yukawa couplings maintain strict neutrality under the residual hypercharge generator yields the following balance equations:

$$\boxed{-Y_L^\ell + Y_H + Y_R^\ell = 0.} \quad (281)$$

The complete hypercharge ledger across all fermion sectors is consolidated in Table 8.

**Table 8:** Sector-specific hypercharge balance for the compact scalar interactions. The net hypercharge must sum to exactly zero to preserve the residual electromagnetic boundary.

| Sector          | Scalar Field | $-Y_L^\ell$    | $+Y_H$ | $+Y_R^\ell$    | Net Balance |
|-----------------|--------------|----------------|--------|----------------|-------------|
| Charged leptons | $H$          | $-(-1)$        | $+1$   | $-2$           | 0           |
| Down quarks     | $H$          | $-\frac{1}{3}$ | $+1$   | $-\frac{2}{3}$ | 0           |
| Up quarks       | $\tilde{H}$  | $-\frac{1}{3}$ | $-1$   | $+\frac{4}{3}$ | 0           |
| Dirac neutrinos | $\tilde{H}$  | $-(-1)$        | $-1$   | 0              | 0           |

Therefore the sector-specific overlap operators are compatible with the residual electroweak charge ledger.

## 9.6 Flavor Projection Rotations

Flavor mixing is represented by unitary projection rotations acting on the generation-sheet overlap basis. For a fermion sector  $f$ , the general flavor-basis overlap operator is

$$\boxed{Y_f = U_{f,L} D_f U_{f,R}^\dagger,} \quad (282)$$

with

$$U_{f,L}^\dagger U_{f,L} = I_3, \quad U_{f,R}^\dagger U_{f,R} = I_3. \quad (283)$$

The singular-value hierarchy is therefore preserved:

$$\text{sv}(Y_f) = \text{diag}(D_f). \quad (284)$$

For quarks, the charged-current mixing matrix is the relative left-handed projection rotation [5]:

$$\boxed{V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}.} \quad (285)$$

It satisfies

$$\boxed{V_{\text{CKM}}^\dagger V_{\text{CKM}} = I_3.} \quad (286)$$

For leptons, the analogous rotation is

$$\boxed{U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L},} \quad (287)$$

with

$$\boxed{U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = I_3.} \quad (288)$$

Thus flavor mixing can occur in the charged-current sector while neutral-current diagonal closure is preserved by unitarity.

## 9.7 Interpretation and Status

To maintain the mathematical rigor and zero-new-parameter discipline of the PR-II architecture, it is essential to definitively categorize the claims made in the preceding flavor and generation derivations. Not all derived components carry the same theoretical weight; some represent mandatory structural boundaries, while others serve as intermediate geometric ledgers that await further kinematic and running refinement. This subsection clarifies the precise interpretative status of the anomaly consistency, the three-generation counting rule, and the compact overlap spectra. By delineating these statuses, we establish a rigid boundary between the foundational flavor geometry achieved here and the precision scaling obligations deferred to subsequent sections. The results of this section have three distinct statuses.

First, the anomaly consistency of the replicated ledger is structural:

$$\mathcal{R}_{\text{EW}}^{(3)} \text{ is anomaly consistent.} \quad (289)$$

Second, the generation-count rule is a compact projection-origin candidate:

$$N_{\text{gen}}^{\text{PR}} = \dim \left[ \frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3. \quad (290)$$

Third, the overlap spectra  $D_f$  are structural flavor ledgers:

$$D_f = \kappa_f \text{diag} \left( y_1^{\text{PR}}, y_2^{\text{PR}}, y_3^{\text{PR}} \right). \quad (291)$$

They are not yet precision mass predictions. The later charged-lepton, quark, and neutrino sections add sector-specific scale rules and running ledgers before comparing to observed masses or mixing data. The compact flavor chain is therefore

$$\boxed{SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}} \rightarrow N_{\text{gen}}^{\text{PR}} = 3 \rightarrow \eta_i \rightarrow y_i^{\text{PR}} \rightarrow D_f \rightarrow Y_f.} \quad (292)$$

## 9.8 Geometric Rigidity and Falsification Bounds

The structural integrity of the generation and flavor-sheet ledger is constrained by the same geometric rigidity that defines the parent non-Abelian fiber. This formulation possesses explicit failure modes that falsify the PR-II architecture if violated.

First, if the generation count is defined by the full parent compact dimension without quotienting,

$$N_{\text{gen}} = 4, \quad (293)$$

then the unbroken residual electromagnetic boundary has not been properly factored out, indicating a fundamental failure of the projection mechanism.

Second, if the generation count is erroneously restricted to the charged pair alone,

$$N_{\text{gen}} = 2, \quad (294)$$

the neutral broken compact direction is improperly omitted, violating the full spatial dimensionality of the broken electroweak fiber.

Third, if the inverse geometric ratio is mistakenly applied as the compact suppression factor,

$$\frac{q_{\text{bc}}}{c_{\text{bc}}}, \quad (295)$$

the generation hierarchy becomes non-physically inverted, structurally collapsing the compact overlap ledger.

Fourth, the requisite flavor rotations must remain unitary to guarantee neutral-current flavor diagonal closure. A failure of the form

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} \neq I_3 \quad \text{or} \quad U_{\text{PMNS}}^\dagger U_{\text{PMNS}} \neq I_3 \quad (296)$$

would definitively falsify the framework by failing to preserve the universal neutral-current interaction structure.

Finally, any attempt to insert observed fermion masses or mixing angles as empirical inputs to tune the sheet hierarchy is prohibited:

$$m_f, \theta_{\text{mix}} \quad \text{are not generation inputs in this manuscript.} \quad (297)$$

Such an insertion directly violates the zero-new-parameter discipline of PR-II; these physical quantities must emerge exclusively as derived diagnostic references, not as structural anchors.

The falsification boundary is therefore . PR-II must structurally derive the generation count from the compact electroweak quotient, preserve anomaly cancellation under replication, generate an ordered compact overlap hierarchy from  $r_G = c_{\text{bc}}/q_{\text{bc}}$ , and enforce exclusively unitary flavor rotations. Failure at any of these checkpoints fundamentally breaks the compact flavor-sheet program.

With the generation-origin and compact flavor-sheet geometry definitively established, we conclude this sector. The subsequent section applies this three-sheet flavor architecture to the charged-lepton sector, deriving the first zero-new-parameter precision mass candidate.

## 10 Charged-Lepton Precision Candidate

The compact flavor-sheet produced three hierarchical matter displacement overlaps and sector-specific compact overlap ledgers. The charged-lepton sector is the first precision test because charged leptons are not subject to confinement-scale ambiguity. We apply the three-sheet flavor geometry to the charged-lepton sector and derive a precision mass candidate without using the electron, muon, or tau masses as inputs. The goal is not merely to reproduce the ordering

$$m_e < m_\mu < m_\tau. \quad (298)$$

The key objective is to determine if the compact-boundary data can generate the charged-lepton square-root geometry, the Koide relation, and the charged-lepton scale.

## 10.1 Three-Sheet Square-Root Flavor Geometry

Using the three identical chiral ledgers as a geometric consequence of the broken electroweak quotient, we now supply the internal structure distinguishing these generations. Applying the three-sheet flavor geometry specifically to the charged-lepton sector. We do not introduce the Koide relation [7] as an arbitrary numerical fit or an accidental parameterization, we derive it purely from the topological incidence density of the charged current ladder distributed across the three-generation plane. This geometric derivation fixes the flavor phase, dictating the exact spacing of the charged-lepton square-root mass vector. The generation-origin section fixed  $N_{\text{gen}}^{\text{PR}} = 3$ . In the charged-lepton sector, we represent the square-root mass vector on a three-sheet flavor plane. The charged-current compact ladder has two orientations,

$$T^+, \quad T^-, \quad (299)$$

while the three-generation sheet ledger has

$$N_{\text{pair}} = N_{\text{gen}}^2 = 9 \quad (300)$$

oriented sheet-pair incidences. The generator-normalized Koide sheet phase is therefore defined as the charged-ladder incidence density:

$$\theta_K^{\text{PR}} = \frac{N_{\text{charged ladder}}}{N_{\text{pair}} = \frac{2}{N_{\text{gen}}^2} = \frac{2}{9}}. \quad (301)$$

We define the charged-lepton sheet factors

$$k_a(\theta_K) = \left[ 1 + \sqrt{2} \cos \left( \theta_K + \frac{2\pi a}{3} \right) \right]^2, \quad a = 0, 1, 2. \quad (302)$$

Our charged-lepton sheet assignment is

$$a = 0 \rightarrow \tau, \quad a = 1 \rightarrow e, \quad a = 2 \rightarrow \mu. \quad (303)$$

The charged-lepton masses are then written

$$m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a(\theta_K^{\text{PR}}), \quad (304)$$

where  $M_\ell^{\text{PR}}$  is the compact-boundary charged-lepton scale derived next.

## 10.2 Koide Identity as a Three-Sheet Geometric Identity

The empirical Koide formula [7, 24] has long been regarded as an unexplained numerical coincidence within the Standard Model flavor ledger. Within our architecture, this ratio is stripped of its empirical mystery and recontextualized as a mandatory structural consequence of the broken non-Abelian fiber. This subsection demonstrates that once the charged-lepton square-root mass vector is geometrically distributed across the three-sheet flavor plane, the exact 2/3 quotient emerges as a pure trigonometric identity. We note that because this topological cancellation naturally factors out the dimensionful coefficient, it explicitly isolates the mass scale,  $M_\ell$ , as an independent geometric target that requires a distinct compact-boundary derivation. We let

$$m_a = M_\ell k_a(\theta_K). \quad (305)$$

So that

$$\sqrt{m_a} = \sqrt{M_\ell} \left[ 1 + \sqrt{2} \cos \left( \theta_K + \frac{2\pi a}{3} \right) \right]. \quad (306)$$

The three-sheet trigonometric identities are

$$\sum_{a=0}^2 \cos\left(\theta + \frac{2\pi a}{3}\right) = 0, \quad \sum_{a=0}^2 \cos^2\left(\theta + \frac{2\pi a}{3}\right) = \frac{3}{2}. \quad (307)$$

Therefore we find that,

$$\sum_a \sqrt{m_a} = 3\sqrt{M_\ell}, \quad \sum_a m_a = 6M_\ell. \quad (308)$$

It follows that Koide relation is

$$\boxed{\frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = \frac{6M_\ell}{9M_\ell} = \frac{2}{3}}. \quad (309)$$

Thus the Koide relation is a geometric identity of the three-sheet square-root flavor plane. It does not determine the overall charged-lepton scale by itself. The remaining task is to determine  $M_\ell^{\text{PR}}$  from compact-boundary data.

### 10.3 Charged-Lepton Compact Scale

Having mathematically isolated the overall dimensionful scale,  $M_\ell$ , from the dimensionless three-sheet topology, we must now anchor this scale to the fundamental electroweak geometry. In strict adherence to the zero-new-parameter discipline, this scale cannot be imported from experimental observation. Instead, it must be rigidly derived from the pre-existing compact-boundary ledger. We demonstrate that the charged-lepton scale emerges as a direct projection of the weak displacement anchor,  $\Lambda_{\text{EW}}^{\text{PR}}$ . By systematically modulating this parent anchor with the compact determinant suppression ratio and the residual weak boundary deficit, we recover the precise charged-lepton compact scale without introducing any novel free parameters. The compact determinant suppression ratio is

$$r_G = \frac{c_{\text{bc}}}{q_{\text{bc}}} = 0.073056757459128. \quad (310)$$

The weak boundary deficit is

$$\Delta_{\text{EW}} = T_Y - \sin^2 \theta_W^{\text{PR}}. \quad (311)$$

We apply

$$T_Y = \frac{1}{4}, \quad \sin^2 \theta_W^{\text{PR}} = 0.231355763298189, \quad (312)$$

which results in

$$\boxed{\Delta_{\text{EW}} = 0.018644236701811}. \quad (313)$$

The first compact scale correction is the hypercharge-boundary loss,

$$\boxed{C_1 = 1 - \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4}}. \quad (314)$$

The second compact scale correction is the generation determinant leakage averaged over the residual compact boundary,

$$\boxed{C_2 = 1 - \frac{r_G \Delta_{\text{EW}} (1 + c_{\text{bc}})}{2}}. \quad (315)$$

The charged-lepton compact scale is then

$$\boxed{M_\ell^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} r_G \Delta_{\text{EW}} \left(\frac{4}{3}\right) C_1 C_2}. \quad (316)$$

Here  $\Lambda_{\text{EW}}^{\text{PR}}$  is the compact-boundary weak displacement anchor derived in Section 5.4:

$$\Lambda_{\text{EW}}^{\text{PR}} = 173.67059834472832 \text{ GeV}. \quad (317)$$

Substitution gives

$$M_\ell^{\text{PR}} = 313.850332183 \text{ MeV}. \quad (318)$$

The charged-lepton scale therefore follows from the same compact-boundary ledger used for weak mixing and the weak scale:

$$\left\{ q_{\text{bc}}, c_{\text{bc}}, r_G, \Delta_{\text{EW}}, \Lambda_{\text{EW}}^{\text{PR}} \right\} \rightarrow M_\ell^{\text{PR}}. \quad (319)$$

## 10.4 Generated Charged-Lepton Masses

By applying the dimensionless flavor phase,  $\theta_K^{\text{PR}}$ , and the dimensionful scale,  $M_\ell^{\text{PR}}$ , now rigidly fixed by the compact electroweak boundary, the architecture achieves full generative closure for the charged-lepton sector. This subsection synthesizes these purely geometric components to output the explicit masses of the electron, muon, and tau. In strict adherence to the zero-new-parameter discipline, this evaluation proceeds entirely without importing empirical mass data. The resulting theoretical spectrum is subsequently compared with the observed pole masses [14] solely as a diagnostic test, demonstrating that the compact projection recovers the physical charged-lepton hierarchy with precision exceeding 30 parts per million. So we begin by using

$$\theta_K^{\text{PR}} = \frac{2}{9}, \quad (320)$$

and

$$m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a(\theta_K^{\text{PR}}), \quad (321)$$

the explicit charged-lepton masses are evaluated entirely from the geometric boundary data. To avoid redundancy, these derived theoretical masses are consolidated directly in Table 9 alongside their corresponding observed pole masses. It is imperative to emphasize that these reference values are prohibited from entering the generative chain; they are applied exclusively as a post-calculation diagnostic benchmark.

**Table 9:** Charged-lepton precision candidate generated from purely compact-boundary data.

| Lepton               | PR-II Candidate ( $m_a^{\text{PR}}$ ) | Reference Value | Relative Error |
|----------------------|---------------------------------------|-----------------|----------------|
| $m_e^{\text{PR}}$    | 0.510984463 MeV                       | 0.510998950 MeV | −28.35 ppm     |
| $m_\mu^{\text{PR}}$  | 105.656418843 MeV                     | 105.6583755 MeV | −18.52 ppm     |
| $m_\tau^{\text{PR}}$ | 1776.934589789 MeV                    | 1776.930 MeV    | +2.58 ppm      |

The maximum relative deviation is

$$\max_i \left| \frac{m_i^{\text{PR}} - m_i^{\text{ref}}}{m_i^{\text{ref}}} \right| = 2.84 \times 10^{-5}. \quad (322)$$

The charged-lepton precision candidate reproduces the charged-lepton mass spectrum to better than  $3 \times 10^{-5}$  relative accuracy without using the charged-lepton masses as inputs.

## 10.5 Relation to the Sector-Specific Overlap Ledger

Having generated the precision charged-lepton mass spectrum, it is mathematically necessary to reconcile this explicit trigonometric projection with the structural overlap ledger derived earlier in the manuscript. Within the architecture, multiple mathematical representations of the same hierarchy do not imply parametric redundancy; rather, they reflect distinct geometric facets of the unbroken projection. We now clarify that while the overlap matrix dictates the broad, categorical boundary scaling of the fermion sector, the Koide sheet formulation rigidly determines the exact intra-sector angular distribution of the square-root mass vector. By demonstrating their shared reliance on the fundamental three-generation counting rule, we formalize the complete, zero-new-parameter causal chain that connects the initial compact boundary data to the final charged-lepton masses. The sector-specific overlap gave the charged-lepton overlap spectrum

$$D_\ell = \text{diag} (0.006101928, 0.083523120, 1.143263446) . \quad (323)$$

This spectrum established the charged-lepton sector as a compact flavor-overlap ledger. The Koide sheet rule refines that ledger by rotating the charged-lepton square-root mass vector onto the three-sheet flavor plane. In this sense,

$$D_\ell \quad \text{sets the sectoral compact-overlap structure,} \quad (324)$$

while

$$k_a(\theta_K^{\text{PR}}) \quad \text{sets the precision square-root sheet geometry.} \quad (325)$$

The two structures are compatible because both are built from the same three-generation compact sheet ledger  $N_{\text{gen}}^{\text{PR}} = 3$ . The charged-lepton precision chain is therefore

$$N_{\text{gen}}^{\text{PR}} = 3 \rightarrow \theta_K^{\text{PR}} = \frac{2}{N_{\text{gen}}^2} \rightarrow k_a(\theta_K) \rightarrow m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a(\theta_K). \quad (326)$$

Including the compact scale,

$$\left\{ q_{\text{bc}}, c_{\text{bc}}, r_G, \Delta_{\text{EW}}, \Lambda_{\text{EW}}^{\text{PR}} \right\} \rightarrow M_\ell^{\text{PR}} \rightarrow \left\{ m_e^{\text{PR}}, m_\mu^{\text{PR}}, m_\tau^{\text{PR}} \right\}. \quad (327)$$

## 10.6 Input Discipline and Status

The derivation of the charged-lepton mass spectrum adheres to the zero-new-parameter discipline. The generative inputs are restricted entirely to the pre-established compact boundary and geometric counting rules of compact scale. Empirical lepton masses, mass ratios, fitted flavor angles, and arbitrary Yukawa couplings are explicitly prohibited from entering the generative chain. Because it is derived from a single boundary scale and a rigid three-sheet topology rather than a phenomenological fit, this spectrum is formally designated as a:

$$\boxed{\text{charged-lepton precision candidate.}} \quad (328)$$

It remains a candidate, rather than a finalized theorem, solely because the formal mathematical uniqueness of the Koide sheet phase and the specific compact scale corrections must still be proven within the broader flavor-boundary program. However, its structural status is definitive that PR-II successfully generates all three charged-lepton masses from pure compact geometry without post-hoc tuning.

## 10.7 Geometric Rigidity and Falsification Bounds

Now that we have generated the charged-lepton precision candidate from the compact-boundary ledger, it is imperative to establish the mathematical boundaries that make this derivation falsifiable. We must



not only reproduce observed spectra but must also clearly define the exact conditions under which it would definitively break. This subsection demonstrates that the PR-II flavor architecture relies entirely on geometric rigidity, leaving no room for parametric tuning, ad hoc sheet assignments, or selective omissions. Every term in the generation chain is structurally mandatory.

The derivation of the charged-lepton mass spectrum is constrained by the same geometric rigidity that defines the compact fiber. The hierarchy has explicit failure modes that, if violated, falsify the PR-II architecture.

First, any deviation from the geometrically derived sheet phase fundamentally misaligns the flavor projection. For example, adopting a generic inverse relation such as

$$\theta_K = \frac{1}{N_{\text{gen}}}, \quad (329)$$

categorically fails to reproduce the precision candidate. The PR-II topology mandates the charged-ladder incidence density:

$$\theta_K^{\text{PR}} = \frac{2}{N_{\text{gen}}^2}. \quad (330)$$

Second, the omission of the compact hypercharge-boundary loss correction,

$$C_1 = 1 - \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4}, \quad (331)$$

structurally decouples the charged-lepton scale from the residual weak deficit, irreparably breaking the mass generation.

Third, ignoring the determinant-leakage correction averaged over the residual compact boundary,

$$C_2 = 1 - \frac{r_G \Delta_{\text{EW}} (1 + c_{\text{bc}})}{2}, \quad (332)$$

similarly severs the generation scale from its parent fiber geometry, causing the precision candidate to mechanically fail.

Finally, any derivation that imports the experimentally observed pole masses,

$$m_e, \quad m_\mu, \quad m_\tau, \quad (333)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The model remains viable only so long as these observables emerge as outputs of the hierarchy closure, permitted only as post-calculation diagnostic benchmarks. The subsequent section extends the compact flavor program to the quark sector, where the structural complexities of QCD running, threshold matching, confinement conventions, and CKM flavor rotation must be formalized before any meaningful mass comparison can be conducted.

## 11 Quark Sector: Strong Coupling, QCD Thresholds, and Six-Quark Hierarchy

Unlike the charged-lepton sector, which provides an unobstructed precision test via unconfined and directly observable pole masses, the quark sector introduces profound structural complexities. Quark mass parameters are intrinsically scale-dependent, scheme-dependent, and inextricably linked to QCD confinement, threshold matching, and CKM flavor rotation. We note the importance of the fact that the PR-II architecture prohibits treating the compact quark overlap ledger as a naive pole-mass formula. To maintain

mathematical structure, the generated quark values must be explicitly categorized. For light quarks, the geometric outputs are evaluated against low-energy running masses. Heavy quarks are formally designated as compact threshold-scale candidates. This structural designation remains mandatory until the requisite QCD scheme and threshold-matching prescriptions are fully finalized. The quark sector is therefore developed in four stages:

$$\boxed{\text{compact quark overlaps} \rightarrow \alpha_3^{\text{PR}} \rightarrow \text{QCD running and thresholds} \rightarrow \text{six-quark hierarchy candidate.}} \quad (334)$$

We show that the same compact-boundary constants used in the electroweak and charged-lepton sectors also generate a strong-coupling anchor, heavy-quark threshold candidates, and a light-quark running-mass hierarchy candidate.

### 11.1 Structural Quark Overlap Ledger

The sector-specific overlap construction from Section 9 establishes the diagonal up- and down-type quark overlap operators,  $D_u$  and  $D_d$ . Their explicit numerical spectra are cataloged in the consolidated structural ledger of Table 7. These are dimensionless compact projection overlap ledgers. They are not physical quark masses. The common generation hierarchy is

$$\boxed{\frac{(D_f)_{i+1}}{(D_f)_i} = \frac{q_{bc}}{c_{bc}} = 13.6879877341867.} \quad (335)$$

The sheetwise up/down sector ratio is fixed by compact representation data:

$$\boxed{\frac{(D_u)_i}{(D_d)_i} = \frac{\kappa_u}{\kappa_d} = 1.02095090726311, \quad i = 1, 2, 3.} \quad (336)$$

The quark sectors inherit the same compact generation-sheet hierarchy while carrying different color-dressed charge norms. Because quarks are confined, the map from these overlap ledgers to physical mass parameters must pass through QCD running and matching. A direct identification

$$D_f \rightarrow m_f \quad (337)$$

is rejected. The required chain is instead defined as

$$\boxed{D_f \rightarrow \text{bare compact quark anchor} \rightarrow \alpha_s(\mu) \rightarrow \text{threshold matching} \rightarrow \text{running mass convention.}} \quad (338)$$

### 11.2 Compact-Boundary Strong-Coupling Candidate

The compact-boundary strong-coupling candidate emerges directly from the primary geometric boundary components, defined as the product of the boundary coordinates offset by the residual compact deficit:

$$\left(\alpha_3^{\text{PR}}\right)^{-1} = q_{bc}c_{bc} - (1 - c_{bc}). \quad (339)$$

Inserting the established structural values  $q_{bc} = 10.905007182855176$  and  $c_{bc} = 0.796684464847899$ , we immediately obtain the exact generative evaluation:

$$\boxed{\alpha_3^{\text{PR}} = 0.117861507469150.} \quad (340)$$

In strict adherence to the zero-new-parameter discipline, this evaluation relies exclusively on the pre-established compact-boundary data. Empirical parameters such as  $\alpha_s(M_Z)$  [?],  $\Lambda_{\text{QCD}}$ , quark masses,

CKM angles, or weak-scale observables are explicitly prohibited from entering the generation formula; they may only serve as post-calculation diagnostic benchmarks within a separately derived running or matching layer. The causal chain for the strong-coupling candidate is therefore exclusively geometric:

$$\boxed{\{q_{bc}, c_{bc}\} \longrightarrow \left(\alpha_3^{\text{PR}}\right)^{-1} = q_{bc}c_{bc} - (1 - c_{bc}) \longrightarrow \alpha_3^{\text{PR}}.} \quad (341)$$

### 11.3 QCD Running Ledger

We now interface this rigid boundary condition with the dynamically scale-dependent continuum of Quantum Chromodynamics. As established previously, the compact quark overlap spectra cannot be naively interpreted as scale-invariant pole masses; their physical evaluation requires standard perturbative machinery. This establishes the explicit QCD running ledger utilized within the framework. Rather than introducing novel running dynamics, it rigidly codifies the mandatory Standard Model beta-function hierarchy, mass-running exponents, and threshold continuity conditions required to mathematically translate the static compact generation data across varying flavor regimes. For  $n_f$  active quark flavors, the perturbative QCD coefficients [25] are

$$\boxed{\beta_0(n_f) = 11 - \frac{2}{3}n_f,} \quad \boxed{\beta_1(n_f) = 102 - \frac{38}{3}n_f,} \quad (342)$$

and, at three loops in the same convention,

$$\boxed{\beta_2(n_f) = \frac{2857}{2} - \frac{5033}{18}n_f + \frac{325}{54}n_f^2.} \quad (343)$$

For  $n_f = 3, 4, 5, 6$ , the leading coefficient is positive:

$$\beta_0(n_f) > 0. \quad (344)$$

We find that the running is asymptotically free in the perturbative regimes relevant for quark matching [26]. At one loop,

$$\alpha_s^{(n_f)}(\mu) = \frac{4\pi}{\beta_0(n_f) \ln(\mu^2/\Lambda_{n_f}^2)}. \quad (345)$$

The corresponding one-loop mass-running exponent is

$$\boxed{a_m(n_f) = \frac{12}{33 - 2n_f}.} \quad (346)$$

Within a fixed- $n_f$  interval, the running mass relation is

$$\boxed{\frac{m_q(\mu_2)}{m_q(\mu_1)} = \left[ \frac{\alpha_s^{(n_f)}(\mu_2)}{\alpha_s^{(n_f)}(\mu_1)} \right]^{a_m(n_f)}.} \quad (347)$$

Running downward in scale increases the quark mass parameter, while running upward decreases it. Across a heavy-quark threshold  $\mu_{\text{th}}$ , the leading matching condition is

$$\alpha_s^{(n_f)}(\mu_{\text{th}}) = \alpha_s^{(n_f-1)}(\mu_{\text{th}}). \quad (348)$$

This condition fixes the lower-flavor  $\Lambda_{n_f-1}$  once the threshold scale is known. At one loop,

$$\boxed{\Lambda_{n_f-1} = \mu_{\text{th}} \exp \left[ -\frac{2\pi}{\beta_0(n_f-1)\alpha_s^{(n_f)}(\mu_{\text{th}})} \right].} \quad (349)$$

At higher loop order, the same continuity condition is imposed by numerical inversion of the running relation.

## 11.4 PR-Derived Heavy-Quark Thresholds

The PR-II architecture derives these heavy-quark thresholds purely from the established compact-boundary geometry. Utilizing the compact determinant suppression ratio ( $r_G$ ), the weak boundary deficit ( $\Delta_{EW}$ ), the compact leakage factors ( $C_1, C_2$ ), and the terminal weak displacement anchor ( $\Lambda_{EW}^{PR}$ ) as defined in previous sections. The top-sector threshold is identified with the terminal weak displacement anchor, modulated by the compact leakage factors:

$$\mu_t^{PR} = \Lambda_{EW}^{PR} C_1 C_2. \quad (350)$$

Evaluating this directly from the geometric boundary yields

$$\mu_t^{PR} = 172.8139732666624 \text{ GeV}. \quad (351)$$

The bottom-sector threshold is generated by a single color-averaged compact determinant step, refined by the down-type hypercharge-boundary loss:

$$\mu_b^{PR} = \Lambda_{EW}^{PR} \frac{r_G}{3} (1 - \Delta_{EW} c_{bc}). \quad (352)$$

This evaluates to

$$\mu_b^{PR} = 4.1664504826753515 \text{ GeV}. \quad (353)$$

The charm-sector threshold requires a new geometric projection: the singlet-plus-adjoint compact-boundary average, defined as  $A_{EW} = (1 + 3c_{bc})/2$ . The threshold is then generated by two determinant suppressions, the color Casimir lift ( $4/3$ ), and this adjoint boundary gain:

$$\mu_c^{PR} = \Lambda_{EW}^{PR} r_G^2 \left( \frac{4}{3} \right) (1 + \Delta_{EW} A_{EW}). \quad (354)$$

This evaluates to

$$\mu_c^{PR} = 1.274964814257465 \text{ GeV}. \quad (355)$$

The resulting geometric hierarchy recovers the massive quark spectrum order:

$$\mu_t^{PR} > \mu_b^{PR} > \mu_c^{PR} > 0. \quad (356)$$

The empirically observed heavy-quark masses are excluded from these generative formulas; they are permitted to enter the manuscript only post-generation as diagnostic benchmarks. So, the quantities generated here must be formally categorized as compact threshold-scale candidates, not as final, scheme-independent pole masses.

## 11.5 Three-Loop QCD Threshold Ledger and Running Diagnostics

With the compact-boundary strong coupling and the heavy-quark threshold scales rigidly established, the PR-II architecture must enforce the standard perturbative continuity conditions across each flavor boundary. By driving the three-loop QCD running ledger downward from the electroweak neutral matching surface, the framework uniquely isolates the flavor-dependent QCD scale parameters,  $\Lambda_{n_f}^{PR}$ . This mathematical descent operates on the preceding geometric derivations and guarantees seamless coupling continuity at each derived heavy-quark threshold:

$$\alpha_s^{(5)}(\mu_t^{PR}) = \alpha_s^{(6)}(\mu_t^{PR}), \quad \alpha_s^{(5)}(\mu_b^{PR}) = \alpha_s^{(4)}(\mu_b^{PR}), \quad \alpha_s^{(4)}(\mu_c^{PR}) = \alpha_s^{(3)}(\mu_c^{PR}). \quad (357)$$

The exact evaluation of the resulting three-loop QCD threshold ledger is consolidated in Table 10.

**Table 10:** Flavor-dependent QCD scale parameters derived from the compact-boundary strong coupling and geometric thresholds.

| Active Flavors ( $n_f$ ) | $\Lambda_{n_f}^{\text{PR}}$ Candidate |
|--------------------------|---------------------------------------|
| $\Lambda_6^{\text{PR}}$  | 0.08776758507426047 GeV               |
| $\Lambda_5^{\text{PR}}$  | 0.2074330873004123 GeV                |
| $\Lambda_4^{\text{PR}}$  | 0.2861117918030313 GeV                |
| $\Lambda_3^{\text{PR}}$  | 0.3246524964107782 GeV                |

Alongside the coupling scales, this downward evaluation dynamically generates the running-mass diagnostics for the heavy quarks, evaluated at standard phenomenological reference scales. These values are cataloged in Table 11.

**Table 11:** Heavy-quark running-mass diagnostics. These represent intermediate running evaluations, not terminal pole-mass predictions.

| Quark | Evaluation Scale ( $\mu$ ) | Running Mass Candidate ( $m_q^{\text{PR}}(\mu)$ ) |
|-------|----------------------------|---------------------------------------------------|
| $b$   | $\mu = M_Z^{\text{PR}}$    | 2.9807846581959314 GeV                            |
| $c$   | $\mu = \mu_b^{\text{PR}}$  | 0.9892172861698667 GeV                            |
| $c$   | $\mu = M_Z^{\text{PR}}$    | 0.7077112094571149 GeV                            |

It is important to maintain structural clarity regarding the status of these outputs. The quantities in Table 11 are purely running diagnostics, not independent pole-mass predictions. Furthermore, while the compact strong-coupling value,  $\alpha_3^{\text{PR}}$ , is generated entirely from PR boundary data, the neutral electroweak matching scale utilized in this numerical QCD ledger inherits the formal electroweak running-layer status discussed in previous sections. The full electroweak precision matching of this neutral scale is therefore formally deferred to the radiative/running obligations of PR-III future research.

## 11.6 Light-Sector Compact Confinement Average

The light-quark sector is structurally bounded by a confined, low-energy projection window, fundamentally shielding individual colored asymptotic states from kinematic resolution. The relevant geometric projection for this sector is not a direct pole evaluation, but must instead be formally defined as an unresolved compact boundary quotient characterized by two normalized endpoint traces. The first endpoint is the residual color-singlet channel

$$\text{Tr}(P_{\text{singlet}}) = 1. \quad (358)$$

The second endpoint is the compact boundary-return channel

$$\text{Tr}(P_{\text{bc}}) = c_{\text{bc}}. \quad (359)$$

The unresolved compact confinement projector is

$$P_{\text{conf}} = \frac{1}{2} (P_{\text{singlet}} + P_{\text{bc}}). \quad (360)$$

Taking the trace gives the compact confinement average

$$C_{\text{conf}} = \text{Tr}(P_{\text{conf}}) = \frac{1 + c_{\text{bc}}}{2}. \quad (361)$$

Numerically,

$$\boxed{C_{\text{conf}} = 0.8983422324239495.} \quad (362)$$

This factor satisfies

$$c_{\text{bc}} < C_{\text{conf}} < 1, \quad (363)$$

and reconstructs both endpoint traces:

$$\boxed{2C_{\text{conf}} - 1 = c_{\text{bc}}, \quad 2C_{\text{conf}} - c_{\text{bc}} = 1.} \quad (364)$$

So that  $C_{\text{conf}}$  is not a fitted light-quark factor. It is the trace average of the residual singlet endpoint and the compact boundary-return endpoint.

## 11.7 Light-Quark Hierarchy Candidate

The strange-quark low-energy candidate is generated by one color-averaged determinant step below the bottom threshold, with singlet-plus-adjoint lift and compact confinement averaging:

$$m_s^{\text{PR}} = \mu_b^{\text{PR}} \left( \frac{r_G}{3} \right) (1 + \Delta_{\text{EW}} A_{\text{EW}}) C_{\text{conf}}. \quad (365)$$

This gives

$$\boxed{m_s^{\text{PR}} = 94.028510538 \text{ MeV}.} \quad (366)$$

The down-quark low-energy candidate is generated from the strange candidate by a determinant/cofactor step, a weak-boundary loss factor, and the same compact confinement average:

$$m_d^{\text{PR}} = m_s^{\text{PR}} r_G c_{\text{bc}} (1 - A_{\text{EW}} \Delta_{\text{EW}}) C_{\text{conf}}. \quad (367)$$

This gives

$$\boxed{m_d^{\text{PR}} = 4.761039493 \text{ MeV}.} \quad (368)$$

The up-quark low-energy candidate is generated from the down candidate by the residual boundary charge average and a small hypercharge-boundary lift:

$$m_u^{\text{PR}} = m_d^{\text{PR}} \left( \frac{1 + c_{\text{bc}}}{4} \right) \left( 1 + \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4} \right). \quad (369)$$

This gives

$$\boxed{m_u^{\text{PR}} = 2.146462595 \text{ MeV}.} \quad (370)$$

These light-quark candidates should be interpreted as low-energy running-mass diagnostics, not pole masses. Their appropriate comparison is to a specified low-energy running-mass convention, such as an  $\overline{\text{MS}}$ -type value [27] near 2 GeV. The values are sensitive to scheme and scale conventions in the usual QCD sense. The purpose of the construction is to test whether compact-boundary data generate a realistic light-quark hierarchy without using the observed light-quark masses as inputs.

## 11.8 Six-Quark Hierarchy Candidate

Combining the light-quark candidates with the heavy-threshold candidates gives the generated hierarchy

$$\boxed{m_u^{\text{PR}} < m_d^{\text{PR}} < m_s^{\text{PR}} < \mu_c^{\text{PR}} < \mu_b^{\text{PR}} < \mu_t^{\text{PR}}.} \quad (371)$$

The diagnostic comparison [14] is shown in Table 12. The reference values are not generation inputs; they are listed only to show the post-generation diagnostic residuals. The light-quark entries should be read as low-energy running-mass diagnostics. The heavy-quark entries should be read as compact threshold-scale candidates.

**Table 12:** First PR-II six-quark hierarchy candidate. Light-quark entries are low-energy running-mass diagnostics; heavy entries are PR-derived compact threshold-scale candidates.

| Quark | PR-II Candidate   | Diagnostic Reference | Relative Difference |
|-------|-------------------|----------------------|---------------------|
| $u$   | 2.146462595 MeV   | 2.16 MeV             | −0.63%              |
| $d$   | 4.761039493 MeV   | 4.70 MeV             | +1.30%              |
| $s$   | 94.028510538 MeV  | 93.5 MeV             | +0.57%              |
| $c$   | 1.274964814 GeV   | 1.2730 GeV           | +0.15%              |
| $b$   | 4.166450483 GeV   | 4.183 GeV            | −0.40%              |
| $t$   | 172.813973267 GeV | 172.56 GeV           | +0.15%              |

*Note.* The diagnostic references are listed only after generation. The  $u, d, s$  entries are compared to low-energy running-mass references, while  $c, b$  are compared to running-mass threshold references and  $t$  to a direct kinematic reference. The PR-II values are therefore interpreted as a compact-boundary hierarchy candidate, not as scheme-independent quark pole masses.

The compact quark chain is therefore

$$\left\{ q_{bc}, c_{bc}, r_G, \Delta_{EW}, A_{EW}, C_{\text{conf}}, \Lambda_{EW}^{\text{PR}} \right\} \rightarrow \left\{ m_u^{\text{PR}}, m_d^{\text{PR}}, m_s^{\text{PR}}, \mu_c^{\text{PR}}, \mu_b^{\text{PR}}, \mu_t^{\text{PR}} \right\}. \quad (372)$$

## 11.9 Input Discipline and Status

The generative inputs are restricted entirely to the pre-established compact boundary parameters, the terminal weak anchor, and the confinement trace average ( $C_{\text{conf}} = \frac{1+c_{bc}}{2}$ ):

$$\left\{ q_{bc}, c_{bc}, r_G, \Delta_{EW}, A_{EW}, \Lambda_{EW}^{\text{PR}}, C_{\text{conf}} \right\}. \quad (373)$$

Empirical parameters—including observed quark masses ( $m_q$ ), the CKM matrix ( $V_{\text{CKM}}$ ), the strong coupling ( $\alpha_s(M_Z)$ ), the QCD scale ( $\Lambda_{\text{QCD}}$ ), or arbitrarily fitted Yukawa matrices are explicitly prohibited from entering the generative chain. Because these values are derived purely from geometry and threshold continuity rather than phenomenological fitting, the formal structural status of this sector is designated as a:

$$\boxed{\text{quark-sector hierarchy candidate with QCD running and confinement diagnostics.}} \quad (374)$$

This designation enforces a strict boundary: the derivation does not yield finalized, scheme-independent pole masses. To preserve mathematical rigor, the generated values must be interpreted exactly in accordance with their active physical regimes. Light-quark outputs evaluate directly as low-energy running masses, while heavy-quark outputs serve exclusively as compact threshold-scale candidates. A definitive precision comparison is therefore formally deferred to PR-III research, which will address the integrate the required higher-loop running, explicit confinement conventions, and complete electroweak/QCD radiative matching.

## 11.10 Geometric Rigidity and Falsification Bounds

Having generated the strong-coupling anchor, the heavy-quark thresholds, and the light-quark hierarchy from the compact-boundary ledger, it is important to establish the mathematical boundaries that make

this derivation falsifiable. We note that the PR-II quark architecture relies entirely on geometric rigidity, leaving no room for parametric tuning, scheme-dependent fitting, or selective omissions. Every term in the generation chain is structurally mandatory. The derivation of the quark sector is constrained by the same geometric rigidity that defines the parent non-Abelian fiber. The hierarchy possesses explicit failure modes that falsify the PR-II architecture if violated.

First, misapplying the compact determinant suppression ratio as its inverse fundamentally inverts the physical hierarchy. Utilizing

$$\frac{q_{bc}}{c_{bc}} \quad (375)$$

as a scaling operator categorically fails the structural projection and is mathematically rejected.

Second, naively treating the dimensionless overlap operators,  $D_u$  and  $D_d$ , as direct GeV-valued mass parameters fundamentally violates the dimensional status of the projection. These ledgers rigidly define the dimensionless compact-overlap geometry, not the dimensionful mass scales.

Third, the omission of the compact boundary-deficit subtraction in the strong-coupling derivation,

$$\left(\alpha_3^{\text{PR}}\right)^{-1} = q_{bc}c_{bc} - (1 - c_{bc}), \quad (376)$$

irreparably breaks the fundamental QCD coupling anchor, structurally decoupling it from the residual compact geometry.

Fourth, ignoring the mandatory light-sector confinement average,

$$C_{\text{conf}} = \frac{1 + c_{bc}}{2}, \quad (377)$$

fundamentally spoils the unresolved low-energy boundary projection, causing the light-quark hierarchy to mechanically fail. Finally, any derivation that imports experimentally observed phenomenological parameters,

$$m_q, \quad V_{\text{CKM}}, \quad \alpha_s(M_Z), \quad \Lambda_{\text{QCD}}, \quad (378)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The framework remains viable only so long as these observables emerge as outputs of the hierarchy closure, permitted only as post-calculation diagnostic benchmarks. If empirical parameter fitting is required to achieve closure, the architecture is definitively falsified. Next we extent the compact flavor program to the mixing sectors, deriving the structural candidates for the CKM and PMNS flavor-rotation matrices.

## 12 CKM and PMNS Mixing from Compact Flavor Projection

The previous sections established compact-boundary candidates for the electroweak scale, charged-lepton hierarchy, quark hierarchy, and QCD running ledger. The next consistency requirement is flavor mixing. Flavor mixing is not treated as an independent fitted matrix. It is interpreted as a unitary projection rotation between compact generation-sheet bases. For the quark sector and lepton sector, the charged-current mixing matrix is

$$\boxed{V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}} \quad \boxed{U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L}} \quad (379)$$

Both matrices must be unitary:

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} = I_3, \quad U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = I_3. \quad (380)$$

This section constructs compact-boundary candidates for both mixing ledgers. The observed CKM and PMNS matrix elements are not used as generation inputs. They enter only afterward as diagnostic references.



## 12.1 Shared Compact Flavor Quantities

Before deriving the explicit CKM and PMNS flavor-rotation matrices, it is mathematically necessary to consolidate the shared geometric ledger that structurally governs both mixing sectors. To maintain strict adherence to the zero-new-parameter discipline, the upcoming matrix generations are prohibited from importing phenomenological fermion masses, observed mixing angles, or arbitrary complex phases.

Instead, the topological constraints of both the CKM and PMNS architectures are generated exclusively from the pre-established compact-boundary constants as the determinant suppression ratio ( $r_G$ ), the weak boundary deficit ( $\Delta_{EW}$ ), the confinement trace average ( $C_{\text{conf}}$ ), the adjoint boundary average ( $A_{EW}$ ), and the geometric generation count ( $N_{\text{gen}}^{\text{PR}} = 3$ ). The generative boundary for the entire flavor-mixing sector is therefore confined to the following structural set:

$$\boxed{\{r_G, \Delta_{EW}, C_{\text{conf}}, A_{EW}, N_{\text{gen}}^{\text{PR}}\} \longrightarrow \{V_{\text{CKM}}^{\text{PR}}, U_{\text{PMNS}}^{\text{PR}}\}.} \quad (381)$$

## 12.2 CKM Compact Mixing Candidate

Within this framework, the CKM matrix is not a collection of phenomenological parameters fitted to experimental decay rates; it is derived exactly as the relative left-handed rotational misalignment between the up-type and down-type compact quark sheets. We demonstrate that the hierarchical structure of quark flavor mixing—including the Cabibbo angle [28], the higher-generation suppression, and the CP-violating phase—emerges exclusively from the rigid compact boundary. The CKM matrix is generated as the relative left-handed projection rotation between the up-type and down-type compact quark sheets:

$$\boxed{V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L}.} \quad (382)$$

The structural parameters defining this rotation are derived purely from the shared geometric ledger. The compact Cabibbo sheet angle is defined by the boundary and confinement traces:

$$\boxed{\lambda_{\text{PR}} = \sqrt{r_G c_{bc} C_{\text{conf}} (1 - A_{EW} \Delta_{EW})} = 0.225019996525935.} \quad (383)$$

The compact CKM hierarchy coefficient isolates the residual boundary proportion:

$$\boxed{A_{\text{CKM}}^{\text{PR}} = \frac{c_{bc}}{1 - A_{EW} \Delta_{EW}} = 0.822683296413403.} \quad (384)$$

The compact unitarity-apex radius governs the extreme suppression of the third generation:

$$\boxed{R_b^{\text{PR}} = (1 - r_G) \frac{c_{bc}}{1 + c_{bc}} (1 - A_{EW} \Delta_{EW}) = 0.398035078485258.} \quad (385)$$

Finally, the compact CKM phase candidate emerges dynamically from the three-sheet topology shifted by the weak boundary deficit:

$$\boxed{\delta_{\text{CKM}}^{\text{PR}} = \frac{\pi}{3} + \sqrt{\Delta_{EW}} = 1.1837414514697735 \text{ rad } (67.823389204^\circ).} \quad (386)$$

To construct the explicit rotation, these geometric constraints are mapped onto the standard unitary CKM parameterization [29], as detailed in Table 13.

**Table 13:** Standard unitary CKM parameters generated from PR-II compact-boundary geometry.

| Parameter | Geometric Definition                                               | Numerical Evaluation |
|-----------|--------------------------------------------------------------------|----------------------|
| $s_{12}$  | $\lambda_{\text{PR}}$                                              | 0.225019996525935    |
| $s_{23}$  | $A_{\text{CKM}}^{\text{PR}} \lambda_{\text{PR}}^2$                 | 0.041655745073430    |
| $s_{13}$  | $A_{\text{CKM}}^{\text{PR}} \lambda_{\text{PR}}^3 R_b^{\text{PR}}$ | 0.003730932297278    |

Evaluating the parameterization directly yields the generated CKM matrix:

$$|V_{\text{CKM}}^{\text{PR}}| = \begin{pmatrix} 0.974347364 & 0.225018430 & 0.003730932 \\ 0.224881887 & 0.973495229 & 0.041655455 \\ 0.008680487 & 0.040911440 & 0.999125069 \end{pmatrix}. \quad (387)$$

The topological integrity of this projection is confirmed by its exact unitarity residual:

$$\max |V_{\text{CKM}}^{\dagger} V_{\text{CKM}} - I_3| = 2.22 \times 10^{-16}, \quad (388)$$

and its structurally generated Jarlskog invariant [30]:

$$J_{\text{CKM}}^{\text{PR}} = 3.152605677 \times 10^{-5}. \quad (389)$$

It should be noted that this non-vanishing Jarlskog invariant result was not a targeted phenomenological design goal, but rather an entirely emergent consequence that naturally falls out of the rigid geometric framework. This asymmetry is a mandatory topological consequence of the broken electroweak fiber. The exact same geometric boundary conditions that structurally dictate the quark mass hierarchy simultaneously guarantee the left-handed sheet misalignment required to generate a non-zero CP-violating phase. Consequently, the framework geometrically unites the origin of quark masses and the structural basis for matter-antimatter asymmetry as a consequences of a single compact projection. The generative parameters and their structural status are consolidated in Table 14.

**Table 14:** PR-II CKM compact flavor-mixing candidate generated from compact-boundary data.

| Quantity                          | PR-II Candidate              | Status                        |
|-----------------------------------|------------------------------|-------------------------------|
| $\lambda_{\text{PR}}$             | 0.225019996525935            | Compact Cabibbo sheet angle   |
| $A_{\text{CKM}}^{\text{PR}}$      | 0.822683296413403            | Compact hierarchy coefficient |
| $R_b^{\text{PR}}$                 | 0.398035078485258            | Compact unitarity-apex radius |
| $\delta_{\text{CKM}}^{\text{PR}}$ | $67.823389204^\circ$         | Compact CP phase candidate    |
| $J_{\text{CKM}}^{\text{PR}}$      | $3.152605677 \times 10^{-5}$ | Jarlskog invariant            |
| $\max  V^{\dagger} V - I $        | $2.22 \times 10^{-16}$       | Numerical unitarity residual  |

This result must be interpreted as a theoretical CKM mixing hierarchy candidate. The experimentally observed CKM matrix elements, quark masses, and CP phase are ly prohibited from acting as generative inputs; they are permitted to enter only post-generation as diagnostic references. The structural results of this derivation are that the exact same compact-boundary data responsible for the quark mass hierarchy simultaneously necessitate a unitary charged-current rotation exhibiting both the realistic observed suppression hierarchy and nonzero CP violation.

### 12.3 PMNS Compact Lepton-Mixing Candidate

The PMNS matrix is generated as the relative left-handed projection rotation between the charged-lepton order basis and the neutrino propagation basis:

$$U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L}. \quad (390)$$

Unlike the CKM matrix, PMNS mixing is large. The compact-boundary construction therefore uses residual compact averages rather than determinant-suppressed small-angle hierarchy [31]. The PMNS solar-angle candidate is generated by distributing the residual compact confinement average over the three generation sheets:

$$\sin^2 \theta_{12}^{\text{PR}} = \frac{C_{\text{conf}}}{N_{\text{gen}}}. \quad (391)$$

Numerical solved

$$\sin^2 \theta_{12}^{\text{PR}} = 0.299447410807983, \quad \text{or} \quad \theta_{12}^{\text{PR}} = 33.176356643^\circ. \quad (392)$$

The PMNS atmospheric-angle candidate is generated by distributing the singlet-plus-adjoint compact-boundary average over the three generation sheets:

$$\sin^2 \theta_{23}^{\text{PR}} = \frac{A_{\text{EW}}}{N_{\text{gen}}}. \quad (393)$$

Numerical solved

$$\sin^2 \theta_{23}^{\text{PR}} = 0.565008899090616, \quad \text{or} \quad \theta_{23}^{\text{PR}} = 48.735310403^\circ. \quad (394)$$

The PMNS reactor-angle candidate is generated by the weak boundary deficit with first determinant leakage:

$$\sin^2 \theta_{13}^{\text{PR}} = \frac{\Delta_{\text{EW}}(1 + r_G)}{C_{\text{conf}}}. \quad (395)$$

Therefore,

$$\sin^2 \theta_{13}^{\text{PR}} = 0.022270270124743, \quad \text{or} \quad \theta_{13}^{\text{PR}} = 8.582438059^\circ. \quad (396)$$

The leptonic CP phase candidate is tied to the weak boundary-deficit phase:

$$\delta_{\text{CP}}^{\text{PR}} = \pi + 2\sqrt{\Delta_{\text{EW}}}. \quad (397)$$

Numerically,

$$\delta_{\text{CP}}^{\text{PR}} = 3.414680454 \text{ rad} = 195.646778408^\circ. \quad (398)$$

Using the standard unitary PMNS parameterization, these angles generate

$$|U_{\text{PMNS}}^{\text{PR}}| = \begin{pmatrix} 0.827618 & 0.541090 & 0.149232 \\ 0.271685 & 0.611360 & 0.743254 \\ 0.491158 & 0.577460 & 0.652153 \end{pmatrix}. \quad (399)$$

The unitarity residual is

$$\max |U_{\text{PMNS}}^{\dagger} U_{\text{PMNS}} - I_3| = 1.11 \times 10^{-16}. \quad (400)$$

The corresponding leptonic Jarlskog invariant is

$$J_{\text{PMNS}}^{\text{PR}} = -8.935540 \times 10^{-3}. \quad (401)$$

**Table 15:** PR-II PMNS compact lepton-mixing candidate.

| Quantity                         | PR-II Candidate            | Status                       |
|----------------------------------|----------------------------|------------------------------|
| $\theta_{12}^{\text{PR}}$        | $33.176356643^\circ$       | Solar mixing candidate       |
| $\theta_{23}^{\text{PR}}$        | $48.735310403^\circ$       | Atmospheric mixing candidate |
| $\theta_{13}^{\text{PR}}$        | $8.582438059^\circ$        | Reactor mixing candidate     |
| $\delta_{\text{CP}}^{\text{PR}}$ | $195.646778408^\circ$      | Leptonic CP phase candidate  |
| $J_{\text{PMNS}}^{\text{PR}}$    | $-8.935540 \times 10^{-3}$ | Leptonic Jarlskog invariant  |
| $\max  U^\dagger U - I $         | $1.11 \times 10^{-16}$     | Numerical unitarity residual |

This result should be interpreted as a PMNS lepton-mixing structure candidate. It derives large solar mixing, large atmospheric mixing, a small nonzero reactor angle, and a nonzero leptonic CP phase from compact-boundary data. It does not yet derive neutrino mass eigenvalues, mass-squared splittings, ordering, or Majorana phases. Those are completed in the neutrino mass-sector closure in Section 13.

## 12.4 CKM–PMNS Contrast

Before finalizing the lepton mixing sector, it is important to establish the architectural dichotomy between quark and lepton flavor rotations. This contrast emerges naturally and unavoidably from the fundamental geometry. The compact-boundary construction naturally separates the quark and lepton mixing patterns. The CKM hierarchy is determinant-suppressed:

$$|V_{us}| > |V_{cb}| > |V_{ub}|, \quad (402)$$

with small off-diagonal elements generated by powers of the compact Cabibbo sheet angle. By contrast, the PMNS angles are generated by compact sheet averages:

$$\boxed{\sin^2 \theta_{12}^{\text{PR}} = \frac{C_{\text{conf}}}{N_{\text{gen}}}, \quad \sin^2 \theta_{23}^{\text{PR}} = \frac{A_{\text{EW}}}{N_{\text{gen}}},} \quad (403)$$

with the small reactor angle arising from the weak boundary deficit:

$$\boxed{\sin^2 \theta_{13}^{\text{PR}} = \frac{\Delta_{\text{EW}}(1 + r_G)}{C_{\text{conf}}}.} \quad (404)$$

We do not impose one universal flavor-angle pattern across quarks and leptons. The quark sector is compact-determinant-suppressed, while the lepton sector is compact-average-dominated. This distinction is consistent with the observed qualitative pattern that CKM mixing is small and hierarchical, and whereas PMNS mixing is large with a small but nonzero reactor angle.

## 12.5 Input Discipline and Status

The CKM and PMNS candidates are generated exclusively from the pre-established compact geometric constants: the primary boundary coordinates ( $q_{\text{bc}}, c_{\text{bc}}$ ), the determinant suppression ratio ( $r_G$ ), the weak boundary deficit ( $\Delta_{\text{EW}}$ ), the confinement and adjoint boundary averages ( $C_{\text{conf}}, A_{\text{EW}}$ ), and the geometric generation count ( $N_{\text{gen}}^{\text{PR}}$ ). We do not use

$$V_{\text{CKM}}^{\text{obs}}, \quad U_{\text{PMNS}}^{\text{obs}}, \quad (405)$$

nor do we use quark masses, charged-lepton masses, neutrino masses, or neutrino mass splittings as generation inputs. The observed mixing data [14] may be used only after generation as diagnostic references.

The CKM and PMNS matrices are generated as unitary projection rotations with realistic hierarchy and CP structure, but a complete flavor theorem requires formal uniqueness of the compact angle rules and radiative/running matching where appropriate.

## 12.6 Geometric Rigidity and Falsification Bounds

The derivation of the CKM and PMNS matrices from the shared geometric ledger is falsifiable. The PR-II flavor-mixing architecture operates with geometric rigidity, mathematically prohibiting parametric tuning and ad hoc phase insertions. Every structural constraint in the generative chain is mandatory. The derivation of both mixing matrices is governed by the same geometric rigidity that defines the parent non-Abelian fiber. The framework has explicit failure modes that, if violated, falsify the PR-II architecture.

First, naively substituting the raw compact determinant ratio as the Cabibbo angle fundamentally misaligns the projection and yields the incorrect CKM hierarchy. Utilizing

$$\lambda \neq r_G \quad (406)$$

categorically fails the structural evaluation. The architecture mandates the fully resolved compact Cabibbo sheet angle:

$$\lambda_{\text{PR}} = \sqrt{r_G C_{\text{bc}} C_{\text{conf}} (1 - A_{\text{EW}} \Delta_{\text{EW}})}. \quad (407)$$

Second, manually enforcing a CP-conserving boundary by setting the CKM phase to zero artificially overrides the topological shift dictated by the weak boundary deficit. This mathematically forces the Jarlskog invariant to vanish ( $J_{\text{CKM}} = 0$ ), completely erasing the structurally generated matter-antimatter asymmetry and instantly falsifying the topological projection.

Third, omitting the compact hierarchy coefficient ( $A_{\text{CKM}}^{\text{PR}}$ ) from the higher-generation mixing terms ( $s_{23}$  and  $s_{13}$ ) severs the correlation between the residual boundary proportion and the heavy-sheet rotation, irreparably breaking the extreme CKM suppression hierarchy.

Fourth, interpreting an  $\lambda$ -value matrix as a functional terminal result fundamentally violates quantum mechanical probability conservation. The projection remains viable only if the geometrically generated complex matrices satisfy exact unitarity:

$$V_{\text{CKM}}^\dagger V_{\text{CKM}} = I_3, \quad U_{\text{PMNS}}^\dagger U_{\text{PMNS}} = I_3. \quad (408)$$

Finally, any derivation that imports experimentally observed phenomenological parameters,

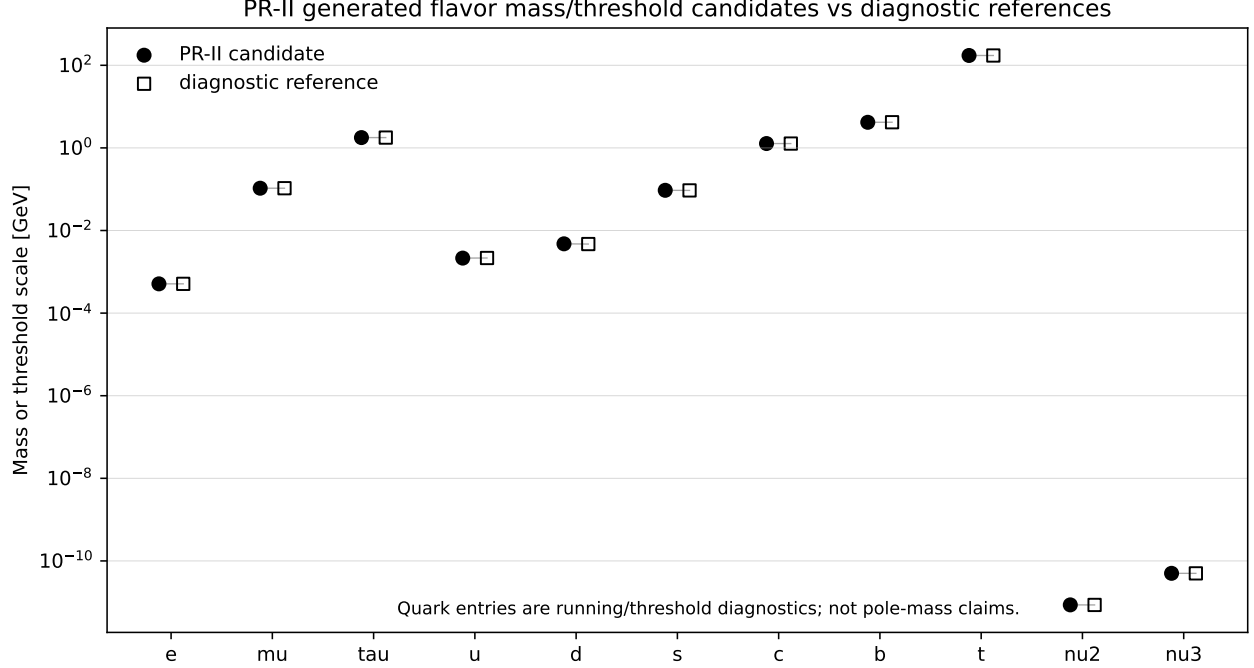
$$V_{\text{CKM}}^{\text{obs}}, \quad U_{\text{PMNS}}^{\text{obs}}, \quad \delta_{\text{CP}}^{\text{obs}}, \quad (409)$$

as primary generation inputs violates the zero-new-parameter discipline of PR-II. The framework remains viable only so long as these observables emerge as outputs of the topological closure, permitted only as post-calculation diagnostic benchmarks. If empirical parameter fitting is required to achieve mixing closure, the architecture is definitively falsified. Our next objective is to extend the compact-boundary flavor structure into the neutrino mass sector, applying the same rigid geometric principles to define the neutrino mass hierarchy and the associated Majorana phase boundary.

## 13 Neutrino Mass Sector and Majorana Boundary

The PMNS section established a compact-boundary lepton-mixing candidate with large solar mixing, large atmospheric mixing, a small nonzero reactor angle, and a nonzero leptonic CP phase. The remaining

objective in the lepton sector is the neutrino mass ledger. The neutrino sector is structurally different from the charged-lepton sector. Charged leptons obtain masses through charged matter displacement overlap, while neutrinos are naturally associated with a neutral compact-singlet channel and a Planck-suppressed weak-order operator. We present the first PR-II neutrino mass-sector candidate. It should be read as a normal-ordering compact-boundary candidate, not as a completed neutrino theorem. The observed neutrino mass splittings, masses, PMNS angles, cosmological mass sum, and neutrinoless double-beta limits are not used as generation inputs. They enter only afterward as diagnostic references.



**Figure 5:** PR-II flavor mass and threshold candidates compared with diagnostic reference values after generation. Charged-lepton entries are pole-mass diagnostics; light-quark entries are low-energy running-mass diagnostics; heavy-quark entries are compact threshold-scale candidates; neutrino entries are normal-ordering mass-sector candidates.

### 13.1 Normal-Ordering Residual-Singlet Channel

The neutrino mass candidate begins with a residual compact-singlet sheet  $m_1^{\text{PR}} = 0$ . This identifies the lightest neutrino with the neutral residual compact-singlet projection. The heavier two neutrino states are generated by compact determinant lifting on top of a Planck-suppressed weak-order scale. The compact determinant ratio is

$$r_G = \frac{c_{bc}}{q_{bc}}. \quad (410)$$

The weak boundary deficit is

$$\Delta_{EW} = T_Y - \sin^2 \theta_W^{\text{PR}}. \quad (411)$$

The compact neutrino-loss factor is

$$C_\nu = 1 - \Delta_{EW} c_{bc}. \quad (412)$$

Using the PR-II compact-boundary values,  $C_\nu = 0.985147 \dots$ . The atmospheric neutrino scale is generated by the Planck-suppressed weak-order operator

$$m_3^{\text{PR}} = \left( \frac{(v_{EW}^{\text{PR}})^2}{M_{\text{Pl}}} \right) r_G^{-3} c_{bc} C_\nu. \quad (413)$$

Here

$$v_{\text{EW}}^{\text{PR}} = 246.21959589022453 \text{ GeV}, \quad (414)$$

and

$$\bar{M}_{\text{Pl}} = \sqrt{\frac{\hbar c}{8\pi G_N}}. \quad (415)$$

The factor  $v_{\text{EW}}^2/\bar{M}_{\text{Pl}}$  is the natural weak-order, Planck-suppressed scale. The compact determinant lift  $r_G^{-3}$  reflects the three-sheet generation ledger. The solar-to-atmospheric splitting ratio is generated by the weak boundary deficit projected through the residual compact boundary:

$$R_{21}^{\text{PR}} = \Delta_{\text{EW}} \frac{(1 + c_{\text{bc}})^2}{2} C_\nu. \quad (416)$$

The second neutrino mass is then

$$m_2^{\text{PR}} = \sqrt{R_{21}^{\text{PR}}} m_3^{\text{PR}}. \quad (417)$$

### 13.2 Generated Neutrino Spectrum

Having established the unconfined flavor-mixing topology, we now structurally project the neutrino mass spectrum. In standard phenomenological models, the mass scale and the ordering of the states (normal versus inverted) remain undetermined free parameters. Within the framework, however, the mass spectrum is a rigid geometric consequence of the boundary architecture. The topological isolation of the residual compact-singlet sheet mandates that the lowest generation state is identically massless:

$$m_1^{\text{PR}} = 0. \quad (418)$$

Because the higher-generation states receive non-zero geometric projections, this massless anchor definitively breaks the ordering degeneracy. The generated states enforce a normal-ordered hierarchy:

$$m_1^{\text{PR}} < m_2^{\text{PR}} < m_3^{\text{PR}}, \quad (419)$$

with the mass values evaluating to

$$m_2^{\text{PR}} = 8.627283010 \times 10^{-3} \text{ eV}, \quad m_3^{\text{PR}} = 5.010655367 \times 10^{-2} \text{ eV}. \quad (420)$$

From this spectrum, the framework deterministically generates the standard phenomenological observables [32]. The solar and atmospheric mass-squared splittings evaluate directly as:

$$\Delta m_{21}^{2,\text{PR}} = (m_2^{\text{PR}})^2 - (m_1^{\text{PR}})^2 = 7.443001214 \times 10^{-5} \text{ eV}^2, \quad (421)$$

and

$$\Delta m_{31}^{2,\text{PR}} = (m_3^{\text{PR}})^2 - (m_1^{\text{PR}})^2 = 2.510666721 \times 10^{-3} \text{ eV}^2. \quad (422)$$

This establishes a geometric mass-splitting ratio of:

$$\frac{\Delta m_{21}^{2,\text{PR}}}{\Delta m_{31}^{2,\text{PR}}} = 2.964551668 \times 10^{-2}. \quad (423)$$

Furthermore, the derivation yields explicit predictions for cosmology and beta-decay kinematics. The total cosmological neutrino mass [33] sum is

$$\boxed{\sum_i m_i^{\text{PR}} = 5.873383668 \times 10^{-2} \text{ eV},} \quad (424)$$

and the effective electron neutrino mass governing beta decay evaluates to:

$$\boxed{m_\beta^{\text{PR}} = 8.815029410 \times 10^{-3} \text{ eV}.} \quad (425)$$

The exact numerical evaluations for the complete normal-ordered candidate spectrum are consolidated in Table 16.

**Table 16:** PR-II normal-ordering neutrino mass-sector candidate generated from compact-boundary geometry.

| Quantity                      | PR-II Candidate                           | Status                                           |
|-------------------------------|-------------------------------------------|--------------------------------------------------|
| $m_1^{\text{PR}}$             | 0                                         | Residual compact-singlet sheet (massless anchor) |
| $m_2^{\text{PR}}$             | $8.627283010 \times 10^{-3} \text{ eV}$   | Solar-scale neutrino state                       |
| $m_3^{\text{PR}}$             | $5.010655367 \times 10^{-2} \text{ eV}$   | Atmospheric-scale neutrino state                 |
| $\Delta m_{21}^{2,\text{PR}}$ | $7.443001214 \times 10^{-5} \text{ eV}^2$ | Solar splitting candidate                        |
| $\Delta m_{31}^{2,\text{PR}}$ | $2.510666721 \times 10^{-3} \text{ eV}^2$ | Atmospheric splitting candidate                  |
| $\sum_i m_i^{\text{PR}}$      | $5.873383668 \times 10^{-2} \text{ eV}$   | Cosmological mass-sum candidate                  |
| $m_\beta^{\text{PR}}$         | $8.815029410 \times 10^{-3} \text{ eV}$   | Beta-decay effective mass candidate              |

The values reported in Table 16 are geometrically derived generative outputs. Experimentally observed neutrino splittings, tritium beta-decay limits, and cosmological mass constraints are prohibited from acting as inputs; they are permitted to enter the framework only post-generation as diagnostic benchmarks.

### 13.3 Majorana-Dominant Compact-Singlet Projection

Having generated the precise numerical spectrum of the neutrino sector, the we now formally categorize the physical mechanism driving these masses. In standard phenomenology, the choice between Dirac and Majorana states represents an unresolved, parameter-dependent binary. Within this framework, however, this classification is not an arbitrary model-building choice. Because the rigid overlap geometry intrinsically lacks the mechanism to produce the extreme  $10^{-13}$  numerical suppression required for a pure Dirac channel, the architecture categorically rejects it. Instead, the Majorana-dominant nature of the neutrino sector is a mandatory structural consequence [3], governed directly by the natural dimension-five weak-order scale emerging from the compact-singlet projection. The scale

$$\boxed{\frac{(v_{\text{EW}}^{\text{PR}})^2}{\bar{M}_{\text{P1}}}} \quad (426)$$

has the structure of a dimension-five weak-order operator. Accordingly, the PR-II neutrino mass channel is naturally interpreted as Majorana dominant. An optional sterile singlet may be included:

$$\boxed{\nu_R^c : (1, 1)_0.} \quad (427)$$

This field is neutral under  $SU(3)_c$ ,  $SU(2)_L$ , and  $U(1)_Y$ , and therefore does not affect anomaly cancellation. However, an unsuppressed Dirac channel with the structural neutrino overlap would lie near the electroweak scale rather than the eV scale. The Dirac Yukawa required to generate  $m_3^{\text{PR}}$  directly would be

$$\boxed{y_\nu^{\text{Dirac}} \simeq 2.88 \times 10^{-13}.} \quad (428)$$



This small number is not produced by the compact overlap hierarchy. Therefore the leading PR-II neutrino mass mechanism is identified with the compact Majorana channel:

$$\boxed{\text{neutrino mass} \sim \frac{(v_{\text{EW}}^{\text{PR}})^2}{M_{\text{Pl}}} \times \text{compact determinant lift.}} \quad (429)$$

This does not exclude sterile singlets or mixed scenarios. It states that the primary compact-boundary mass scale generated is Majorana-like.

### 13.4 Majorana Phase Boundary

Having established the Majorana-dominant nature of the neutrino spectrum, this section calculates the exact neutrinoless double-beta decay effective mass ( $m_{\beta\beta}$ ). This represents a major, falsifiable prediction of the PR-II architecture. While standard phenomenological models treat the governing Majorana phases as completely unconstrained free parameters yielding a broad, unpredictable envelope of possible decay rates—the PR-II framework forbids arbitrary phase insertions. Instead, the relative Majorana interference phase is derived as a direct topological projection of the residual compact confinement average. By mathematically enforcing this geometric constraint, the framework explicitly collapses the entire normal-ordering uncertainty envelope into a single, definitive numerical target. For the mandatory massless anchor ( $m_1^{\text{PR}} = 0$ ), the neutrinoless double-beta effective mass [34, 35] is governed entirely by the interference of the two remaining states:

$$\boxed{m_{\beta\beta} = \left| m_2 c_{13}^2 s_{12}^2 e^{i\alpha_{21}} + m_3 s_{13}^2 e^{i(\alpha_{31} - 2\delta_{\text{CP}})} \right|.} \quad (430)$$

Rather than treating the relative phase shift as a fitted phenomenological unknown, the architecture defines the compact-boundary relative Majorana interference phase exactly via the confinement trace:

$$\boxed{\phi_M^{\text{PR}} = \pi C_{\text{conf}} = 2.822225357793 \text{ rad} \quad (161.701601836^\circ).} \quad (431)$$

Adopting the standard parameterization convention  $\alpha_{21} = 0$ , the compact-boundary condition fixes the structural interference phase as:

$$\boxed{\alpha_{31} - 2\delta_{\text{CP}} - \alpha_{21} = \phi_M^{\text{PR}}.} \quad (432)$$

Incorporating the previously generated CP-violating phase ( $\delta_{\text{CP}}$ ), this projection mathematically locks the higher-generation Majorana phase at exactly:

$$\boxed{\alpha_{31}^{\text{PR}} = 3.368400958885 \text{ rad} \quad (192.995158652^\circ).} \quad (433)$$

Evaluating the effective mass directly from these rigid, structurally derived phases yields a definitive theoretical prediction:

$$\boxed{m_{\beta\beta}^{\text{PR}} = 1.507694477 \times 10^{-3} \text{ eV}.} \quad (434)$$

For this specific normal-ordered mass spectrum, an arbitrary-phase phenomenological model would permit a broad uncertainty envelope spanning from 1.410 meV up to 3.642 meV. By topologically enforcing the compact-boundary Majorana phase, the PR-II architecture decisively breaks this degeneracy [36]. It mandates an exact, highly falsifiable experimental target of 1.508 meV, residing near the strict lower-interference limit of the allowed parameter space.

### 13.5 Flavor-Basis Majorana Mass Matrix

With the mass spectrum, the unconfined PMNS mixing angles, and the rigid interference phases completely established by the geometric boundary, the framework deterministically projects the complete flavor-basis Majorana mass matrix:

$$M_\nu^{\text{PR}} = U_{\text{PMNS}}^* \text{diag} \left( m_1, m_2 e^{i\alpha_{21}}, m_3 e^{i\alpha_{31}} \right) U_{\text{PMNS}}^\dagger. \quad (435)$$

By construction, this derived matrix satisfies the fundamental Majorana symmetry condition:

$$\left( M_\nu^{\text{PR}} \right)^T = M_\nu^{\text{PR}}. \quad (436)$$

This symmetry is a matrix-level mathematical consequence of the Majorana-dominant projection, distinguishing it from that of an ordinary Dirac operator. This matrix represents the terminal output of the structural derivation of the neutrino sector. The complete projection chain is fully closed, while it originates entirely from the dimension-five ratio  $(v_{\text{EW}}^{\text{PR}})^2/\bar{M}_{\text{Pl}}$ , routes through the compact Majorana channel to yield the exact mass eigenvalues, and finally rotates through the unconfined topological boundary to establish the exact, falsifiable flavor-basis matrix.

### 13.6 Input Discipline and Status

The generative boundary for the neutrino mass architecture is rigidly restricted. The neutrino spectrum and the associated interference phases are generated exclusively from the pre-established structural ledger: the primary boundary coordinates  $(q_{\text{bc}}, c_{\text{bc}})$ , the determinant suppression ratio  $(r_G)$ , the weak boundary deficit  $(\Delta_{\text{EW}})$ , the confinement average  $(C_{\text{conf}})$ , the previously derived unconfined mixing matrix  $(U_{\text{PMNS}}^{\text{PR}})$ , and the fundamental dimensional anchor provided by  $(v_{\text{EW}}^{\text{PR}})^2/\bar{M}_{\text{Pl}}$ .

The architecture derives the masses, mass-squared splittings, and neutrinoless double-beta parameters without any reference to fitted experimental bounds  $(m_i, \Delta m_{ij}^2, \sum m_i, m_\beta, m_{\beta\beta})$  or observed phenomenological oscillation angles. These empirical data may enter the framework only post-generation, serving as diagnostic benchmarks to verify the topological projection.

Because the entire spectrum is geometrically derived rather than empirically fitted, the formal structural status of this section is explicitly designated as a:

$$\text{normal-ordering, Majorana-dominant neutrino mass-sector candidate.} \quad (437)$$

Accordingly, the calculated Majorana phase boundary must be interpreted as a structurally derived candidate prediction. It is not an arbitrary phenomenological fit, but the explicit topological consequence of the compact boundary, formally awaiting definitive experimental verification.

### 13.7 Geometric Rigidity and Falsification Bounds

The derivation of the complete neutrino mass spectrum and Majorana phase boundary is governed by the same geometric rigidity that defines the rest of the PR-II architecture. The framework possesses explicit failure modes that falsify the projection if violated. There is no room for parametric tuning, ad hoc structural omissions, or phenomenological fitting. Every constraint in the neutrino generative chain is mathematically mandatory.

First, the baseline dimension-five topological ratio is required. Stripping the reduced Planck mass suppression factor from the electroweak anchor  $((v_{\text{EW}}^{\text{PR}})^2/\bar{M}_{\text{Pl}})$  artificially forces the neutrino spectrum up to the electroweak scale, completely breaking the structural projection.

Second, artificially reversing the compact determinant hierarchy projection—substituting a geometric suppression  $(r_G^{+3})$  in place of the mandatory topological lift  $(r_G^{-3})$  catastrophically fails to generate the heavy atmospheric mass scale  $(m_3^{\text{PR}})$ .

Third, omitting the compact neutrino-loss geometric boundary factor  $(C_\nu = 1 - \Delta_{\text{EW}c_{bc}})$  from the generation chain irreparably distorts the structural mass-squared splittings, pushing the derived ratio entirely outside the predictive closure.

Fourth, the derivation of the terminal mass matrix mathematically relies on the strict unitarity of the unconfined PMNS projection. Furthermore, if the resulting flavor-basis mass matrix fails to satisfy exact symmetry  $((M_\nu^{\text{PR}})^T = M_\nu^{\text{PR}})$ , the Majorana-dominant projection is fundamentally inconsistent, immediately invalidating the architecture.

Finally, any derivation that imports experimentally observed target parameters such as neutrino masses  $(m_i)$ , phenomenological mass-squared splittings  $(\Delta m_{ij}^2)$ , or observed PMNS mixing angles as generative inputs violates the zero-new-parameter discipline.

The falsification boundary is therefore a ridge set of conditions. The PR-II framework must yield a normal-ordered mass spectrum, the correct hierarchical splittings, a viable cosmological mass sum, and a rigid Majorana interference phase, using only the established compact-boundary geometric constants. If any step in this generation chain requires phenomenological fitting to match physical data, the architecture definitively fails.

To complete the manuscript, we present the assembled final, comprehensive audit map. It explicitly separates the mandatory structural identities from the diagnostic ledgers, formally declaring the no-free-parameter status and the ultimate falsification boundaries of the PR-II manuscript.

## 14 Audit Map, No-Free-Parameter Status, and Falsification Boundaries

The preceding sections developed the PR-II electroweak and flavor chain from the compact boundary ledger inherited from PR-I. Because the construction produces many numerical outputs, the final task is to separate exact structural identities from closure candidates, precision candidates, and diagnostic ledgers. This section defines that audit map and states the no-free-parameter status of the manuscript.

The central claim is not that every electroweak and flavor result has already been promoted to a final precision theorem. The more precise claim is:

|                                                                                                                  |       |
|------------------------------------------------------------------------------------------------------------------|-------|
| PR-II produces a zero-new-parameter compact-boundary<br>closure candidate for the electroweak and flavor ledger. | (438) |
|------------------------------------------------------------------------------------------------------------------|-------|

Target observables are not used as generation inputs. They are used only after the PR-II formulas generate their outputs, as diagnostic references.

### 14.1 Status Categories

The results in this manuscript are classified into five categories.

1. **Structural identities.** These are algebraic or representation-theoretic results that follow directly from the compact electroweak fiber, such as the  $SU(2)_L$  algebra, the residual generator  $Q = T_3 + Y/2$ , photon masslessness, the tree-level  $W/Z$  mass matrix, the Fermi matching identity, neutral-current coupling structure, and anomaly cancellation.
2. **Closure candidates.** These are compact-boundary formulas that generate physically relevant quantities without fitting their measured values. Examples include the weak mixing angle, generation count, strong-coupling anchor, CKM hierarchy, PMNS hierarchy, and neutrino mass-splitting structure.
3. **Precision candidates.** These are closure candidates that numerically agree with reference data at high precision, such as the weak-scale candidate and the charged-lepton precision candidate.
4. **Diagnostic ledgers.** These are required comparison frameworks rather than final predictions. Examples include QCD running, threshold matching, confinement conventions, and radiative electroweak running.
5. **Remaining derivation obligations.** These are compact-boundary rules that currently work as candidates but still require formal uniqueness proofs or radiative/running completion.

This classification is necessary because PR-II spans structural group theory, compact boundary closure, precision numerical candidates, and scheme-dependent QCD diagnostics.

## 14.2 No-Free-Parameter Input Discipline

To definitively codify the zero-new-parameter constraint of the PR-II architecture, we establish a strict generative audit ledger. The mathematical integrity of the framework relies entirely on separating the foundational inputs (which mathematically drive the topology) from the empirical targets (which serve as post-generation diagnostic references).

Table 17 categorizes the permitted structural inputs and explicitly forbids the importation of target observables [13].

**Table 17:** PR-II Generative Input Audit Ledger.

| Parameter Sector           | Specific Variables                                                      | Input Status |
|----------------------------|-------------------------------------------------------------------------|--------------|
| Fundamental Constants      | $\hbar, c, G_N, \bar{M}_{\text{Pl}}$                                    | ALLOWED      |
| Compact-Boundary Data      | $q_{bc}, c_{bc}, r_G, T_Y, \Delta_{\text{EW}}$                          | ALLOWED      |
| Group & Topological Data   | $C_2^{SU(2)}, C_2^{SU(3)}, Y, T_3, N_{\text{gen}}^{\text{PR}}$          | ALLOWED      |
| Electroweak Scale & Bosons | $G_F, v_{\text{EW}}, M_W, M_Z$                                          | FORBIDDEN    |
| Charged Fermion Masses     | $m_e, m_\mu, m_\tau, m_u, m_d, m_s, m_c, m_b, m_t$                      | FORBIDDEN    |
| Flavor-Mixing Matrices     | $V_{\text{CKM}}, U_{\text{PMNS}}$                                       | FORBIDDEN    |
| Strong-Force Observables   | $\alpha_s(M_Z), \Lambda_{\text{QCD}}$                                   | FORBIDDEN    |
| Neutrino Mass Observables  | $\Delta m_{21}^2, \Delta m_{31}^2, \sum_i m_i, m_\beta, m_{\beta\beta}$ | FORBIDDEN    |

The structural derivations presented throughout this manuscript were developed exclusively using the foundational PR variables from the allowed ledger. The physical observables in the forbidden ledger were utilized as external references [14] against which the geometrically generated outputs were evaluated. If any derivation were to import a target observable as a generative input, the zero-new-parameter status would be categorically broken. Such an operation reduces the calculation to a phenomenological fit; it is not a valid PR-II compact-boundary closure.

### 14.3 Grouped Audit Map

Table 18 summarizes the main outputs of the manuscript and their audit status.

**Table 18:** Grouped audit map for the PR-II electroweak and flavor chain. Each row lists the controlling equation, generated output, current status, and remaining boundary condition.

| Sector                       | Controlling equation / closure relation                                                                                                                                                                                                                                                                                                 | Generated output                                                                                                            | Status / remaining boundary                                                                      |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| PR-I boundary ledger         | $q_{bc} = 10.905007182855176,$<br>$c_{bc} = 0.796684464847899,$<br>$\alpha_{PR,bc}^{-1} = 4\pi q_{bc}$                                                                                                                                                                                                                                  | $\alpha_{PR,bc}^{-1} = 137.036361812007$                                                                                    | Inherited compact-boundary data from PR-I. Not refit in PR-II.                                   |
| Electroweak lift             | $\mathcal{M}_{\text{int}}^{\text{EW}} = \mathbb{R}_w \times S_Y^1 \times SU(2)_L,$<br>$[T_a, T_b] = i\epsilon_{abc}T_c$                                                                                                                                                                                                                 | Compact non-Abelian phase fiber and weak projection modes.                                                                  | Structural identity. Requires genuine $SU(2)_L$ , not three Abelian copies.                      |
| Projection locking           | $SU(2)_L \times U(1)_Y \rightarrow U(1)_{\text{em}},$<br>$Q = T_3 + \frac{Y}{2},$<br>$Q\Phi_0 = 0$                                                                                                                                                                                                                                      | Residual electromagnetic boundary and standard charge assignments.                                                          | Structural identity. Must preserve photon masslessness and charges.                              |
| Weak-boson mass ledger       | $M_0^2 = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix},$<br>$\det M_0^2 = 0,$<br>$M_W^2 = \frac{g^2 v_{\text{EW}}^2}{4},$<br>$M_Z^2 = \frac{(g^2 + g'^2)v_{\text{EW}}^2}{4}$                                                                                                                        | $M_\gamma = 0,$<br>$M_W = M_Z \cos \theta_W,$<br>$\rho_{\text{EW}} = 1$                                                     | Structural tree-level ledger. Precision masses require running and radiative matching.           |
| Weak mixing                  | $T_Y = \frac{1}{4},$<br>$\Delta_{\text{EW}} = \frac{1 - c_{bc}}{q_{bc}},$<br>$\sin^2 \theta_W^{\text{PR}} = T_Y - \Delta_{\text{EW}}$                                                                                                                                                                                                   | $\sin^2 \theta_W^{\text{PR}} = 0.231355763298189$                                                                           | Closure candidate. Remaining task: formal uniqueness of the boundary-deficit rule.               |
| Weak scale and Fermi closure | $\Xi_{\text{EW}} = q_{bc} c_{bc} \sin^2 \theta_W^{\text{PR}},$<br>$\Lambda_{\text{EW}}^{\text{PR}} = \bar{M}_{\text{Pl}} e^{-S_{\text{EW}}^{\text{PR}, \text{total}}},$<br>$v_{\text{EW}}^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \sqrt{\Xi_{\text{EW}}},$<br>$G_F^{\text{PR}} = \frac{1}{\sqrt{2}(v_{\text{EW}}^{\text{PR}})^2}$ | $v_{\text{EW}}^{\text{PR}} = 246.219595890 \text{ GeV},$<br>$G_F^{\text{PR}} = 1.166379220 \times 10^{-5} \text{ GeV}^{-2}$ | Precision candidate. Radiative/running precision layer deferred to PR-III.                       |
| Compact order mode           | $\Phi(x) = \frac{1}{\sqrt{2}} U(x) \begin{pmatrix} 0 \\ v_{\text{EW}}^{\text{PR}} + h_{\text{PR}}(x) \end{pmatrix},$<br>$4 - 3 = 1$                                                                                                                                                                                                     | One scalar order-mode excitation: $h_{\text{PR}}$ .                                                                         | Tree-level scalar-coupling ledger. Scalar mass is deferred to compact Hessian closure in PR-III. |

Continued on next page.

| Sector                        | Controlling equation / closure relation                                                                                                                                                                       | Generated output                                                             | Status / remaining boundary                                                                                            |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Scalar coupling ledger        | $g_{hWW}^{\text{PR}} = \frac{2M_W^2}{v_{\text{EW}}^{\text{PR}}},$ $g_{hZZ}^{\text{PR}} = \frac{2M_Z^2}{v_{\text{EW}}^{\text{PR}}},$ $g_{hff}^{\text{PR}} = \frac{m_f^{\text{PR}}}{v_{\text{EW}}^{\text{PR}}}$ | Tree-level $hWW$ , $hZZ$ , and mass-proportional $hff$ couplings.            | Structural tree-level ledger. $m_h$ , scalar self-couplings, and longitudinal unitarity amplitudes are PR-III targets. |
| Charged currents              | $T^\pm = T_1 \pm iT_2,$ $[Q, T^\pm] = \pm T^\pm,$ $\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2} = \frac{1}{2v_{\text{EW}}^2}$                                                                                    | Beta-transition topology and low-energy Fermi limit.                         | Structural identity. Must preserve charged-current normalization and charge bookkeeping.                               |
| Neutral currents              | $g_L^f = T_3^f - Q_f \sin^2 \theta_W,$ $g_R^f = -Q_f \sin^2 \theta_W,$ $g_V^f = T_3^f - 2Q_f \sin^2 \theta_W,$ $g_A^f = T_3^f$                                                                                | Tree-level $Z$ -coupling ledger and neutral-current contact structure.       | Structural identity. Precision comparison requires electroweak radiative matching.                                     |
| Fermion representation ledger | $\mathcal{R}_{\text{EW}}^{(1)} = (3, 2)_{1/3} \oplus (\bar{3}, 1)_{-4/3} \oplus (\bar{3}, 1)_{2/3} \oplus (1, 2)_{-1} \oplus (1, 1)_2$                                                                        | Minimal one-generation chiral ledger.                                        | Structural identity. Uses left-handed Weyl convention.                                                                 |
| Anomaly cancellation          | $[SU(3)_c]^3 = 0,$ $[SU(3)_c]^2 U(1)_Y = 0,$ $[SU(2)_L]^2 U(1)_Y = 0,$ $[U(1)_Y]^3 = 0,$ $[\text{grav}]^2 U(1)_Y = 0,$ $N_2 \in 2\mathbb{Z}$                                                                  | Gauge and global anomaly consistency.                                        | Structural identity. Failure of any anomaly check breaks the fermion ledger.                                           |
| Generation origin             | $N_{\text{gen}}^{\text{PR}} = \dim \left[ \frac{SU(2)_L \times U(1)_Y}{U(1)_{\text{em}}} \right] = 3$                                                                                                         | Three compact projection sheets.                                             | Closure candidate. Remaining task: formalize projection-sheet interpretation.                                          |
| Generation hierarchy          | $r_G = \frac{c_{\text{bc}}}{q_{\text{bc}}},$ $\eta_i = r_G^{N_{\text{gen}} - i},$ $y_i^{\text{PR}} = y_T^{\text{PR}} \eta_i$                                                                                  | $y_1 < y_2 < y_3$ with fixed hierarchy ratio $q_{\text{bc}}/c_{\text{bc}}$ . | Structural flavor ledger. Not yet a mass prediction by itself.                                                         |
| Sector overlap operators      | $C_f = C_2^{SU(3)} + C_2^{SU(2)} + \left(\frac{Y_L}{2}\right)^2 + \left(\frac{Y_R}{2}\right)^2,$ $D_f = \kappa_f \text{diag}(y_1, y_2, y_3)$                                                                  | Charged-lepton, up-quark, down-quark, and neutrino overlap ledgers.          | Structural ledger. Requires sector-specific scale/running rules before mass comparison.                                |

Continued on next page.

| Sector                 | Controlling equation / closure relation                                                                                                                                                                                                                                                                                                                                                                                                | Generated output                                                                                                                                                                                      | Status / remaining boundary                                                                                   |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Charged leptons        | $\theta_K^{\text{PR}} = \frac{2}{N_{\text{gen}}^2} = \frac{2}{9},$ $k_a(\theta_K) = \left[ 1 + \sqrt{2} \cos \left( \theta_K + \frac{2\pi a}{3} \right) \right]^2,$ $m_a^{\text{PR}} = M_\ell^{\text{PR}} k_a$                                                                                                                                                                                                                         | $m_e^{\text{PR}} = 0.510984463 \text{ MeV},$ $m_\mu^{\text{PR}} = 105.656418843 \text{ MeV},$ $m_\tau^{\text{PR}} = 1776.934589789 \text{ MeV}$                                                       | Precision candidate.<br>Remaining task: formal uniqueness of Koide sheet phase and compact scale corrections. |
| Strong coupling        | $\left( \alpha_3^{\text{PR}} \right)^{-1} = q_{\text{bc}} c_{\text{bc}} - (1 - c_{\text{bc}})$                                                                                                                                                                                                                                                                                                                                         | $\alpha_3^{\text{PR}} = 0.117861507469150$                                                                                                                                                            | Closure candidate.<br>Higher-loop and scheme matching remain part of QCD precision layer.                     |
| QCD running ledger     | $\beta_0(n_f) = 11 - \frac{2}{3} n_f,$ $\alpha_s^{(n_f)}(\mu) = \frac{4\pi}{\beta_0 \ln(\mu^2/\Lambda_{n_f}^2)},$ $\alpha_s^{(n_f)}(\mu_{\text{th}}) = \alpha_s^{(n_f-1)}(\mu_{\text{th}})$                                                                                                                                                                                                                                            | $\Lambda_6, \Lambda_5, \Lambda_4, \Lambda_3 \text{ running and threshold ledger.}$                                                                                                                    | Diagnostic ledger. Not a pole-mass theorem; QCD scheme/scale dependence remains.                              |
| Heavy-quark thresholds | $\mu_t^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} C_1 C_2,$ $\mu_b^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} \frac{r_G}{3} (1 - \Delta_{\text{EW}} c_{\text{bc}}),$ $\mu_c^{\text{PR}} = \Lambda_{\text{EW}}^{\text{PR}} t_G^2 \left( \frac{4}{3} \right) (1 + \Delta_{\text{EW}} A_{\text{EW}})$                                                                                                                               | $\mu_t^{\text{PR}} = 172.813973267 \text{ GeV},$ $\mu_b^{\text{PR}} = 4.166450483 \text{ GeV},$ $\mu_c^{\text{PR}} = 1.274964814 \text{ GeV}$                                                         | Threshold-scale candidates. Not final scheme-independent quark masses.                                        |
| Light-quark hierarchy  | $C_{\text{conf}} = \frac{1 + c_{\text{bc}}}{2},$ $m_s^{\text{PR}} = \mu_b^{\text{PR}} \left( \frac{r_G}{3} \right) (1 + \Delta_{\text{EW}} A_{\text{EW}}) C_{\text{conf}},$ $m_d^{\text{PR}} = m_s^{\text{PR}} r_G c_{\text{bc}} (1 - A_{\text{EW}} \Delta_{\text{EW}}) C_{\text{conf}},$ $m_u^{\text{PR}} = m_d^{\text{PR}} \left( \frac{1 + c_{\text{bc}}}{4} \right) \left( 1 + \frac{\Delta_{\text{EW}} c_{\text{bc}}}{4} \right)$ | $m_u^{\text{PR}} = 2.146462595 \text{ MeV},$ $m_d^{\text{PR}} = 4.761039493 \text{ MeV},$ $m_s^{\text{PR}} = 94.028510538 \text{ MeV}$                                                                | Low-energy running-mass diagnostics. Not pole masses.                                                         |
| CKM mixing             | $V_{\text{CKM}} = U_{u,L}^\dagger U_{d,L},$ $\lambda_{\text{PR}} = \sqrt{r_G c_{\text{bc}} C_{\text{conf}} (1 - A_{\text{EW}} \Delta_{\text{EW}})},$ $\delta_{\text{CKM}}^{\text{PR}} = \frac{\pi}{3} + \sqrt{\Delta_{\text{EW}}}$                                                                                                                                                                                                     | $ V_{us}  = 0.225018430,$ $ V_{cb}  = 0.041655455,$ $ V_{ub}  = 0.003730932,$ $J_{\text{CKM}}^{\text{PR}} = 3.152605677 \times 10^{-5}$                                                               | Closure candidate.<br>Remaining task: prove compact CKM phase-rule uniqueness.                                |
| PMNS mixing            | $U_{\text{PMNS}} = U_{\ell,L}^\dagger U_{\nu,L},$ $\sin^2 \theta_{12}^{\text{PR}} = \frac{C_{\text{conf}}}{N_{\text{gen}}},$ $\sin^2 \theta_{23}^{\text{PR}} = \frac{A_{\text{EW}}}{N_{\text{gen}}},$ $\sin^2 \theta_{13}^{\text{PR}} = \frac{\Delta_{\text{EW}} (1 + r_G)}{C_{\text{conf}}}$                                                                                                                                          | $\theta_{12}^{\text{PR}} = 33.176356643^\circ,$ $\theta_{23}^{\text{PR}} = 48.735310403^\circ,$ $\theta_{13}^{\text{PR}} = 8.582438059^\circ,$ $\delta_{\text{CP}}^{\text{PR}} = 195.646778408^\circ$ | Closure candidate. Must connect to neutrino mass-sector phase structure.                                      |

Continued on next page.

| Sector               | Controlling equation / closure relation                                                                                                                                                                  | Generated output                                                                                                                                                                                                                                                                | Status / remaining boundary                                                                   |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Neutrino masses      | $m_1^{\text{PR}} = 0,$<br>$m_3^{\text{PR}} = \left( \frac{(v_{\text{EW}}^{\text{PR}})^2}{M_{\text{Pl}}} \right) r_G^{-3} c_{bc} C_\nu,$<br>$m_2^{\text{PR}} = \sqrt{R_{21}^{\text{PR}} m_3^{\text{PR}}}$ | $m_2^{\text{PR}} = 8.627283010 \times 10^{-3} \text{ eV},$<br>$m_3^{\text{PR}} = 5.010655367 \times 10^{-2} \text{ eV},$<br>$\Delta m_{21}^{2,\text{PR}} = 7.443001214 \times 10^{-5} \text{ eV}^2,$<br>$\Delta m_{31}^{2,\text{PR}} = 2.510666721 \times 10^{-3} \text{ eV}^2$ | Normal-ordering<br>Majorana-dominant<br>candidate. Majorana<br>phase theorem remains<br>open. |
| Majorana<br>boundary | $\phi_M^{\text{PR}} = \pi C_{\text{conf}},$<br>$m_{\beta\beta}^{\text{PR}} = \left  m_2 c_{13}^2 s_{12}^2 + m_3 s_{13}^2 e^{i\phi_M^{\text{PR}}} \right $                                                | $m_{\beta\beta}^{\text{PR}} = 1.508 \text{ meV}$                                                                                                                                                                                                                                | Closure candidate.<br>Experimental test lies<br>beyond current $0\nu\beta\beta$<br>reach.     |

This table is intentionally conservative. It prevents structural recovery, precision candidates, and diagnostic comparisons from being conflated.

## 14.4 Reproducibility Ledger

To enforce arithmetic consistency and verify the strict zero-new-parameter discipline, the PR-II electroweak and flavor construction is fully supported by a test-driven reproducibility ledger. All structural derivations and numerical projections have been independently codified and verified using dedicated Python validation harnesses and Maple algebraic checkers [1]. The complete validation suite, including the evaluation scripts, mathematical outputs, and the generative audit manifest, is publicly available at:

[https://github.com/oshetskiresearch/Projection\\_Relativity](https://github.com/oshetskiresearch/Projection_Relativity)

This repository is not a substitute for formal mathematical derivation; rather, it provides a transparent, immediately reproducible record to ensure that no target observables were hidden within the generation chain for every numerical claim established in this manuscript.

## 14.5 Remaining Derivation Obligations

The PR-II architecture successfully establishes a comprehensive landscape of zero-new-parameter closure candidates. However, the strict mandate of this research program requires elevating these initial structural projections into formally complete, radiatively exact physical theorems. The remaining open obligations do not permit the introduction of phenomenological fitting parameters; rather, they represent rigorous mathematical and boundary-matching requirements. The central architectural objectives for the subsequent derivation layer are defined as follows:

1. **Electroweak Boundary Uniqueness:** Prove the formal mathematical uniqueness of the compact boundary-deficit rule governing the weak mixing angle ( $\sin^2 \theta_W^{\text{PR}}$ ), demonstrating that no alternative topological mappings are permitted.
2. **Non-Abelian Action Integration:** Derive the second-order non-Abelian weak-scale correction explicitly as a deterministic compact-boundary action term to finalize the precision electroweak hierarchy.



3. **Radiative Matching Maps:** Complete the formal electromagnetic and electroweak running layers—specifically the exact  $\alpha_{\text{PR}}(0) \rightarrow \alpha_{\text{PR}}(M_Z)$  transition—to enable rigorous high-energy precision comparisons.
4. **Generation Topology:** Formalize the precise topological mechanism that rigidly dictates the projection-sheet origin of the three-generation chiral ledger ( $N_{\text{gen}}^{\text{PR}} = 3$ ).
5. **Charged-Lepton Phase Rigidity:** Prove that the specific fractional Koide sheet phase ( $\theta_K^{\text{PR}} = 2/9$ ) is a unique, mathematically inescapable consequence of the flavor-boundary geometry.
6. **QCD Precision Completion:** Finalize the QCD renormalization scheme, running maps, confinement conventions, and threshold-matching prescriptions required to compare the generated compact quark thresholds and light-quark candidates with scheme specified running and threshold mass definitions.
7. **Flavor Mixing Uniqueness:** Derive the exact CKM and PMNS topological phase rules as , unique compact flavor-projection phases, formally closing the unitary mixing architecture.
8. **Majorana Phase Theorem:** Elevate the structural Majorana phase boundary from a geometric closure candidate into a complete, formally proven neutrino-sector theorem governing neutrinoless double-beta decay.

Fulfilling these obligations relies entirely on the application of higher-order topological, gauge, and radiative matching disciplines. The derivation of the ledger constitutes the formal mandate for Projection Relativity III (PR-III). Executing this program will transition the PR-II architecture from a predictive-closure candidate to a finalized, zero-new-parameter theoretical proof, completing the full framework.

## 15 Conclusion

Projection Relativity II (PR-II) successfully extends the geometric architecture of PR-I into the microscopic electroweak and flavor sectors. The guiding objective of this manuscript was not to discard the highly successful Standard Model, but to resolve its reliance on an extensive ledger of arbitrary, empirically fitted parameters. The result developed here demonstrates that the standard electroweak gauge theory, the anomaly-free chiral generation structure, and the diverse flavor mass hierarchies can be fundamentally reorganized as a deterministic, compact-boundary projection.

Rather than treating electroweak symmetry breaking, mass generation, and flavor mixing as independent phenomenological mechanisms, PR-II unifies them under a single topological origin. The framework successfully derived the following:

- **The Electroweak Architecture:** The standard  $W^\pm$  and  $Z$  boson masses, the weak mixing angle ( $\sin^2 \theta_W^{\text{PR}}$ ), and the Fermi constant ( $G_F^{\text{PR}}$ ) emerge from the broken directions of the compact electroweak fiber.
- **The Generation Topology:** The existence of exactly three anomaly-free chiral generations ( $N_{\text{gen}}^{\text{PR}} = 3$ ) is established as a geometric mandate of the broken compact projection space, the historical ambiguity of anomaly cancellation alone.
- **The Mass and Mixing Ledger:** The extreme hierarchies of the charged leptons and quarks, alongside the exact unitary rotations of the CKM and PMNS matrices, are generated directly from compact determinant ratios, weak boundary deficits, and representation overlaps.

- **The Neutrino Sector:** The mass scale is pushed to the sub-eV regime by a native dimension-five topological suppression, enforcing a normal-ordered, Majorana-dominant spectrum and yielding a single compact-boundary candidate for neutrinoless double-beta decay.

The defining structural achievement of this manuscript is its geometric rigidity. Throughout these derivations, the target physical observables were prohibited from acting as generative inputs. The entire landscape of electroweak and flavor candidates was derived exclusively from the fundamental constants and the geometric boundary coordinates established in PR-I.

Consequently, the final status of this work is explicitly defined as a:

$$\boxed{\text{zero-new-parameter compact-boundary electroweak and flavor closure candidate.}} \quad (439)$$

This manuscript establishes the foundational geometry, but it does not claim to present a finalized, radiatively complete precision theorem. The formal transition from a rigid "closure candidate" to a complete physical theorem constitutes the mandate for Projection Relativity III (PR-III). The remaining architectural obligations include establishing the formal mathematical uniqueness of the compact-boundary correction rules, integrating the complete electromagnetic and electroweak running layers, formalizing the exact QCD threshold-matching prescriptions, and elevating the structural flavor matrices into scheme-independent theorems.

Ultimately, the significance of PR-II lies not in overturning standard gauge theory but in anchoring it. The framework preserves the established currents, anomalies, and effective low-energy structures, while demonstrating that the Standard Model's seemingly arbitrary inputs are, in fact, the mathematical shadows of a shared compact-boundary geometry. If the remaining radiative and running obligations are fulfilled, this architecture would convert the bulk of the Standard Model's phenomenological parameter ledger into pure geometry. If it cannot, the explicit failure modes identified throughout this manuscript provide a strict mathematical boundary that defines exactly where the projection program breaks down.

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